Vitaran Edutech

HINTS & SOLUTIONS

INTEGRATED PROGRAM

MATHEMATICS

Class – 10

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1. REAL NUMBERS

Solutions

Worksheet – 1

1. **Correct Answer:180**
   Given
   $a = 2^2 \times 3^3 \times 5^4, b = 2^3 \times 3^2 \times 5$
   To find HCF, take the smallest power of each common prime factor.
   Common primes are 2, 3 and 5.
   $\text{HCF} = 2^{\min(2,3)} \times 3^{\min(3,2)} \times 5^{\min(4,1)}$
2. **Correct Answer: 60**
   Given numbers:
   $2^3 \times 3^2 \times 5, 2^2 \times 3^3 \times 5^2, 2^4 \times 3 \times 5^3 \times 7$
   For HCF, take only common prime factors with the smallest powers.
   Common primes in all three numbers are 2, 3 and 5.
   $\text{HCF} = 2^{\min(3,2,4)} \times 3^{\min(2,3,1)} \times 5^{\min(1,2,3)}$
   $= 2^2 \times 3 \times 5$
   $= 4 \times 3 \times 5 = 60$
3. **Correct Answer: 1680**
   Given:
   $2^3 \times 3 \times 5$ and $2^4 \times 5 \times 7$
   To find LCM, take all prime factors with the highest powers.
   $\text{LCM} = 2^4 \times 3 \times 5 \times 7$
   $= 16 \times 3 \times 5 \times 7$
   =1680
4. **Correct Answer: 81**
   Given:
   HCF = 27
   LCM = 162
   One number = 54
   We use the formula:
   Product of two numbers = HCF × LCM
   Product $= 27 \times 162 = 4374$
   Let the other number be x.
   $54 \times x = 4374$
   $x = \frac{4374}{54} = 81$
5. **Correct Answer: C) 320**
   The product of two numbers is 1600 and their HCF = 5. Find the LCM.
   Solution:
   We know the formula:
   Product of two numbers = HCF × LCM
   $1600 = 5 \times \text{LCM}$
   $\text{LCM} = \frac{1600}{5} = 320$
6. **Correct Answer: D)** $\frac{15}{16}$
   Find the simplest form of $\frac{1095}{1168}$
   Solution:
   Find the HCF of 1095 and 1168.
   1095=3×5×73
   $1168 = 16 \times 73$
   Common factor = 73
   Divide numerator and denominator by 73:
   $\frac{1095}{1168} = \frac{15}{16}$
7. **Correct Answer: C)** $0 \leq r < b$
   According to Euclid's Division Lemma,
   $a = bq + r$
   What condition must r satisfy?
   Solution:
   Euclid's Division Lemma states that the remainder r is always:
   $0 \leq r < b$
8. **Correct Answer: D) 5**
   A number when divided by 143 leaves a remainder 31. Find the remainder when the same number is divided by 13.
   Solution:
   $143 = 11 \times 13$
   The number can be written as:
   $143k + 31$
   When divided by 13:
   143k is divisible by 13
   So remainder depends only on 31:
   $31 \div 13 = 2$ remainder 5
9. **Correct Answer: B) 22**
   Find the number of natural numbers up to 150 that are divisible by 7.
   Solution:
   Smallest number divisible by 7 is 7.
   Largest multiple of 7 less than or equal to 150:
   $150 \div 7 = 21.42$
   So, largest multiple = 21×7

Total numbers:= 21

10. Correct Answer: B) 22

LCM of two numbers = 105

HCF = 1

Difference of numbers = 8

Find the sum of the numbers.

Let the numbers be $a$and b.

$$a \times b = \text{LCM} \times \text{HCF} = 105$$

Factorize 105:

$$105 = 15 \times 7$$

Difference:$15 - 7 =$ 8(matches given condition)

Sum:$15 + 7 = 22$

WORKSHEET-2

1. Correct Answer: C

Which of the following rational number is expressible as a terminating decimal?

A rational number $\frac{p}{q}$(in lowest form) has a terminating decimal if the prime factors of $q$are only 2 and/or 5.

- $\frac{124}{165} \rightarrow 165 = 3 \times 5 \times 11$
- $\frac{131}{30} \rightarrow 30 = 2 \times 3 \times 5$
- $\frac{2027}{625} \rightarrow 625 = 5^4$
- $\frac{1625}{462} \rightarrow 462 = 2 \times 3 \times 7 \times 11$

2. Correct Answer: B) Two decimal places

The decimal expansion of the rational number $\frac{37}{2^2 \times 5}$will terminate after

$$2^2 \times 5 = 20 \Rightarrow \frac{37}{20} = 1.85$$

It has 2 decimal places.

3. Correct Answer: C) A rational number

The number 1.732 is

$$1.732 = \frac{1732}{1000}$$

It can be written as a ratio of integers.

Answer: B) A rational number

4. Correct Answer: C) An irrational number

$\frac{1}{\sqrt{2}}$is

$\sqrt{2}$is irrational, and dividing 1 by an irrational number gives an irrational number.

5. Correct Answer: B)-288

$(-2)^5 \times 3^2$

$(-2)^5 = -32, 3^2 = 9$

$-32 \times 9 = -288$

6. Correct Answer: B) D) 0

$(-1)^{2001} + (-1)^{2002} + \cdots + (-1)^{2010}$

- Odd powers of (-1) = −1
- Even powers of (-1)= +1

From 2001 to 2010 ⟶ 10 terms

There are 5 pairs of (−1 + 1) = 0

$$\text{Total sum} = 0$$

7. Correct Answer: B) $-\frac{128}{675}$

The product of the cube of $-\frac{2}{3}$and the square of $-\frac{4}{5}$

$\left(-\frac{2}{3}\right)^3 = -\frac{8}{27}, \left(-\frac{4}{5}\right)^2 = \frac{16}{25}$

$\text{Product} = -\frac{8}{27} \times \frac{16}{25} = -\frac{128}{675}$

8. Correct Answer: C) 125

$\left[(3^2 - 2^2) \div \left(\frac{1}{5}\right)^2\right]$

$3^2 - 2^2 = 9 - 4 = 5$

$\left(\frac{1}{5}\right)^2 = \frac{1}{25}$

$5 \div \frac{1}{25} = 5 \times 25 = 125$

9. Correct Answer: B) 5^5

The exponential form of 3125

$3125 = 5 \times 5 \times 5 \times 5 \times 5 = 5^5$

10. Correct Answer: D) 1

$\frac{3^5 \times 10^5 \times 25}{5^7 \times 6^5}$

Write in prime factors:

$10^5 = (2 \times 5)^5 = 2^5 \times 5^5$

$25 = 5^2, 6^5 = (2 \times 3)^5 = 2^5 \times 3^5$

Cancel common terms:

$= \frac{3^5 \times 2^5 \times 5^7}{5^7 \times 2^5 \times 3^5} = 1$

COMPITETIVE CORNER

SINGLE OPTION CORRECT TYPE

1. Correct Answer: C) 25

Natural numbers divisible by 8 up to 200 are

8,16,24,...,200

Number of such terms

=200/8=25

2. Correct Answer: A) 10

Numbers between 11 and 111 that are multiples of both 2 and 5

A number divisible by both 2 and 5 must be divisible by LCM(2,5) = 10
Multiples of 10 between 11 and 111:
20, 30, 40, ..., 110
Number of terms
$\frac{110-20}{10}+1=10$

3. **Correct Answer: B) 5**

Unit digit in the product of first 50 odd natural numbers
Odd numbers include 5 and multiples of 5.
Product will contain a factor 5, but no factor 2 → unit digit 5

4. **Correct Answer: D) 2, 4, 6**

LCM = 12, numbers in the ratio 1 : 2 : 3
Let numbers be $x, 2x, 3x$
LCM of $x, 2x, 3x = 6x$
Given $6x = 12 \Rightarrow x = 2$
Numbers: 2, 4, 6

5. **Correct Answer: B) 34**

Let the two numbers be 2x and 2y (since HCF = 2), where gcd(x,y)=1.
LCM =2xy=120⇒xy=60
Difference:
2y−2x=14⇒y−x=7
So,x(x+7)=60⇒x=5,y=12
Numbers:
2x=10,2y=24
Sum:
10+24=34

6. **Correct Answer: C)**

If $x^{1/3}+y^{1/3}+z^{1/3}=0$
Using log property:
$\log\left(\frac{x+y+z}{3}\right)=\frac{1}{3}(\log x+\log y+\log z)$
$\frac{1}{3}(\log x+\log y+\log z)$

7. **Correct Answer: B)** $x^3-x-1=0$

The given expression is interpreted as
$\log_e(x-1)+\log_e(x)+\log_e(x+1)=0$
Step 1: Use log property
$\log a+\log b+\log c=\log(abc)$
So,
$\log_e[(x-1)x(x+1)]=0$
Step 2: Convert log equation to exponential form
$(x-1)x(x+1)=e^0=1$
Step 3: Simplify
$x(x^2-1)=1$
$x^3-x-1=0$

8. **Correct Answer: C) 22338**

Given that HCF (306, 657) = 9, find LCM (306, 657)
Formula:
HCF × LCM = Product of the numbers
$\text{LCM}=\frac{306\times 657}{9}$
$306\times 657=201042$
$\text{LCM}=\frac{201042}{9}=22338$

9. **Correct Answer: A)** $a^{15}b^6$

Using laws of exponents, simplify $(a^5b^2)^3$
$(a^5b^2)^3=a^{5\times 3}b^{2\times 3}$
$=a^{15}b^6$

10. **Correct Answer: C)** $3(\log 7-\log 5)$

Expand $\log\frac{343}{125}$
$\log\frac{343}{125}=\log 343-\log 125$
$343=7^3, 125=5^3$
$=3\log 7-3\log 5$
$=3(\log 7-\log 5)$

ONE OR MORE THAN ONE OPTION TYPE

11. **Correct Answer: A, B, C**

Rational numbers between $-\frac{1}{2}$ and $\frac{1}{2}$
Check each option:

- $\frac{3}{10}=0.3\rightarrow$ lies between
- $0\rightarrow$ lies between
- $\frac{21}{100}=0.21\rightarrow$ lies between
- $\frac{2}{3}=0.66\rightarrow$ not between

12. **Correct Answer: A, B**

The terminating decimals among the following

A fraction terminates if the denominator has only 2 and/or 5 as prime factors.

- $\frac{5}{32}\rightarrow 32=2^5$
- $\frac{27}{80}\rightarrow 80=2^4\times 5$
- $\frac{11}{24}\rightarrow 24=2^3\times 3$
- $\frac{5}{12}\rightarrow 12=2^2\times 3$

Correct Answers: A, B

13. Correct Answer: D) $2\frac{13}{33}$

Find the value of

$$0.\bar{2} + 0.\bar{3} + 0.\bar{4} + 0.\bar{9} + 0.3\bar{9}$$

Convert to fractions:

$$0.\bar{2} = \frac{2}{9},\ 0.\bar{3} = \frac{3}{9},\ 0.\bar{4} = \frac{4}{9},\ 0.\bar{9} = 1,\ 0.3\bar{9} = \frac{39}{99}$$

$$= \frac{2+3+4}{9} + 1 + \frac{39}{99}$$

$$= 1 + 1 + \frac{39}{99}$$

$$= 2 + \frac{13}{33} = 2\frac{13}{33}$$

14. Correct Answer:

The problem asks:
If $\log_a x = b$, which statement can be correct regarding the rationality of x?

First, convert the logarithmic equation into exponential form:

$$x = a^b$$

We now examine whether $x$can be a rational number under each given condition.

Evaluation of Options

A) If $a$and $b$are irrational, then $x$can be rational.

True.
Example:
Let $a = \sqrt{2}$(irrational) and $b = \log_{\sqrt{2}} 3$(irrational).
Then,

$$x = (\sqrt{2})^{\log_{\sqrt{2}} 3} = 3$$

Since 3is rational, this case is possible.

B) If $a$is rational and $b$is irrational, then $x$can be rational.

True.
Example:
Let $a = 2$(rational) and $b = \log_2 3$(irrational).
Then,

$$x = 2^{\log_2 3} = 3$$

Since 3is rational, this case is possible.

C) If $a$is irrational and $b$is rational, then $x$can be rational.

True.
Example:
Let $a = \sqrt{2}$(irrational) and $b = 2$(rational).
Then,

$$x = (\sqrt{2})^2 = 2$$

Since 2is rational, this case is possible.

D) If $a$is rational and $b$is rational, then $x$can be rational.

True.
Example:
Let $a = 2$and $b = 2$.
Then,

$$x = 2^2 = 4$$

Since 4 is rational, this case is possible.

Conclusion

All four options A, B, C, and D describe situations that can be correct.

15. Correct Answer: log5

If

$$\log(x+y) = \log\left(\frac{3x-3y}{2}\right)$$

find $\log x - \log y$.

Solution:

Since logarithms are equal, their arguments are equal:

$$x + y = \frac{3x-3y}{2}$$

Multiply both sides by 2:

$$2x + 2y = 3x - 3y$$

Rearranging,

$$2x - 3x = -3y - 2y$$
$$-x = -5y \Rightarrow x = 5y$$

Now, $\log x - \log y = \log\left(\frac{x}{y}\right) = \log 5$

16. Correct Answer: A $\boxed{-25a^8}$**, C)** -5^2a^8

Evaluate$(-1)^3 \times (a^2)^3 \times (5a)^2$
Solution:

$$(-1)^3 = -1$$
$$(a^2)^3 = a^6$$
$$(5a)^2 = 25a^2$$

Multiply all terms:

$$-1 \times a^6 \times 25a^2 = -25a^8$$

17. Correct Answer: $C)\ \boxed{5^4}$**, D) 625**

Evaluate $\left[\left\{\left(\frac{1}{5}\right)^{-2}\right\}^2\right]^1$

Solution:

$\left(\frac{1}{5}\right)^{-2} = 5^2$
$(5^2)^2 = 5^4$
$(5^4)^1 = 5^4$
Answer:

18. Correct Answer:

Solution:

$2\log\ 5 = \log\ 25$
$\frac{1}{2}\log\ 9 = \log\ 3$
Substitute:

$\log\ 25 + \log\ 3 - \log\ 3 = \log\ x$
$\log\ 25 = \log\ x$
So,$x = 25$

ASSERTION AND REASON TYPE

19. Correct Answer:

The correct option is: (C) √2 + √3 is irrational, but the sum of two irrationals can be rational (counterexample: √2 + (−√2) = 0).

20. Correct Answer:

The correct option is: (B) property "p | $a^2 \Rightarrow$ p | a" is true, uniqueness of prime factorization is true, but it is not the direct explanation.

21. Correct Answer:

The correct option is: (A) proof uses Euclid's method extended to primes of the form 4n + 3.

22. Correct Answer:

The correct option is: (A) direct application of the theorem on decimal expansion of rationals.

23. Correct Answer:

The correct option is: (A) because $n^5 - n$ is divisible by 2 (consecutive product), by 3 (consecutive product), and by 5 (Fermat's theorem), hence divisible by 30

MATRIX MATCHING TYPE

24. Correct Answer:

(a) If $2^{a+1} \times 5^a = 200$, find a

$2^{a+1} \times 5^a = 2 \times (2^a 5^a) = 2 \times 10^a$
$2 \times 10^a = 200$
$10^a = 100 = 10^2 \Rightarrow a = 2$
Match: $a \rightarrow$ Q (2)

(b)$\frac{2^{x-1} \times 4^{2x+1}}{8^{x-1}} = 64$

Write all terms in base 2:

$4 = 2^2, 8 = 2^3, 64 = 2^6$
$\frac{2^{x-1} \times (2^2)^{2x+1}}{(2^3)^{x-1}} = 2^6$
$\frac{2^{x-1} \times 2^{4x+2}}{2^{3x-3}} = 2^6$
$2^{(x-1+4x+2)-(3x-3)} = 2^6$
$2^{2x+4} = 2^6 \Rightarrow 2x + 4 = 6 \Rightarrow x = 1$

Match: $b \rightarrow$ R (1)

(c) $\frac{2^1+2^0+2^{-1}}{2^{-2}+2^{-3}+2^{-4}}$

Numerator:$2 + 1 + \frac{1}{2} = \frac{7}{2}$

Denominator:

$\frac{1}{4} + \frac{1}{8} + \frac{1}{16} = \frac{7}{16}$
$\frac{\frac{7}{2}}{\frac{7}{16}} = 8$
Match: $c \rightarrow$ S (8)

(d)

$5^{-5} \times 5^{2n+1} = 5^{13} \div 5^5$
$5^{2n-4} = 5^8 \Rightarrow 2n - 4 = 8 \Rightarrow n = 6$
Match: $d \rightarrow$ P (6)

25. Correct Answer:

1) $\log_{0.5} 0.0625$

$0.5 = \frac{1}{2}, 0.0625 = \frac{1}{16} = \left(\frac{1}{2}\right)^4$
$\log_{1/2}\left(\frac{1}{16}\right) = 4$

Value = 4

2) $\log_2 8$

$8 = 2^3 \Rightarrow \log_2 8 = 3$
Value = 3 (Prime number)

3) $\log_4 3 \cdot \log_5 4 \cdot \log_3 5$
Using property:

$\log_a b \cdot \log_b c = \log_a c$

$(\log_4 3 \cdot \log_3 5) \cdot \log_5 4 = \log_4 5 \cdot \log_5 4$
$= 1$

Value = 1 (Odd number)

4) $\log_3(\log_2 512)$

$512 = 2^9 \Rightarrow \log_2 512 = 9$
$\log_3 9 = 2$
Value = 2 (Perfect square)

COMPREHENSION TYPE

Passage – I

26. Correct Answer: $\boxed{\frac{125}{4}}$

Simplify

$$\frac{8^{-2} \times 5^3}{2^{-4}}$$

Solution:

Write all terms with the same base:

$8 = 2^3 \Rightarrow 8^{-2} = (2^3)^{-2} = 2^{-6}$
So the expression becomes:

$$\frac{2^{-6} \times 5^3}{2^{-4}}$$

Using the law $\frac{a^m}{a^n} = a^{m-n}$:

$= 2^{-6-(-4)} \times 5^3 = 2^{-2} \times 125$
$2^{-2} = \frac{1}{4}$
$= \frac{125}{4}$

Option: B

27. Correct Answer: C $\boxed{2}$

If

$5^m \div 5^{-3} = 5^5$
find m.

Solution:

Using the law:

$a^m \div a^n = a^{m-n}$
$5^{m-(-3)} = 5^5$
$5^{m+3} = 5^5$
Comparing powers:

$m + 3 = 5$
$m = 2$

28. Correct Answer: B $\boxed{36}$

Simplify

$$\frac{12^4 \times 3^{-2}}{4^3}$$

Solution:

Write each term in prime factors:

$12 = 2^2 \times 3 \Rightarrow 12^4 = 2^8 \times 3^4$
$3^{-2} = \frac{1}{3^2}, 4^3 = (2^2)^3 = 2^6$
Substitute:

$$\frac{2^8 \times 3^4 \times 3^{-2}}{2^6}$$

Simplify powers:

$= 2^{8-6} \times 3^{4-2} = 2^2 \times 3^2 = 4 \times 9 = 36$

Passage – II

29. Correct Answer: A) 0

Number of solutions of

$x^{\log_x (x+3)^2} = 16$
Using the identity $a^{\log_a b} = b$,

$x^{\log_x (x+3)^2} = (x+3)^2$
So,

$(x+3)^2 = 16$
$x + 3 = \pm 4 \Rightarrow x = 1$ or $x = -7$
Domain conditions for $\log_x$:
$x > 0$and $x \neq 1$

- $x = 1$ ✗ (log base 1 is not defined)
- $x = -7$ ✗ (base must be positive)

No valid solution

30. Correct Answer: B) {25}

Solution set of

$3x^{\log_5 2} + 9^{\log_5 x} = 64$
Test the given options (all are powers of 5):

Check $x = 25$:

$\log_5 25 = 2$
$x^{\log_5 2} = (5^2)^{\log_5 2} = 2^2 = 4$
$9^{\log_5 x} = 9^2 = 81$
$3(4) + 81 = 12 + 52 = 64$
Equation is satisfied.

Solution set = {25}

31. Correct Answer: B) 1

Number of solutions of

$2^{\log_6(-4x)} = \log_7(2401)$
Evaluate RHS:

$2401 = 7^4 \Rightarrow \log_7(2401) = 4$

So, $2^{\log_6(-4x)} = 4 = 2^2$
Equating powers: $\log_6(-4x) = 2 \Rightarrow -4x = 6^2 = 36 \Rightarrow x = -9$
Domain check: $-4x > 0 \Rightarrow x < 0$ (valid)
One valid solution

INTEGER TYPE

32. Correct Answer:

$\log_2[\log_3(8 + \log_3 \log_2(x^2 + 4))] = 1$
$\Rightarrow \log_3(8 + \log_3 \log_2(x^2 + 4)) = 2$
$\Rightarrow 8 + \log_3 \log_2(x^2 + 4) = 9$
$\Rightarrow \log_3 \log_2(x^2 + 4) = 1$
$\Rightarrow \log_2(x^2 + 4) = 3$
$\Rightarrow x^2 + 4 = 8 \Rightarrow x^2 = 4$

Positive value: x=2

33. Correct Answer:

$$2^{2x-1} = \frac{1}{8^{x-3}}$$

Since $8 = 2^3$, $\frac{1}{8^{x-3}} = 2^{-3(x-3)} = 2^{-3x+9}$
Equate exponents: $2x - 1 = -3x + 9 \Rightarrow 5x = 10 \Rightarrow \boxed{x = 2}$

34. Correct Answer:

$$\frac{p}{q} = \left(\frac{4}{3}\right)^2 \div \left(\frac{3}{4}\right)^{-2}$$
$$= \left(\frac{4}{3}\right)^2 \times \left(\frac{3}{4}\right)^2 = 1$$
$$\Rightarrow \left(\frac{p}{q}\right)^{-10} = 1^{-10} = \boxed{1}$$

35. Correct Answer:

$$\frac{3^{n+5} - 6 \cdot 3^{n+2}}{3^{n-1} \times 7}$$

Factor 3^{n+2} in numerator: $3^{n+2}(27 - 6) = 21 \cdot 3^{n+2}$
Denominator:
$7 \cdot 3^{n-1}$
$= \frac{21 \cdot 3^{n+2}}{7 \cdot 3^{n-1}} = 3 \cdot 3^3 = \boxed{81}$

36. Correct Answer:

$3^{x-1} = 27 = 3^3 \Rightarrow x - 1 = 3 \Rightarrow x = 4$
$4^{y+2} = 256 = 4^4 \Rightarrow y + 2 = 4 \Rightarrow y = 2$
$\frac{x}{y} = \frac{4}{2} = \boxed{2}$

PREVIOUS YEAR QUESTIONS

1. Correct Answer:

Option A:
The inverse (reciprocal) of $2 + \sqrt{3}$ is

$$\frac{1}{2+\sqrt{3}} = \frac{2-\sqrt{3}}{(2+\sqrt{3})(2-\sqrt{3})} = 2 - \sqrt{3}$$

Now, sum: $(2 + \sqrt{3}) + (2 - \sqrt{3}) = 4$
4 is a rational number.
✓ Rational

Option B:

$\sqrt{18} = \sqrt{9 \times 2} = 3\sqrt{2}$
Since $\sqrt{2}$ is irrational, $3\sqrt{2}$ is irrational.
✗ Not rational

Option C:

$$\sqrt{7 + 4\sqrt{3}}$$

This expression is irrational.
✗ Not rational

Option D:
Since option A is rational, this option is incorrect.

2. Correct Answer: Option C 80

Let
H.C.F. = x

Given,
L.C.M. = $63x$

According to the question:

H.C.F. + L.C.M. = 320
$x + 63x = 320$
$64x = 320$
$x = 5$
So,
H.C.F. = 5
L.C.M. = $63 \times 5 = 315$

Now, using the formula:

Product of two numbers = H.C.F. × L.C.M.
$= 5 \times 315 = 1575$
One number is given as 35.
Let the other number be n.

$$35 \times n = 1575$$
$$n = \frac{1575}{35} = 45$$

Sum of the two numbers: $35 + 45 = 80$

3. Correct Answer: D) 0

Step 1: Rationalise each term

$$\frac{3\sqrt{2}}{\sqrt{6}-\sqrt{3}}=\frac{3\sqrt{2}(\sqrt{6}+\sqrt{3})}{6-3}=\sqrt{2}(\sqrt{6}+\sqrt{3})$$
$$=\sqrt{12}+\sqrt{6}$$
$$\frac{4\sqrt{3}}{\sqrt{6}-\sqrt{2}}=\frac{4\sqrt{3}(\sqrt{6}+\sqrt{2})}{6-2}=\sqrt{3}(\sqrt{6}+\sqrt{2})$$
$$=\sqrt{18}+\sqrt{6}$$
$$\frac{2\sqrt{3}}{\sqrt{6}+2}=\frac{2\sqrt{3}(\sqrt{6}-2)}{6-4}=\sqrt{3}(\sqrt{6}-2)$$
$$=\sqrt{18}-2\sqrt{3}$$

Step 2: Substitute

$$\left(\sqrt{12}+\sqrt{6}\right)-(\sqrt{18}+\sqrt{6})+(\sqrt{18}-2\sqrt{3})$$

Step 3: Simplify

$$\sqrt{12}-2\sqrt{3}=2\sqrt{3}-2\sqrt{3}=0$$

4. Correct Answer: 166400 (Option B)

Let the HCF = x
Then LCM = x + 8300

Given:
LCM + HCF = 8340

So,

$x+(x+8300)=8340$
$2x+8300=8340$
$2x=40$
$x=20$
Hence,
HCF = 20
LCM = 20 + 8300 = 8320

Formula used:

Product of two numbers = LCM × HCF
$= 8320 \times 20$
$= 166400$

5. Correct Answer:

A rational number $\frac{p}{q}$(in lowest terms) has

- a terminating decimal if the prime factors of $q$are only 2and/or 5.
- a non-terminating repeating decimal if $q$has any prime factor other than 2or 5.

(i) $\frac{125}{441}$

$$441=3^2\times7^2$$

Since the denominator has prime factors other than 2and 5,
$\frac{125}{441}$is non-terminating repeating.

(ii) $\frac{129}{2^2\times5^7\times17^7}$

The prime factors of the denominator are 2, 5, and 17.

Since the factor 17 is present, which is not 2 or 5, this fraction has a non-terminating repeating decimal expansion.

(iii) $\frac{29}{343}$

$$343=7^3$$

Since the denominator has a prime factor other than 2and 5,
$\frac{29}{343}$ is non-terminating repeating.

Conclusion:

All three fractions (i), (ii), and (iii) are non-terminating repeating decimal numbers.

6. Correct Answer: 16 (Option C)

$$\frac{\left[\ (676)^{1/2}\right]^4}{\left[\ (2197)^{1/3}\right]^4}$$

$=\frac{(676)^2}{(2197)^{4/3}}$[Using $(a^m)^n=a^{mn}$]

Now,

$676=26^2\Rightarrow676^2=(26^2)^2=26^4$
$2197=13^3\Rightarrow2197^{4/3}=(13^3)^{4/3}=13^4$
$=\frac{26^4}{13^4}$
$=\left(\frac{26}{13}\right)^4$
$=2^4=16$

7. Correct Answer: 7(Option A)

First, find the denominator:

$$(3-2\sqrt{2})(3+2\sqrt{2})=3^2-(2\sqrt{2})^2$$
$$=9-8=1$$

Now, $x+y=\frac{(3+2\sqrt{2})+(3-2\sqrt{2})}{1}=\frac{6}{1}=6$

Also, $xy=\frac{1}{1}=1$

Therefore, $x+y+xy=6+1=7$

8. Correct Answer: C) 2

First, find the HCF of 135 and 216.

$135=3^3\times5$
$216=2^3\times3^3$
Common factor is 3^3.

$$\text{HCF}=27$$

According to the question,

$$87x-147=27$$

$$87x = 27 + 147$$
$$87x = 174$$
$$x = \frac{174}{87} = 2$$

9. Correct Answer: C $\boxed{p^2q}$

Given,

$$a = p^2q^2$$
$$b = p^3q$$

To find the HCF, we take the common prime factors with the least powers.

- Common prime p: powers are 2and 3⟶ least power = 2
- Common prime q: powers are 2and 1⟶ least power = 1

$$\text{HCF}(a, b) = p^2q$$

10. Correct Answer: B $\boxed{xy^2}$

Given,

$$P = x^2y^3, Q = xy^2$$

To find the HCF, take the common prime factors with the smallest powers.

- Common factor x:
  Powers are 2(in P) and 1(in Q)
  ⇒smallest power $= 1$
- Common factor y:
  Powers are 3(in P) and 2(in Q)
  ⇒smallest power $= 2$

$$\text{HCF}(P, Q) = x^1y^2 = xy^2$$

2. SETS

Solutions

Worksheet – 1

1. **Correct Answer: A.** $\{2, 4, 6, 8\}$

   $A = \{x \mid x$ is an even natural number less than $10\}$
   Even natural numbers $<10$are $2,4,6,8$.

2. **Correct Answer: C)** $\{-1,0,1\}$

   Since $x$is an integer and $-1 \leq x < 2$, the possible values of x are $-1,0,1$.

   So, the roster form is $\{-1,0,1\}$.

3. **Correct Answer:**

   Given: $x$is a multiple of 3 and $x < 15$.
   Multiples of 3 less than 15 are 3,6,9,12.
   Correct Answer: A. $\{3, 6, 9, 12\}$

4. **Correct Answer: B**
   $\{ x \mid x$ **is a square number less than 20** $\}$

   Set $\{1, 4, 9, 16\}$ are square numbers less than 20.

5. **Correct Answer: A.** $\{3\}$

   $A = \{x \mid x^2 = 9, x \in \mathbb{N}\}$
   $x^2 = 9 \Rightarrow x = 3$(only natural number)

6. **Correct Answer: B**

   6. $A = \{1,2,3,4\}$
   $\{x \mid x \leq 4, x \in \mathbb{N}\} = \{1,2,3,4\}$

7. **Correct Answer: D. Both B and C**

   Empty set representations:
   $\{\ \}$ is empty, and $\{x \mid x < 0, x \in \mathbb{N}\} = \emptyset$

8. **Correct Answer: B**

   Subset must contain only elements of $A = \{a, b, c\}$
   $\{a, b\} \subset A$

9. **Correct Answer: C**

   Number of subsets of $\{1,2,3\} = 2^3 = 8$

10. **Correct Answer: B.** $A \subseteq B$

    "All elements of A are in B" means $A$is a subset of B

Worksheet -2

1. **Correct Answer: C)** $\{1, 2, 3, 4, 6, 8\}$
   Union means all distinct elements from both sets.
   $A \cup B = \{1,2,3,4,6,8\}$

2. **Correct Answer: B)** $\{3, 6, 9, 12\}$
   Multiples of 3 less than or equal to 12 are $3,6,9,12$.
   So, $A = \{3,6,9,12\}$

3. **Correct Answer: B)** $\{2, 4\}$
   Intersection means common elements.
   Common elements are 2 and 4.

4. **Correct Answer: C)** $\{15\}$
   $A = \{3,6,9,12,15\}$, $B = \{5,10,15\}$
   Common element is 15.

5. **Correct Answer: C)** $\{1, 2\}$
   Set difference means elements in A not in B.
   So, $A - B = \{1,2\}$

6. **Correct Answer: C) They have exactly same elements**
   Two sets are equal if
   Two sets are equal only when every element of one set is also an element of the other, and vice-versa.

7. **Correct Answer: B) Set of natural numbers**
   The set of natural numbers $\{1,2,3, \ldots\}$has no end, so it is infinite.
   All other options are limited.

8. **Correct Answer: B) 4**
   If $A = \{2,4,6,8\}$, then $n(A) =$
   Cardinality means the number of elements in a set. Here, there are 4 elements.

9. **Correct Answer: B) 5**
   Vowels in English alphabet are $\{a, e, i, o, u\}$.
   Total number of vowels = 5.

10. **Correct Answer: C)** $\{2, 4, 6\}$**and** $\{1, 3, 5\}$
    Disjoint sets have no common elements.
    $\{2, 4, 6\}$and $\{1, 3, 5\}$have no common elements.

11. Correct Answer: C) Finite but not disjoint

Set A is finite.
Set B(set of all even numbers) is infinite.
They have common elements $\{2, 4\}$, so they are not disjoint.

12. Correct Answer: B) Set of prime numbers

The set of prime numbers has infinitely many elements.
All other sets listed are limited.

COMPITETIVE CORNER

SINGLE OPTION CORRECT TYPE

1. Correct Answer: (A)

$A = \{x: x \in \mathbb{N}, x < 8\}$
Natural numbers less than 8 are 1,2,3,4,5,6,7.

$$A = \{1,2,3,4,5,6,7\}$$

2. Correct Answer: (B)

Set: $\{5, 10, 15, 20\}$
Each element can be written as $5n$ where $n \in \mathbb{N}$ and $n \leq 4$.

$\{x: x = 5n, n \in \mathbb{N}, n \leq 4\}$

3. Correct Answer: (C)

Set: $\{2, 4, 6, 8\}$
These are even numbers of the form $2n$ with $n \in \mathbb{N}$ and $n \leq 4$.

$\{x: x = 2n, n \in \mathbb{N}, n \leq 4\}$

4. Correct Answer: (A)

$A = \{3,4,5,6\}$, $B = \{4,6,8,10\}$
Union contains all distinct elements of both sets.

$$A \cup B = \{3,4,5,6,8,10\}$$

5. Correct Answer: (B)

$A = \{1,3,5,7,9\}$, $B = \{2,3,5,7,11,13\}$

Common elements in both sets are 3,5,7.

$$A \cap B = \{3,5,7\}$$

6. Correct Answer: (C)

$A = \{3,5,7,9,11\}$
$B = \{7,9,11,13\}$
$C = \{11,13,15\}$

First find: $B \cup C = \{7,9,11,13,15\}$

Now find common elements with A:$A \cap (B \cup C) = \{7,9,11\}$

$$A \cap (B \cup C) = \{7,9,11\}$$

7. Correct Answer: (D)

$A = \{a, b, c, d, e\}$, $B = \{a, c, e, g\}$, $C = \{b, e, f, g\}$

First find:$B \cap C = \{e, g\}$

Now subtract from A:$A - (B \cap C) = \{a, b, c, d\}$

$$A - (B \cap C) = \{a, b, c, d\}$$

8. Correct Answer: (C)

$A = \{1,2,3,4,5\}$
Number of elements $= 5$

Total subsets $= 2^5 = 32$
Proper subsets $= 32 - 1 = 31$

Number of proper subsets $= 31$

9. Correct Answer: (C)

If two sets have no common elements, they are disjoint sets.
Their Venn diagram shows no overlap.

Two separate circles

10. Correct Answer: (C)

Word: BOOKKEEPER
Distinct letters are: $\{B, O, K, E, P, R\}$

Number of distinct letters $= 6$

$$n(A) = 6$$

11. Correct Answer: (C) Finite set

$A = \{x: x \in \mathbb{N}, x < 80\}$

Natural numbers less than 80 are 1,2,3, …,79.
The number of elements is limited.

A has a finite number of elements.

12. Correct Answer:(C)

:$U = \{a, b, c, d, e, f\}$
$A = \{a, c, e\}$, $B = \{b, d, f\}$

Union:$A \cup B = \{a, b, c, d, e, f\} = U$

Intersection:$A \cap B = \emptyset$

$$A \cup B = U$$

13. Correct Answer: (B)

$A = \{x \mid x^2 = 4\} \Rightarrow A = \{-2,2\}$
$B = \{-2,2\}$

Both sets have exactly the same elements.

$$A = B$$

ONE OR MORE THAN ONE OPTION TYPE

14. Correct Answer: (A)
(A) $A = \{x: x \in \mathbb{N}, x^2 < 36\}$
Here, $x^2 < 36 \Rightarrow x = 1,2,3,4,5$.
So, $A = \{1,2,3,4,5\} \rightarrow$ finite

15. Correct Answer: (A), (D)
Which of the following is an empty set?
(A) $x \in \mathbb{N}, x + 6 = 6 \Rightarrow x = 0$
But $0 \notin \mathbb{N} \rightarrow$ empty
(B) $x^2 + 3x + 2 = 0 \Rightarrow (x + 1)(x + 2) = 0$
Solutions: $x = -1, -2 \in \mathbb{Z} \rightarrow$ not empty
(C) $x^2 = x \Rightarrow x(x - 1) = 0 \Rightarrow x = 0,1$
$1 \in \mathbb{N} \rightarrow$ not empty
(D) $x < 5$and $x > 8$cannot be true together $\rightarrow$ empty

16. Correct Answer: (A), (B)
Which of the following are disjoint sets?
(A)$A = \{3,4,5,6\}, B = \{2,8,7,9\}$
No common elements. (Disjoint)
(B)$C = \{1,2,3,4,5\}, D = \{6,7,9,11\}$
No common elements(Disjoint)

17. Correct Answer: (A), (D)
If $A = \{2,4,6,8,10,12\}, B = \{3,4,5,6,7,8,10\}$
Common elements: $\{4, 6, 8, 10\}$

- $A - B = \{2,12\}$
- $B - A = \{3,5,7\}$

18. Correct Answer: (C), (D)
Let$A = \{2,4,6\}$
$B = \{x: x$ is even and $x < 8\} = \{2,4,6\}$

- $A \subset B$
- $B \subset A$
- $A = B$ ✔
- $A \subseteq B$ ✔

ASSERTION AND REASON TYPE

19. Correct Answer: (D) A is false, R is true
Assertion (A): Set $D = \{x: x$ is a prime number which is a divisor of 60$\}$)
Prime divisors of 60 are 2,3,5.
Correct roster form is $\{2, 3, 5\}$, not $\{1, 2, 3, 4, 5\}$.
Assertion (A) is false.
Reason (R): The word TRIGONOMETRY has distinct letters: $\{T, R, I, G, O, N, M, E, Y\}$
Reason (R) is true.

20. Correct Answer: (A) Both A and R are true, and R is the correct explanation of A
Assertion (A):"The collection of all natural numbers less than 100 is a set."
Natural numbers less than 100 are clearly defined and distinct.
Assertion (A) is true.
Reason (R):"A set is a well-defined collection of distinct objects."
This is the definition of a set.
Reason (R) is true and correctly explains A.

21. Correct Answer: (C) A is true, R is false
Assertion (A):
A = set of prime numbers less than 6 = $\{2, 3, 5\}$
B = set of prime factors of 30 = $\{2, 3, 5\}$
So, $A = B$.
Assertion (A) is true.
Reason (R):$P = \{1,2,3\}$ and $Q = \{2,2,1,3,3\}$
In sets, repetition of elements is not considered.
So, $Q = \{1,2,3\} = P$.
Reason (R) is false.

22. Correct Answer: (A) Both A and R are true, and R is the correct explanation of A
Assertion (A):
$A = \{$Sita, Mohan, Ali$\}$
$B = \{$Mohan, Reena, Farooq$\}$
Union contains all distinct elements:
$A \cup B = \{$Sita, Mohan, Ali, Reena, Farooq$\}$
Assertion (A) is true.
Reason (R):"The union of two sets contains all elements that belong to either set or both sets."
This is the definition of union and explains A correctly.
Reason (R) is true and correct.

23. Correct Answer: (D) A is false, R is true
Assertion (A): $A = \{1,2,3,4,5\}, B = \{4,5,6,7\}$
Elements in A but not in B are 1,2,3.
So, $A - B = \{1,2,3\}$, not $\{1, 3\}$.
Assertion (A) is false.
Reason (R):"The difference of two sets contains elements that are in the first set but not in the second set."
This statement is true (correct definition).
Reason (R) is true.

MATRIX MATCHING TYPE

24. Correct Answer:

A) $\{1, 4, 9, 16, 25\}$

These are squares of natural numbers from 1 to 5.

Match: S $\{x: x = n^2, n \in \mathbb{N}, n \leq 5\}$

B) $\{1, 2, 3, 4, 6, 12\}$

These are all factors of 12.

Match: P $\{x: x$ is a factor of $12\}$

C) $\{-1,0,1,2,3\}$

These are integers with $-2 < x \leq 3$.

Match: Q $\{x: x \in \mathbb{Z},\ -2 < x \leq 3\}$

D) $\{1, 3, 5, 7, 9\}$

These are odd natural numbers less than 10.

Match: R $\{x: x$ is an odd natural number $< 10\}$

25. Correct Answer:

A) $A = \{1,2,3\}$, $B = \{2,3,4\}$

Common elements are $\{2, 3\}$.

Match: Q $\{2, 3\}$

B) $A = \{4,5\}$, $B = \{4,5,6,7\}$

Set A itself is a subset of B: $\{4, 5\}$.

Match: P $\{4, 5\}$

C) $A = \{6,7,8\}$, $B = \{7,8,9\}$

Elements of A not in B are $\{6\}$.

Match: S $\{6\}$

D) $A = \{1,2\}$, $B = \{1,2,3,4\}$

Union contains all elements: $\{1, 2, 3, 4\}$.

Match: R $\{1, 2, 3, 4\}$

COMPREHENSION TYPE

Passage – I

$A = \{1,3,5,7,9\}$

$B = \{2,4,6,8\}$

$C = \{2,3,5,7,11\}$

26. Correct Answer: (b) $\emptyset$

Find $A \cap B$.

Set A has odd numbers and set B has even numbers.

There is no common element.

$$A \cap B = \emptyset$$

27. Correct Answer: (a) $\{3, 5, 7\}$

Find $A \cap C$.

Common elements in A and C are 3,5,7.

$$A \cap C = \{3,5,7\}$$

28. Correct Answer: (a) $A \cap B = \emptyset$

Which of the following is correct for two sets A and B to be disjoint?

Two sets are disjoint if they have no common elements, i.e., their intersection is empty.

$$A \cap B = \emptyset$$

Passage – II

29. Correct Answer: (C) $A \subseteq B$ **and** $B \subseteq A$

For two sets to be equal, every element of set A must be in set B and every element of set B must be in set A.

That means:

- $A \subseteq B$ and
- $B \subseteq A$

C. $A \subseteq B$ and $B \subseteq A \rightarrow$ Correct condition for equal sets

30. Correct Answer: (B) Set of even numbers

Which of the following is an infinite set?

A. Set of days in a week $\rightarrow$ 7 days, finite

B. Set of even numbers $\rightarrow$ Even numbers never end, infinite

C. Set of students in a class $\rightarrow$ Finite

D. Set of solutions of $x^2 = 16 \rightarrow$ Solutions are $\{-4,4\}$, finite

31. Correct Answer: (B) 4

Find the cardinality of the set

$C = \{x: x$ is a letter in the word "APPLE"$\}$

Distinct letters in APPLE are: $\{A, P, L, E\}$

Number of distinct letters $= 4$.

INTEGER TYPE

32. Correct Answer:

$A = \{x: x$ is a multiple of 3, $x < 20\}$

Multiples of 3 less than 20 are:

$\{3, 6, 9, 12, 15, 18\}$

Number of elements = 6

33. Correct Answer: 8

If $A = \{1,2,3\}$, then

Number of subsets $= 2^3 = 8$

34. Correct Answer: $\{4, 6\}$

$A = \{2,4,6,8,10\}$, $B = \{4,6\}$

Common elements are $\{4, 6\}$

35. Correct Answer:

$A = \{2,5,6,8\}$, $B = \{5,7,9,1\}$

$A \cup B = \{1,2,5,6,7,8,9\}$

Number of elements = 7

36. Correct Answer:

$A = \{1,2,3,4,5,6\}$, $B = \{2,4,6\}$

All elements of B are already in A, so

$B - A = \emptyset$

Number of elements = 0

3. POLYNOMIALS SOLUTIONS

Solutions

Worksheet – 1

1. Correct Answer: C

Rule:
A polynomial cannot have negative powers of x or x in the denominator.

Check options:

- $x + \frac{1}{x} = x + x^{-1}$
- $\frac{x^3+1}{x^2} = x + x^{-2}$
- $\frac{x^2+x^3}{x} = x + x^2$✓ (all powers are non-negative)
- $x^2 + \frac{1}{x^2} = x^2 + x^{-2}$

2. Correct Answer: A

Simplify each option:

- $\frac{1}{x^{-2}} + \frac{1}{x^{-1}} = x^2 + x$✓
- Contains x^{-2}
- $x^2(x^{-2} + x^{-3}) = 1 + x^{-1}$
- Contains negative and fractional powers

3. Correct Answer: C 5

Degree of the polynomial $x - x^5 + 1$ is
Arrange in descending powers:$-x^5 + x + 1$
Rule:
The degree of a polynomial is the highest power of x.
Highest power = 5

4. Correct Answer:

Degree of a polynomial

$$\frac{\sqrt{2}x^5 - x^{10} + 1}{x}$$

Divide each term by x:

$$= \sqrt{2}x^4 - x^9 + \frac{1}{x}$$

Since $\frac{1}{x} = x^{-1}$(negative power),
It is NOT a polynomial.

D) Not defined

5. Correct Answer:

Number of terms in the polynomial $4x^2 + 5x + 3$
The terms are: $4x^2$, 5,3

Total number of terms = 3

B) 3

6. Correct Answer: C)

Which of the following is a trinomial?

Rule:
A trinomial has exactly 3 terms.

- $x^2 + x \rightarrow$ 2 terms
- $x^3 - 1 \rightarrow$ 2 terms
- $x^2 + 2x + 10 \rightarrow$ 3 terms
- $x^5 + x^3 - 3x^2 + \frac{1}{\sqrt{3}} \rightarrow$ 4 terms

7. Correct Answer: B)

Which of the following is a bi-quadratic polynomial?

Rule:
A bi-quadratic polynomial has degree 4 and only even powers of x.

- Contains odd powers
- Contains x^3(odd power)
- $\frac{x^4-x+1}{x} \rightarrow$ not a polynomial
- $\frac{x+3}{x^4} \rightarrow$ not a polynomial

✓ The correct bi-quadratic polynomial is: $\boxed{x^4 - 30x^2 + 3x^2}$

8. Correct Answer: C) 6

Value of the polynomial $4 + x^3 - x + 2x^2$ at $x = -2$
Substitute $x = -2$:$= 4 + (-2)^3 - (-2) + 2(-2)^2$
$= 4 - 8 + 2 + 8$
$= 6$

9. Correct Answer: B)

The zeroes of the polynomial $3x^2 - 1$
Set $3x^2 - 1 = 0$

$3x^2 = 1$

$x^2 = \frac{1}{3}$

$x = \pm\frac{1}{\sqrt{3}}$

Given options include: $-\frac{1}{\sqrt{3}}$

10. Correct Answer: B) 3

Which of the following is a zero of $p(x) = x^2 - 9$

$x^2 - 9 = 0$

$(x - 3)(x + 3) = 0$

So, zeros are $x = 3$and $x = -3$.

Worksheet -2

1. Correct Answer: B) 1

The maximum number of zeroes of a linear polynomial is

A linear polynomial is of degree 1.
The maximum number of zeroes of a polynomial = its degree.

So, maximum number of zeroes = 1

2. Correct Answer: C) $x = 2$

The graph of $y = 3x - 6$ cuts the x-axis at

At the x-axis, $y = 0$.

$$0 = 3x - 6$$
$$3x = 6$$
$$x = 2$$

3. Correct Answer: A) 0

3. If $7 + 3x$divides $3x^2 + 7x$, then the remainder is

Solution (Remainder Theorem):

For divisor $3x + 7$, put $x = -\frac{7}{3}$

Substitute in the polynomial:$3x^2 + 7x$

$$= 3\left(-\frac{7}{3}\right)^2 + 7\left(-\frac{7}{3}\right)$$
$$= 3 \cdot \frac{49}{9} - \frac{49}{3}$$
$$= \frac{49}{3} - \frac{49}{3} = 0$$

4. Correct Answer: D $\boxed{2020}$

4. If $1 + x + x^2 + x^3 + \cdots + x^{2019}$ is divided by $x - 1$, then the remainder is

By the Remainder Theorem, for divisor $x - 1$, put $x = 1$

Remainder $= 1 + 1 + 1 + \cdots + 1$(2019 + 1 terms)

$= 2020$

5. Correct Answer: C) $-x$

5. To find the remainder when dividing by $x^2 + 1$, we first make the divisor zero:

$$x^2 + 1 = 0 \Rightarrow x^2 = -1$$

Now substitute $x^2 = -1$in the given polynomial

$$x^{19} + x^{17} + x^{13} + x^{11} + x^7 + x^5 + x^3$$

Express each term using powers of x^2:

$$x^{19} = (x^2)^9 \cdot x = (-1)^9 \cdot x = -x$$
$$x^{17} = (x^2)^8 \cdot x = (-1)^8 \cdot x = x$$
$$x^{13} = (x^2)^6 \cdot x = (-1)^6 \cdot x = x$$
$$x^{11} = (x^2)^5 \cdot x = (-1)^5 \cdot x = -x$$
$$x^7 = (x^2)^3 \cdot x = (-1)^3 \cdot x = -x$$
$$x^5 = (x^2)^2 \cdot x = (-1)^2 \cdot x = x$$
$$x^3 = (x^2)^1 \cdot x = (-1)^1 \cdot x = -x$$

Now add all the simplified terms:

$$R(x) = (-x) + (x) + (x) + (-x) + (-x) + (x) + (-x)$$

Group positive and negative terms:

$$R(x) = (x + x + x) - (x + x + x + x)$$
$$R(x) = 3x - 4x$$
$$R(x) = -x$$

The remainder is $-x$.

6. Correct Answer: B)

If $x^4 - 3x^3 - 7x^2 + 27x - 18$ is divided by $x + 3$, find the quotient.

Solution (synthetic division with -3):

-3	1	-3	-7	27	-18
		-3	18	-33	18
	1	-6	11	-6	0

Quotient: $x^3 - 6x^2 + 11x - 6$

7. Correct Answer: B) $9x + 10$

If $3x^3 + x^2 + 2x + 5$ is divided by $x^2 + 2x + 1$, find the remainder.

Remainder is of the form $ax + b$.

Let: $3x^3 + x^2 + 2x + 5 = (x^2 + 2x + 1)(3x - 5) + (9x + 10)$

8. Correct Answer: C) 3

If $2x^2 + 3x + 1$ is divided by $x + 2$, find the remainder.

Solution (Remainder Theorem):
Put $x = -2$

$$2(-2)^2 + 3(-2) + 1 = 8 - 6 + 1 = 3$$

9. Correct Answer: D)

9. If $x^4 - 5x + 6$ is divided by $2 - x^2$, find the quotient and remainder.

Rewrite divisor: $2 - x^2 = -(x^2 - 2)$

Division gives:

Quotient: $-x^2 - 2$

Remainder: $-5x + 10$

10. Correct Answer: A) 0

If $3x^4 + 2x^3 - \frac{x^2}{3} - \frac{x}{9} + \frac{2}{27}$ is divided by $x + \frac{2}{3}$, find the remainder.

By Remainder Theorem, for divisor $x + \frac{2}{3}$,

$x = -\frac{2}{3}$

Substitute: $R = 3\left(-\frac{2}{3}\right)^4 + 2\left(-\frac{2}{3}\right)^3 - \frac{1}{3}\left(-\frac{2}{3}\right)^2 - \frac{1}{9}\left(-\frac{2}{3}\right) + \frac{2}{27}$

$$= \frac{16}{27} - \frac{16}{27} - \frac{4}{27} + \frac{2}{27} + \frac{2}{27}$$
$$= 0$$

11. Correct Answer: B) −1

If the polynomials $px^3 + 4x^2 + 3x - 4$ and $p + x^3 - 4x$ leave the same remainder when divided by $x - 3$, find p.

By Remainder Theorem, substitute $x = 3$.

First polynomial: $p(3)^3 + 4(3)^2 + 3(3) - 4 = 27p + 36 + 9 - 4 = 27p + 41$
Second polynomial: $p + (3)^3 - 4(3) = p + 27 - 12 = p + 15$
Since remainders are equal: $27p + 41 = p + 15$

$26p = -26$
$p = -1$

12. Correct Answer: B) touches the x-axis at $x = 1$and cuts at $x = -2$

A cubic polynomial has a zero of multiplicity 2 at $x = 1$and another zero at $x = -2$.
Geometrically, the graph:

- A zero of multiplicity 2 means the graph touches the x-axis at that point.
- A zero of multiplicity 1 means the graph cuts the x-axis.

So,

- At $x = 1$→ graph touches the x-axis
- At $x = -2$→ graph cuts the x-axis

13. Correct Answer: C) all zeroes are equal

A cubic polynomial has three real zeroes, but the graph intersects the x-axis only at one point. This is possible when:

- If all three zeroes are equal, the graph will touch the x-axis at one point only.
- It does not cut the x-axis at three different points.

Worksheet-3

1. Correct Answer: D)

If $x - 3$ is a factor, then by the Factor Theorem:

$$f(3) = 0$$

Given polynomial: $f(x) = k^2x^2 - kx - 2$
Substitute $x = 3$:

$k^2(3)^2 - k(3) - 2 = 0$
$9k^2 - 3k - 2 = 0$
$9k^2 - 6k + 3k - 2k - 2 = 0$
$(3k - 2)(3k + 1) = 0$
So,

$3k - 2 = 0$ or $3k + 1 = 0$
$k = \frac{2}{3}$ or $k = -\frac{1}{3}$

2. Correct Answer: C) 18

If $(x-1)$and $(x+3)$are factors of $x^3 - ax^2 - 13x + b$
Using Factor Theorem:
For $x = 1$: $1 - a - 13 + b = 0 \Rightarrow b = a + 12$
For $x = -3$: $-27 - 9a + 39 + b = 0 \Rightarrow b = 9a - 12$
Equate:
$a + 12 = 9a - 12 \Rightarrow a = 3$
$b = 15$
$a + b = 18$

3. Correct Answer: A) 0

If $x^2 - 1$ is a factor of $ax^4 + bx^3 + cx^2 + dx + e$

Since $x^2 - 1 = (x - 1)(x + 1)$,

Put $x = 1$: $a + b + c + d + e = 0$ (1)
Put $x = -1$: $a - b + c - d + e = 0$ (2)
Add (1) and (2): $2a + 2c + 2e = 0 \Rightarrow a + c + e = 0$

4. Correct Answer: B) 19

4. If $x^4 - 2x^3 + 3x^2 - nx + 3$ is exactly divisible by $x - 3$, find n.

Since $x - 3$ is a factor, by Factor Theorem:

$f(3) = 0$
$3^4 - 2(3)^3 + 3(3)^2 - n(3) + 3 = 0$
$81 - 54 + 27 - 3n + 3 = 0$
$57 - 3n = 0$
$3n = 57$
$n = 19$

5. Correct Answer: C $\boxed{0}$

Since $x + k$ is a factor, by the Factor Theorem:

$$f(-k) = 0$$

Given polynomial: $f(x) = kx^2 + 2k^2x + m^3$
Substitute $x = -k$:

$k(-k)^2 + 2k^2(-k) + m^3 = 0$
$k^3 - 2k^3 + m^3 = 0$
$m^3 - k^3 = 0$
$(m - k)(m^2 + mk + k^2) = 0$
Since generally $m \neq k$, we get: $m^2 + mk + k^2 = 0$

6. Correct Answer: C) 3

If $kx^3 + 9x^2 + 4x - 8$ is divided by $x + 3$, the remainder is -20. Find k.

By Remainder Theorem:

$f(-3) = -20$
$k(-3)^3 + 9(-3)^2 + 4(-3) - 8 = -20$
$-27k + 81 - 12 - 8 = -20$
$-27k + 61 = -20$
$-27k = -81$
$k = 3$

7. Correct Answer: B) 70

If $2x - 1$ exactly divides $6x^2 + mx - 4$ and $nx^2 - 11x + 3$, find $n^2 - m^2 + m - n$.

For divisor $2x - 1$, put $x = \frac{1}{2}$.

First polynomial:

$6\left(\frac{1}{2}\right)^2 + m\left(\frac{1}{2}\right) - 4 = 0$
$\frac{6}{4} + \frac{m}{2} - 4 = 0$
$\frac{m}{2} - \frac{5}{2} = 0$
$m = 5$
Second polynomial:

$n\left(\frac{1}{2}\right)^2 - 11\left(\frac{1}{2}\right) + 3 = 0$
$\frac{n}{4} - \frac{11}{2} + 3 = 0$
$\frac{n}{4} - \frac{5}{2} = 0$
$n = 10$
Now, $n^2 - m^2 + m - n$
$= 100 - 25 + 5 - 10$
$= 70$

8. Correct Answer: B) $-9\sqrt{3}$, 5

The sum and product of zeroes of $\sqrt{3}x^2 + 27x + 5\sqrt{3} = 0$ are:

For a quadratic equation $ax^2 + bx + c = 0$:

- Sum of zeroes $= -\frac{b}{a}$
- Product of zeroes $= \frac{c}{a}$

Here, $a = \sqrt{3}$, $b = 27$, $c = 5\sqrt{3}$

Sum of zeroes: $-\frac{27}{\sqrt{3}} = -9\sqrt{3}$

Product of zeroes: $\frac{5\sqrt{3}}{\sqrt{3}} = 5$

9. Correct Answer: D) $-\frac{9}{4}$

If α, β are zeroes of $4x^2 - 9$, then the product of zeroes is

Given polynomial: $4x^2 - 9$

Here, $a = 4$, $b = 0$, $c = -9$

Product of zeroes: $\frac{c}{a} = \frac{-9}{4}$

10. Correct Answer: A) $-2, -1, 1$

$p(x) = x^3 + 2x^2 - x - 2$
$= (x^3 + 2x^2) - (x + 2)$
$= x^2(x + 2) - 1(x + 2)$
Take common factor $(x+2)$: $= (x + 2)(x^2 - 1)$
Factor further: $x^2 - 1 = (x - 1)(x + 1)$
So, $p(x) = (x + 2)(x + 1)(x - 1)$
Zeroes are: $-2, -1, 1$

COMPITETIVE CORNER

SINGLE OPTION CORRECT TYPE

1. Correct Answer: B) 98

1. Degree of the polynomial $\frac{x^{100}-3x^2+x^6}{x^2}$
Divide each term by x^2:= $x^{98}-3+x^4$
Highest power of x is 98.

2. Correct Answer: C) x^3-3x

Which of the following is a cubic polynomial?

A) $-\sqrt{2}x^2+x-3\rightarrow$ degree 2
B) $x+\frac{1}{3}\rightarrow$ degree 1
C) $x^3-3x\rightarrow$ degree 3
D) $x^9-x^2+x\rightarrow$ degree 9

3. Correct Answer: C) x^2-x+1

Dividend = x^3-3x^2+x+2
Quotient = $x-2$
Remainder = $-2x+4$

Using:

Dividend = Divisor × Quotient + Remainder
$x^3-3x^2+x+2 = g(x)(x-2)+(-2x+4)$
$g(x)(x-2)=x^3-3x^2+3x-2$

Divide:

```
2 | 1   -3    3   -2
  |     2   -2    2
  ----------------
    1   -1    1    0
```

$$g(x)=x^2-x+1$$

4. Correct Answer: B) 9

If $8x^4+12x^3+16x^2+8x+10$ is divided by $4x+2$

Use Remainder Theorem:
For divisor $4x+2=0\Rightarrow x=-\frac{1}{2}$

Substitute $x=-\frac{1}{2}$:

$$8\left(\frac{1}{16}\right)+12\left(-\frac{1}{8}\right)+16\left(\frac{1}{4}\right)+8\left(-\frac{1}{2}\right)+10$$
$$=0.5-1.5+4-4+10=9$$

5. Correct Answer: A) −1

If $\sqrt{\frac{5}{3}}$and $-\sqrt{\frac{5}{3}}$are zeroes of $3x^4+6x^3-2x^2-10x-5$, then the other zeroes are

Given zeroes:$\pm\sqrt{\frac{5}{3}}$
Corresponding factor:$x^2-\frac{5}{3}$
Multiply by 3:$3x^2-5$
Divide the given polynomial by $3x^2-5$:

$$3x^4+6x^3-2x^2-10x-5=(3x^2-5)(x^2+2x+1)$$
$x^2+2x+1=(x+1)^2$
Other zeroes:$-1,\ -1$

6. Correct Answer: A) $x^5-4x^4-3x^3+8x^2-14x+12$

$x+2$is a factor of

If $x+2$is a factor, then by Factor Theorem:$f(-2)=0$
Check option A:

$$(-2)^5-4(-2)^4-3(-2)^3+8(-2)^2-14(-2)+12$$
$$=-32-64+24+32+28+12=0$$

Hence, option A satisfies the condition.

7. Correct Answer: A) 0

The remainder when x^4-y^4 is divided by $x-y$ is

Use identity:$x^4-y^4=(x-y)(x+y)(x^2+y^2)$
Since $x-y$ is a factor, the remainder is 0.

8. Correct Answer: A

The zeroes of a quadratic polynomial are $\frac{1}{2},\frac{3}{2}$. Find the polynomial.

Sum of zeroes:$\frac{1}{2}+\frac{3}{2}=2$
Product of zeroes:$\frac{1}{2}\times\frac{3}{2}=\frac{3}{4}$
Required polynomial:

x^2-(sum)$x+$(product)
$=x^2-2x+\frac{3}{4}$
Multiply by 4 to remove fraction:$4x^2-8x+3$
Answer: $4x^2-8x+3$

9. Correct Answer: D

For a cubic polynomial with zeroes α,β,γ:

Polynomial $=x^3-(\alpha+\beta+\gamma)x^2+(\alpha\beta+\beta\gamma+\gamma\alpha)x-\alpha\beta\gamma$
Given:$\alpha+\beta+\gamma=2,\alpha\beta+\beta\gamma+\gamma\alpha=-5,\alpha\beta\gamma=-20$
Substitute:$x^3-2x^2-5x+20$

10. Correct Answer: A $\boxed{\sqrt{2}x^2 - x + 2}$

Given: $\alpha + \beta = \frac{1}{\sqrt{2}}, \alpha\beta = \sqrt{2}$

Required: Quadratic polynomial whose zeroes are $\alpha$and β.

Formula

For zeroes α, β, the quadratic polynomial is:

$$x^2 - (\alpha + \beta)x + \alpha\beta$$

Substitute the given values

$$x^2 - \frac{1}{\sqrt{2}}x + \sqrt{2}$$

To remove the denominator, multiply the entire polynomial by $\sqrt{2}$:

$$\sqrt{2}x^2 - x + 2$$

11. Correct Answer:

$$\alpha + \beta = \sqrt{2} + \frac{1}{\sqrt{2}} = \frac{3}{\sqrt{2}}, \alpha\beta = 1$$

Polynomial:$x^2 - \frac{3}{\sqrt{2}}x + 1$

Multiply by $\sqrt{2}$:$\sqrt{2}x^2 - 3x + \sqrt{2}$

Correct Answer: A

12. Correct Answer: D) $\frac{9}{2}$

Given: $\alpha + \beta = 3,\ \alpha\beta = 2$

$$\left(1 + \frac{\alpha}{\beta}\right)(1 + \frac{\beta}{\alpha})$$

$$= 1 + \frac{\alpha}{\beta} + \frac{\beta}{\alpha} + 1$$

$$= 2 + \frac{\alpha^2 + \beta^2}{\alpha\beta}$$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 9 - 4 = 5$$

$$= 2 + \frac{5}{2} = \frac{9}{2}$$

13. Correct Answer: D) Straight line

The graph of a linear polynomial $y = ax + b$is a straight line.

14. Correct Answer: C) A parabola opening downwards

Given polynomial:$y = -2x^2 + 4x + 3$

Degree = 2 ⟶ Parabola

- Coefficient of x^2 is negative ⟶ Opens downwards

15. Correct Answer: C) a curve with at most two bends

A cubic polynomial graph:

- Has degree 3
- Can have at most two bends

ONE OR MORE THAN ONE OPTION CORRECT TYPE

16. Correct Answer: B

Given:$2x^{99} + 3x^{98} + 2x^{97} + 3x^{96} + \cdots + 2x + 3 = 0$

Group terms:$= x^{98}(2x + 3) + x^{96}(2x + 3) + \cdots + (2x + 3)$

$= (2x + 3)(x^{98} + x^{96} + \cdots + 1)$

So,$2x + 3 = 0$ or $x^{98} + x^{96} + \cdots + 1 = 0$

Second factor is always positive, so no real root from it.

$$2x + 3 = 0 \Rightarrow x = -\frac{3}{2}$$

17. Correct Answer: B,C

- $11x^2 - 88x^3 + 14x^4$
  Highest power = 4 ⟶ Bi quadratic
- Number of terms = 3 ⟶ Trinomial

18. Correct Answer: A and B

$x-2,\ x+3$ are factors of $x^3 + ax^2 + bx - 12$

So,

$f(2) = 0, f(-3) = 0$

From $f(2) = 0$:$8 + 4a + 2b - 12 = 0 \Rightarrow 2a + b = 2 \quad (1)$

From $f(-3) = 0$:$-27 + 9a - 3b - 12 = 0 \Rightarrow 3a - b = 13 \quad (2)$

Solving (1) and (2): $a = 3, b = -4$

19. Correct Answer: A, B, C

Given:

$x^3 + 6x^2 + 11x + 6$

Try factors:

$(x + 1)(x + 2)(x + 3)$

All are factors.

20. Correct Answer: A and B

$2x^4 - 3x^3 - 3x^2 + 6x - 2$

Given zeroes: $\sqrt{2}, -\sqrt{2}$

So factor:$x^2 - 2$

Divide polynomial by $x^2 - 2$:

Remaining factor:$(2x - 1)(x - 1)$

Other zeroes:

X=1,x=1/2

21. Correct Answer: A, B, C

Given quadratic: $x^2 - 20x - 30 = 0$
Let zeroes be α, β.

From standard relations:

$$\alpha + \beta = -\frac{b}{a} = -\frac{-20}{1} = 20$$
$$\alpha\beta = \frac{c}{a} = \frac{-30}{1} = -30$$

Now check options:

A) $\alpha + \beta = 20$

B) $\alpha\beta = -30$

C) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 20^2 - 2(-30) = 400 + 60 = 460$

D) $\alpha + \beta = -30$ ✗

22. Correct Answer: A, B, C, D

$$\alpha + \beta = \frac{1}{4}, \alpha\beta = -1$$

Standard quadratic polynomial:

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$
$$x^2 - \frac{1}{4}x - 1 = 0$$

Now check options:

A) $x^2 - \frac{x}{4} - 1 = 0$

B) Multiply A by 4: $4x^2 - x - 4 = 0$

C) Multiply A by 8: $8x^2 - 2x - 8 = 0$

D) Multiply A by 12: $12x^2 - 3x - 12 = 0$

(All represent the same quadratic equation)

ASSERTION AND REASON TYPE

23. Correct Answer: (A)

A quadratic polynomial can have at most two zeroes.

True (degree of quadratic polynomial = 2)

Reason (R):

A quadratic polynomial is always represented by a straight line.
False (it is represented by a parabola, not a straight line)

Correct option: (C) Assertion is true but Reason is false.

24. Correct Answer: (A)

The zeroes of the polynomial $p(x) = x^2 - 5x + 6$ are 2 and 3.

True $x^2 - 5x + 6 = (x - 2)(x - 3)$

Reason (R):

The sum of zeroes of a quadratic polynomial is $-\frac{b}{a}$

True

But the reason only gives a formula, it does not directly explain why the zeroes are 2 and 3.

Correct option: (B) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

25. Correct Answer: (A)

If a polynomial is divided by a linear factor $(x-a)$, the remainder is always zero.
False

The remainder is zero only if $(x-a)$is a factor of the polynomial, i.e., when $p(a) = 0$.

Reason (R):

Remainder Theorem states that the remainder when $p(x)$is divided by $(x-a)$is $p(a)$.
True

Correct option: (D) Assertion is false but Reason is true.

26. Correct Answer: (A)

A cubic polynomial can have at most three zeroes.
True

Reason (R):

The degree of a polynomial decides the maximum number of zeroes it can have.
True

The reason correctly explains the assertion.

Correct option:

✅ (A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

27. Correct Answer: (A)

The graph of a linear polynomial is a curve.
False

The graph of a linear polynomial is a straight line, not a curve.

Reason (R):

A linear polynomial is of the form $ax + b$, which represents a straight line.

True

Correct option:(D) Assertion is false but Reason is true.

MATRIX MATCHING TYPE

28. Correct Answer: (a)

By Remainder Theorem, remainder when x^{10} is divided by $x - 1$is

$$x^{10}\ |_{x=1} = 1^{10} = 1$$

Matches S (1)

(b) If $f(2) = 0$, then by Factor Theorem,

$(x - 2)$ is a factor
Matches P $(x - 2)$

(c) $x^2 - 5x + 6 = (x - 2)(x - 3)$
One factor is $x - 3$
Matches R $(x - 3)$

(d) If $(x+1)$is a factor of $x^2 + px - 1$, then

$f(-1) = 0$
$(-1)^2 + p(-1) - 1 = 0$
$1 - p - 1 = 0 \Rightarrow p = 0$
Matches Q (0)

29. Correct Answer: (a)

Since 2 is a root,

$(2)^2 + p(2) + 26 = 0$
$4 + 2p + 26 = 0 \Rightarrow 2p = -30 \Rightarrow p = -15$

Match: a → S

(b)

Since 5 is a root,

$(5)^2 - (k - 1)(5) + 10 = 0$
$25 - 5k + 5 + 10 = 0 \Rightarrow 40 - 5k = 0 \Rightarrow k = 8$

Match: b → P

(c) $4x^2 - 9 = 0 \Rightarrow 4x^2 = 9 \Rightarrow x = \pm\frac{3}{2}$

Match: c → Q

(d) $(x - a)(x - b)(x - c)(x - d)$
Product of 4 linear factors ⇒ degree = 4

Match: d → R

COMPREHENSION TYPE

Passage – I

30. Correct Answer: B$(x - 1)$.

If $f(x) = x^5 - x^4 + x^3 - x^2 + x - 1$, then the factor of $f(x)$is …

Coefficients: $1, -1, 1, -1, 1, -1$

Check Sum of All Coefficients (Rule 2):

$$1 + (-1) + 1 + (-1) + 1 + (-1) = 0$$

Conclusion:Since the sum of all coefficients is 0, by Rule 2, the factor is

31. Correct Answer: B $\boxed{x + 1}$

If $g(x) = x^3 + x^2 + x + 1$, then the factor of $g(x)$is …

Coefficients: $a_0 = 1,\ a_1 = 1,\ a_2 = 1,\ a_3 = 1$

Check Sum of Alternate Coefficients (Rule 1):

Sum of coefficients of even powers: $a_0 + a_2 = 1 + 1 = 2$

Sum of coefficients of odd powers: $a_1 + a_3 = 1 + 1 = 2$

Conclusion: Since both sums are equal, by Rule 1, the factor is $(x + 1)$.

32. Correct Answer: A $\boxed{x - 1}$

If $f(x) = x^3 - 7x + 6$, then the factor of $f(x)$is …

Coefficients: $a_0 = 1,\ a_1 = 0,\ a_2 = -7,\ a_3 = 6$

Check Sum of All Coefficients (Rule 2): $1 + 0 + (-7) + 6 = 0$

Conclusion: Since the sum of coefficients is 0, the factor is $(x - 1)$.

33. Correct Answer: B $\boxed{128}$

If $(3x - 1)^7 = a_7x^7 + a_6x^6 + \cdots + a_0$, find $a_0 + a_1 + a_2 + \cdots + a_7$.

Concept:
The sum of the coefficients of a polynomial $P(x)$is obtained by substituting $x = 1$.

Calculation:
Let $P(x) = (3x - 1)^7$
Sum of coefficients $= P(1)$

$$P(1) = (3(1) - 1)^7$$

$$P(1) = (2)^7$$
$$P(1) = 128$$

Passage – II

Given:

Zeroes are $\alpha = \frac{1}{2}$, $\beta = \frac{1}{3}$, $\gamma = \frac{1}{5}$

34. Correct Answer: C) $\frac{31}{30}$

$$\alpha + \beta + \gamma$$
$$\frac{1}{2} + \frac{1}{3} + \frac{1}{5} = \frac{15 + 10 + 6}{30} = \frac{31}{30}$$

35. Correct Answer: B) $\frac{4}{12}$

$$\alpha\beta + \beta\gamma + \gamma\alpha$$
$$\frac{1}{2} \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{1}{5} + \frac{1}{5} \cdot \frac{1}{2} = \frac{1}{6} + \frac{1}{15} + \frac{1}{10}$$
$$= \frac{5 + 2 + 3}{30} = \frac{10}{30} = \frac{1}{3}$$

36. Correct Answer: A) $\frac{1}{30}$

$$\alpha\beta\gamma$$
$$\frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{5} = \frac{1}{30}$$

INTEGER TYPE

37. Correct Answer: 0

Given $1 + x + x^2 = 0$

Add the three expressions: $(ax^2 + bx + c) + (a + bx^2 + cx) + (ax + b + cx^2)$

Group like terms:= $(a + b + c)(x^2 + x + 1)$

Since $x^2 + x + 1 = 0$, Sum $= 0$

38. Correct Answer: 0

38.Remainder when $f(x) = x^5 - x^4 + x^3 - x^2 + x - 1$ is divided by $x - 1$

By Remainder Theorem:

Remainder $= f(1)$
$= 1 - 1 + 1 - 1 + 1 - 1 = 0$

39. Correct Answer: 2

If $(x-2)$is a common factor of $x^2 + ax + b$ and $x^2 + cx + d$

Put $x = 2$:

From first: $4 + 2a + b = 0 \Rightarrow b = -4 - 2a$

From second: $4 + 2c + d = 0 \Rightarrow d = -4 - 2c$

Now, $\frac{b-d}{c-a} = \frac{(-4-2a)-(-4-2c)}{c-a} = \frac{2(c-a)}{c-a} = 2$

40. Correct Answer: 3

$$(8m^2 - 6mn) : 2m = 4m - kn$$

LHS:

$$\frac{8m^2 - 6mn}{2m} = 4m - 3n$$

Compare with $4m - kn$:

$$k = 3$$

41. Correct Answer: 3

If $x - 2$ and $x + 3$are factors of $x^3 + ax^2 + bx - 12$

Put $x = 2$:

$$8 + 4a + 2b - 12 = 0 \Rightarrow 2a + b = 2 \quad (1)$$

Put $x = -3$:

$$-27 + 9a - 3b - 12 = 0 \Rightarrow 3a - b = 13 \quad (2)$$

Add (1) and (2):

$$5a = 15 \Rightarrow a = 3$$

PREVIOUS YEAR QUESTIONS

1. Correct Answer: D $\boxed{a = -8,\ b = 12}$

Let $f(x) = 2x^3 + ax^2 + bx - 6$

Step 1: Use factor theorem

Since $(x-1)$is a factor,

$f(1) = 0$
$2(1)^3 + a(1)^2 + b(1) - 6 = 0$
$2 + a + b - 6 = 0$
$a + b = 4$(Equation 1)

Step 2: Use remainder theorem

Remainder when divided by $(x-2)$is 2, so

$f(2) = 2$
$2(2)^3 + a(2)^2 + b(2) - 6 = 2$
$16 + 4a + 2b - 6 = 2$
$4a + 2b = -8$
$2a + b = -4$(Equation 2)

Step 3: Solve equations

From (1): $b = 4 - a$

Substitute in (2):

$2a + (4 - a) = -4$
$a + 4 = -4$
$a = -8$

Then $b = 4 - (-8) = 12$

2. Correct Answer: D $\boxed{a = -7}$

Let $f(x) = 2x^3 + ax^2 + 11x + a + 3$
If the polynomial is exactly divisible by $2x - 1$, then

$$x = \frac{1}{2}$$

is a zero of the polynomial.

So, by the Factor Theorem: $f\left(\frac{1}{2}\right) = 0$

Step 1: Substitute $x = \frac{1}{2}$

$$2\left(\frac{1}{2}\right)^3 + a\left(\frac{1}{2}\right)^2 + 11\left(\frac{1}{2}\right) + a + 3 = 0$$
$$\frac{1}{4} + \frac{a}{4} + \frac{11}{2} + a + 3 = 0$$
$$\frac{1 + a + 22 + 12 + 4a}{4} = 0$$
$$\frac{35 + 5a}{4} = 0$$
$$35 + 5a = 0$$
$$a = -7$$

3. Correct Answer: C $\boxed{320x^2 - 36x + 1 = 0}$

Given quadratic: $x^2 - 12x + 32 = 0$
So, $\alpha + \beta = 12, \alpha\beta = 32$
Step 1: Find sum of new zeroes $S = \frac{1}{2\alpha+\beta} + \frac{1}{\alpha+\beta}$
Since $\alpha + \beta = 12$,

$$S = \frac{1}{2\alpha + \beta} + \frac{1}{12}$$

Now, $2\alpha + \beta = (\alpha + \beta) + \alpha = 12 + \alpha$
We need α. From $\alpha\beta = 32$ and $\alpha + \beta = 12$, roots are $4, 8$.

Take $\alpha = 4,\ \beta = 8$

Then, $2\alpha + \beta = 2(4) + 8 = 16$
So, $S = \frac{1}{16} + \frac{1}{12} = \frac{3+4}{48} = \frac{7}{48}$
Step 2: Find product of new zeroes

$$P = \frac{1}{(2\alpha + \beta)(\alpha + \beta)} = \frac{1}{16 \times 12} = \frac{1}{192}$$

Step 3: Form the required quadratic equation

$$x^2 - Sx + P = 0$$
$$x^2 - \frac{7}{48}x + \frac{1}{192} = 0$$

Multiply throughout by 192:

$$192x^2 - 28x + 1 = 0$$

Multiply by $\frac{5}{3}$ to match options:

$$320x^2 - 36x + 1 = 0$$

4. Correct Answer: D (All of these

Since $f(x)$ is exactly divisible by $g(x)$,

$$f(x) = (x^2 + bx + c)(px + q)$$

Step 1: Expand RHS

$$(x^2 + bx + c)(px + q)$$
$$= px^3 + qx^2 + bpx^2 + bqx + pcx + qc$$
$$= px^3 + (q + bp)x^2 + (bq + pc)x + qc$$

Step 2: Compare with $ax^3 + bx - c$
Matching coefficients:

1. $p = a$
2. $q + bp = 0 \Rightarrow q = -ab$
3. $bq + pc = b$
4. $qc = -c$

Step 3: Use equation (4)

$$(-ab)c = -c$$
$$abc = c$$

Since $c \neq 0$,

$$ab = 1$$

Option B is true.
Step 4: Use equation (3)

$$b(-ab) + ac = b$$
$$-ab^2 + ac = b$$

Using $ab = 1 \Rightarrow a = \frac{1}{b}$:

$$-\frac{1}{b}b^2 + ac = b$$
$$-b + ac = b$$
$$ac = 2b$$

Option C is true.
Step 5: From $ab = 1$ and $ac = 2b$

$$a = \frac{1}{b}$$
$$\frac{c}{b} = 2b \Rightarrow c = 2b^2$$

Option A is true.
Final Answer:
All options A, B, and C are true.

5. Correct Answer: D

Expand the expression: $(x + 3)^3 = (x + 3)(x + 3)(x + 3)$
First expansion:

$$(x + 3)^2 = x^2 + 6x + 9$$

Now multiply by $(x+3)$:

$$(x^2 + 6x + 9)(x + 3)$$
$$= x^3 + 3x^2 + 6x^2 + 18x + 9x + 27$$
$$= x^3 + 9x^2 + 27x + 27$$

The constant term is the term without x.
$\boxed{27}$

6. Correct Answer: C $\boxed{8}$

Given polynomial: $3s^2 - 6s + 4 = 0$
For a quadratic $as^2 + bs + c = 0$: $\alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$
So,

$$\alpha + \beta = \frac{6}{3} = 2$$

$$\alpha\beta = \frac{4}{3}$$

Step 1: Find $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$

$$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 2^2 - 2 \cdot \frac{4}{3}$$

$$= 4 - \frac{8}{3} = \frac{4}{3}$$

$$\frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{\frac{4}{3}}{\frac{4}{3}} = 1$$

Step 2: Find $2\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{2}{\frac{4}{3}} = \frac{3}{2}$$

$$2\left(\frac{3}{2}\right) = 3$$

Step 3: Find $3\alpha\beta$

$$3\alpha\beta = 3 \cdot \frac{4}{3} = 4$$

Step 4: Add all terms

$$1 + 3 + 4 = 8$$

7. Correct Answer: A

Step 1: Use sum of zeroes of the first polynomial

For a quadratic $ax^2 + bx + c$,

$$\text{Sum of zeroes} = -\frac{b}{a}$$

Here, $5x^2 - (3 + k)x + 7$
So, Sum of zeroes $= \frac{3+k}{5}$
Given sum of zeroes is 0: $\frac{3+k}{5} = 0$
$3 + k = 0$
$k = -3$
Step 2: Substitute $k = -3$ in the second polynomial

$$2x^2 - 2(k + 11)x + 30$$
$$= 2x^2 - 2(8)x + 30$$
$$= 2x^2 - 16x + 30$$

Divide by 2:

$$x^2 - 8x + 15 = 0$$

Step 3: Find zeroes

$$x^2 - 8x + 15 = 0$$
$$(x - 3)(x - 5) = 0$$

So, the zeroes are: $\boxed{3, 5}$

8. Correct Answer: D

Let $f(x) = px^2 + 5x + r$
Since $(x-2)$ is a factor,

$f(2) = 0$
$p(2)^2 + 5(2) + r = 0$
$4p + 10 + r = 0$ (1)

Since $\left(x - \frac{1}{2}\right)$ is a factor,

$$f\left(\frac{1}{2}\right) = 0$$

$$p\left(\frac{1}{2}\right)^2 + 5\left(\frac{1}{2}\right) + r = 0$$

$$\frac{p}{4} + \frac{5}{2} + r = 0$$

Multiply by 4: $p + 10 + 4r = 0$ (2)
Solve equations (1) and (2):

From (1): $4p + 10 + r = 0 \Rightarrow r = -4p - 10$
Substitute in (2):

$p + 10 + 4(-4p - 10) = 0$
$p + 10 - 16p - 40 = 0$
$-15p - 30 = 0$
$p = -2$
Now find r: $r = -4(-2) - 10 = 8 - 10 = -2$
Thus, $p = r$
Answer: $\boxed{p = r}$

9. Correct Answer: B

Let the roots be α and β.

If one root is the reciprocal of the other, then: $\alpha\beta = 1$
For a quadratic polynomial $ax^2 + bx + c$,

$$\alpha\beta = \frac{c}{a}$$

Here, $a = 3, c = p$
So,

$$\frac{p}{3} = 1$$

$$p = 3$$

Answer: $\boxed{p = 3}$

4. Pair of Linear Equations

Solutions

Worksheet – 1

1. **Correct Answer: B) 5**
   LHS:$6-6x+9=15-6x$
   RHS:$30-5x-4x=30-9x$
   So,$15-6x=30-9x \Rightarrow x=5$
2. **Correct Answer: B) -1**
   $\frac{2x+3}{3x+4}=\frac{2x-3}{3x-2}$
   Solution:
   Cross multiply:
   $(2x+3)(3x-2)=(2x-3)(3x+4)$
   Expand:
   $6x^2+5x-6=6x^2-x-12$
   Simplify: $6x=-6 \Rightarrow x=-1$
3. **Correct Answer: D) 18.3**
   $0.16(5x-2)=0.4x+7$
   Solution: $0.8x-0.32=0.4x+7$
   $\Rightarrow 0.4x=7.32 \Rightarrow x=18.3$
4. **Correct Answer: A) 4**
   $\frac{2x+1}{3x-2}=\frac{9}{10}$
   Solution:$10(2x+1)=9(3x-2)$
   $20x+10=27x-18 \Rightarrow 7x=28 \Rightarrow x=4$
5. **Correct Answer: C) -10**
   LHS: $\frac{3x-2x-8}{3}=\frac{x-8}{3}$
   RHS: $\frac{1}{4}\left(\frac{7x-2}{6}\right)-3=\frac{7x-2}{24}-3$
   Multiply by 24: $8(x-8)=7x-2-72$
   $8x-64=7x-74 \Rightarrow x=-10$
6. **Correct Answer: D) −3**
   Convert 2 into fraction:$\frac{2(2x+3)}{2x+3}+\frac{2x-3}{2x+3}=\frac{6x+3}{2x+3}$
   So,$\frac{6x+3}{2x+3}=\frac{3x+4}{x+2}$
   Cross multiply:$(6x+3)(x+2)=(3x+4)(2x+3)$
   $6x^2+15x+6=6x^2+17x+12 \Rightarrow -2x=6 \Rightarrow x=-3$
7. **Correct Answer: A) 23**
   Square both sides:$x^2+\frac{1}{x^2}+2=25$
   $$x^2+\frac{1}{x^2}=23$$
8. **Correct Answer: A) 2207**
   Using identities: $x^2+\frac{1}{x^2}=3^2-2=7$
   $$x^4+\frac{1}{x^4}=7^2-2=47$$
   $$x^8+\frac{1}{x^8}=47^2-2=2207$$
9. **Correct Answer: C)** $\frac{27}{10}$
   Take LCM = 60
   $$30x-12=20x+15$$
   $$10x=27 \Rightarrow x=\frac{27}{10}$$
10. **Correct Answer: B) 12**
    $\frac{x}{4}+\frac{x}{6}=x-7$
    LCM = 12 $3x+2x=12x-84$
    $5x=12x-84$
    $7x=84 \Rightarrow x=12$
11. **Correct Answer: B) 60 cm³**
    Volume of a cuboid
    Given: Length l=5 cm, Breadth b=4 cm, Height h=3 cm
    Formula: Volume=l×b×h
    Calculation: 5×4×3=60 cm3
12. **Correct Answer: A) 100 m²**
    Length l=6 m, Breadth b=4 m, Height h=5 m
    Formula:LSA=2h(l+b)
    Calculation:
    2×5×(6+4)=10×10=100 m2

Worksheet – 2

1. **Correct Answer: B) 23, 17**
   Sum of two numbers = 40
   Difference = 6
   Let the numbers be $x$and y. $x+y=40$
   $x-y=6$
   Adding both equations: $2x=46 \Rightarrow x=23$
   $y=40-23=17$
2. **Correct Answer: D)** a
   $\frac{x}{a}+\frac{y}{b}=2andax-by=a^2-b^2$
   From first equation:$x=2a-\frac{ay}{b}$
   Substitute in second: $a\left(2a-\frac{ay}{b}\right)-by=a^2-b^2 \Rightarrow y=b$

Put $y = b$:$x = 2a - a = a$

3. **Correct Answer: B) Age of A $= 5x = 15$years.**
Present age of A
Let ages be $5x$and $3x$.
$\frac{5x+6}{3x+6} = \frac{7}{5} \Rightarrow 25x + 30 = 21x + 42 \Rightarrow x = 3$

4. **Correct Answer: A) Cost of $2x + 3y = 34.50$.**
Cost of pens and notebooks
Let pen $= x$, notebook $= y$.
$5x + 7y = 82, 4x + 4y = 52 \Rightarrow x + y = 13$
$x = 13 - y \Rightarrow 5(13 - y) + 7y = 82 \Rightarrow y = 8.5, x = 4.5$.

5. **Correct Answer: ₱34.50 (Option A)**
$$5x + 7y = 82 \ldots (1)$$
$$4x + 4y = 52 \Rightarrow x + y = 13 \ldots (2)$$
From (2):
$$x = 13 - y$$
Substitute in (1):
$$5(13 - y) + 7y = 82$$
$$65 - 5y + 7y = 82$$
$$2y = 17 \Rightarrow y = 8.5$$
Then:
$$x = 13 - 8.5 = 4.5$$
Cost of 2 pens and 3 notebooks:
$$2x + 3y = 2(4.5) + 3(8.5) = 9 + 25.5 = 34.5$$

6. **Correct Answer: D) 9**
$2x - y = 5$ and $3x + 2y = 11$
Solving: *x=3, y=1*
$$2x + 3y = 6 + 3 = 9$$

7. **Correct Answer: C) 6**
For coincident lines:
$$\frac{3}{9} = \frac{4}{12} = \frac{2}{k} \Rightarrow \frac{1}{3} = \frac{2}{k} \Rightarrow k = 6$$

8. **Correct Answer: A) 212**
Let length = l, breadth = b
From conditions:$5b - 2l = 70$
$$10b - 5l = -50$$
Solving:
L= 78, b=28
Perimeter:$2(l + b) = 2(106) = 212$

9. **Correct Answer: C) Inconsistent**
No solution ⇒ **Inconsistent**

10. **Correct Answer: C) Coincident lines**
10)3x−2y=5
6x−4y=10
Multiply the first equation by 2:
2(3x−2y=5)⇒6x−4y=10
Both equations are identical, so they represent the **same line**.

Competitive Corner

SINGLE CORRECT ANSWER TYPE

1. **Correct Answer: C) 3/2**
Formula: $\frac{4p+9q}{p} = \frac{5q}{p-q}$
Substitution: $(4p + 9q)(p - q) = 5qp$
Result: $4p^2 - 9q^2 = 0 \Rightarrow 2p = 3q \Rightarrow \frac{p}{q} = \frac{3}{2}$

2. **Correct Answer: B) $\frac{x}{y}$**
Formula: $\frac{x-\frac{1}{y}}{y-\frac{1}{x}}$
Substitution: Multiply numerator & denominator by xy
Result: $\frac{x(xy-1)}{y(xy-1)} = \frac{x}{y}$

3. **Correct Answer: 11,36**
$x + y = 47$
$$25x + 50y = 1450$$
Substitution: $x + 2y = 58$and $x + y = 47$
Result: $y = 11, x = 36$

4. **Correct Answer: C) 11**
Formula: $10x + 3y = 75$
Substitution: Take $x = 6 \Rightarrow y = 5$
Result: $x + y = 11$

5. **Correct Answer: C) $\frac{2}{3}$**
Formula: Let numerator $= x$, denominator $= x + 1$
Substitution: $\frac{x-1}{x} = \frac{1}{2}$
Result: $2x - 2 = x \Rightarrow x = 2 \Rightarrow$fraction $= \frac{2}{3}$

6. **Correct Answer: C) Infinitely many solutions**
Formula: Compare ratios $\frac{a_1}{a_2}, \frac{b_1}{b_2}, \frac{c_1}{c_2}$
Substitution: $\frac{2}{4} = \frac{3}{6} = \frac{6}{12}$

Result: All ratios equal ⇒ infinitely many solutions

7. **Correct Answer: B) 6**

Formula: Heads $= h$, goats $= g$
Substitution: $2g + 4h = 2(h + g) + 12$
Result: $2g = 12 \Rightarrow g = 6$

8. **Correct Answer: B) 3**

Formula: Point of intersection satisfies both equations
Substitution: $2(1) + 4b = 6 \Rightarrow b = 1$,
$3(4) = 8 + a \Rightarrow a = 4$
Result: $a - b = 4 - 1 = 3$

9. **Correct Answer: B) –2**

Formula: Substitute $(a + b, a - b)$in equations
Substitution:
$3(a + b) + 2(a - b) = 20 \Rightarrow 5a + b = 20$
$4(a + b) - 5(a - b) = 42 \Rightarrow -a + 9b = 42$
Multiply Eq 2 by 5: -5a + 45b = 210.

Add Eq 1 to this: (5a + b) + (-5a + 45b) = 20 + 210 => 46b = 230 => b = 5.

10. **Correct Answer: B) 8**

Formula: Let fraction $= \frac{x}{y}$
Substitution: $\frac{x+1}{y-1} = 1$, $\frac{x}{y+1} = \frac{1}{2}$
Result: $x = 3, y = 5 \Rightarrow x + y = 8$

11. **Correct Answer: A)** $k = 6$

Formula: Zero solution condition $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
Substitution:

$$\frac{k}{3} = \frac{2}{1} \Rightarrow k = 6, \frac{5}{1} \neq \frac{2}{1}$$

Result: Lines are parallel ⇒ zero solutions

12. **Correct Answer: A) One solution**

Formula:
If $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$, then one solution
Substitution: $\frac{1}{1} \neq \frac{1}{-1}$
Result: Lines intersect at one point ⇒ one solution

ONE OR MORE THAN ONE OPTION CORRECT TYPE

13. **Correct Answer: A) One solution**

Solve the system

$$\frac{ax}{b} - \frac{by}{a} = a + b, ax - by = 2ab$$

14. **Correct Answer: B) 3**

Formula: $\frac{x}{y}$is maximum when $x$is max and $y$is min
Substitution: $x = 6$, $y = 2$
Result: $\frac{x}{y} = 3$

15. **Correct Answer: A and C**

Formula: Solve equations
Substitution: Solution gives $x = 3$, $y = 4$
Result:
A)$19x = 57$(true)
$B)$ $x = 3$, $y = 4$(true)

16. **Correct Answer: B and C**

Formula: Work = Rate × Time
Substitution:$12A + 6B = \frac{1}{18}, A + B = \frac{1}{10}$
Result:
B) A alone = 30 days
C) B alone = 30 days

17. **Correct Answer: A and D**

Formula: Solve simultaneous equations
Substitution: Elimination method
Result:
$x = 5(\sqrt{10} - 3)$
$y = 5(\sqrt{10} - 3)$

ASSERTION AND REASON TYPE

18. **Correct Answer: (A)**

A pair of linear equations
$2x + 3y = 5$, $4x + 6y = 10$
has infinitely many solutions.

Reason (R):
Both equations represent the same line in the plane.

Explanation:
The second equation is a multiple of the first, so both represent the same line ⇒ infinitely many solutions.
Reason correctly explains the assertion.

19. **Correct Answer: A**

Assertion (A):
If two lines are parallel, the pair of linear equations has no solution.

Reason (R):
The slopes of two parallel lines are equal, but their intercepts are different.

Explanation:
Parallel lines never intersect, so no solution. Equal slopes and different intercepts explain why they are parallel.

20. Correct Answer: D

Assertion (A):
The equations $3x + 2y = 7$and $6x + 4y = 14$are inconsistent.

Reason (R):
The ratio of coefficients is

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

Explanation:
Here, the second equation is exactly double the first ⇒ same line ⇒ **infinitely many solutions**, not inconsistent.
Reason states the condition for **no solution**, which is true in general, but not applicable here.

21. Correct Answer (A):

If the pair of linear equations is consistent and independent, the two lines intersect at exactly one point.

Reason (R):
For this case,$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

22. Correct Answer: (A)

The equations $x - y = 5$and $2x - 2y = 10$represent two coincident lines.

Reason (R):
The second equation is a multiple of the first.

Correct Option: (A)

Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

Short Explanation:

- Multiplying the first equation $x - y = 5$by 2 gives $2x - 2y = 10$.
- Hence, both equations represent the same line (coincident lines).
- Therefore, the Reason correctly explains the Assertion.

MATRIX MATCHING TYPE

23. Correct Answer:

A) $2x + 3y + 4 = 0;\ 5x + 6y + 9 = 0$

- Ratios unequal ⇒ **unique solution**
  Solving gives $\left(-1, -\frac{2}{3}\right)$
  Match: P

B) $x + 3y = 1;\ 2x + 6y = 2$

- Second equation is a multiple of the first
  ⇒ **Infinitely many solutions**
  Match: R

C) Solution set for inconsistent system

No common solution
⇒ Null set
Match: S

D) $y = mx,\ m \neq 0$

- Line always passes through origin
  $\Rightarrow (0, 0)$

24. Correct Answer:

A)$\sqrt{3}x + \sqrt{5}y = \sqrt{3}, \sqrt{5}x + \sqrt{3}y = \sqrt{3}$
Solving gives $x = y = \frac{\sqrt{15}-3}{2}$
Match: R

B)$2^x + 3^y = 17, 2^{x+2} - 3^{y+1} = 5$
Checking values gives solution$(x, y) = (2,3)$
Match: S

C)$\frac{2}{x} + \frac{3}{y} = 13, \frac{5}{x} - \frac{4}{y} = -2$
Solving gives$(x, y) = \left(\frac{1}{2}, \frac{1}{3}\right)$
Match: Q

D) $\frac{5}{x-1}+\frac{1}{y-2}=2, \frac{6}{x-1}-\frac{3}{y-2}=1$

Let $u=\frac{1}{x-1}$, $v=\frac{1}{y-2}$
Solution gives

$$(x, y)=(4,5)$$

Match: T

COMPREHENSION TYPE

Passage - I

25. Correct Answer:

1. $\frac{1}{2}a+\frac{12}{7}b=\frac{1}{2} \Rightarrow 7a+24b=7$
2. $7a+4b=2$

Solve: Subtract (2) from (1):$(7a+24b)-(7a+4b)=7-2 \Rightarrow 20b=5 \Rightarrow b=\frac{1}{4}$
Substitute in (2):$7a+4\left(\frac{1}{4}\right)=2 \Rightarrow 7a+1=2 \Rightarrow a=\frac{1}{7}$

Solve:
Multiply (1) by 2: $4x+6y=14$
Multiply (2) by 3: $9x-6y=12$
Add:$13x=26 \Rightarrow x=2$
Substitute:$2(2)+3y=7 \Rightarrow y=1$

26. Correct Answer: 2 (Option A)

The value of x is

27. Correct Answer: 3 (Option C)

The value of x + y is$2+1=3$

28. Correct Answer: 1 (Option B)

The value of x − y is $2-1=1$

Passage-2

29. Correct Answer: -3

Formula (parallel lines):$\frac{a_1}{a_2}=\frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

Substitution:$\frac{2}{4}=\frac{-k}{6} \Rightarrow \frac{1}{2}=\frac{-k}{6}$

Result:$k=-3$

30. Correct Answer: $k \in \mathbb{R} \setminus \left\{\frac{13}{3}\right\}$

Short explanation:
Formula (intersecting lines): $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

Substitution:$\frac{3}{4} \neq \frac{1}{k-3} \Rightarrow k-3 \neq \frac{4}{3}$

Result:$k \neq \frac{13}{3}$

31. Correct Answer:

A) $\boldsymbol{k=\frac{3}{2}}$

Short justification:
For coincident lines,

$$\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$$
$$\frac{k}{3}=\frac{2}{4}=\frac{1}{5} \Rightarrow k=\frac{3}{2}$$

INTEGER TYPE

32. Correct Answer: 4

Formula: Divide by $xy$and put $a=\frac{1}{x}$, $b=\frac{1}{y}$
Substitution:$3a+6b=7$, $9a+3b=11 \Rightarrow b=\frac{2}{3}$, $a=1$
Result:$x=1$, $y=\frac{3}{2} \Rightarrow x+2y=4$

33. Correct Answer: 6

Formula: Three times a number increased by 6 $\rightarrow 3n+6=24$
Substitution: $3n=18$
Result: $n=6$

34. Correct Answer: 2

Formula: No solution condition $\frac{a_1}{a_2}=\frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
Substitution: $\frac{1}{k}=\frac{1}{1} \Rightarrow k=1$
Result: $k+1=2$

35. Correct Answer: 4

Formula:
One bee per flower $\rightarrow B=F+1$
Two bees per flower $\rightarrow B=2(F-1)$
Substitution:

$$F+1=2F-2$$

Result:
$F=3$(number of flowers)

36. Correct Answer: 4

Formula: Clear fractions and solve simultaneous equations
Substitution:$15x+16y=-11$, $10x+9y=-14$
Result:$y=4$

37. Correct Answer: 2

Formula: From $1+\frac{x}{y}=\frac{13}{8} \Rightarrow \frac{x}{y}=\frac{5}{8}$
Substitution: $x=\frac{5y}{8}$in $2x+3y=34$
Result: $x=5$, $y=8 \Rightarrow 2x-y=2$

38. Correct Answer: 4
Formula: Solve simultaneous equations
Substitution: From $x+\frac{4}{y}=4$, $y+\frac{6}{x}=6$
Result: $x=2$, $y=2 \Rightarrow x+y=4$

39. Correct Answer: 2
Formula: Let number $=10a+b$
Substitution:
$$(10a+b)+(10b+a)=66 \Rightarrow a+b=6,\ |a-b|=2$$
Result: Two numbers satisfy the condition

PREVIOUS YEAR QUESTIONS

1. Correct Answer: -5
Formula: Solve the given pair, then use $y=mx+7$
Substitution: Solving gives $(x,y)=(-1,2)$;
$$2=m(-1)+7$$
Result: $m=-5$

2. Correct Answer: A) $x+y=2$
Formula: Equation of line through two points
Substitution: Given points $(0,2)$and $(1,1)$satisfy $x+y=2$
Result: Correct equation is $x+y=2$

3. Correct Answer: C
Short explanation:
Formula: Let cost of a cap = x, cost of a bat = y
Substitution:
$$3x+4y=257,\ 4x+3y=324 \Rightarrow x=59,\ y=16$$
Result:
$x+10y=59+160=$ ₹155 Answer: ₹155
Answer type: Single correct

4. Correct Answer:
Answer: ₹1200
Formula: Let Mohan = M, Sushant = S
Substitution:
$$M+700=2(S-700),\quad S+300=3(M-300)$$
Result: Solving gives $S=$ ₹1200

5. Correct Answer: A
Formula: Let the fraction be $\frac{x}{y}$
Substitution:
$$\frac{x}{y+1}=\frac{2}{3},\frac{x+5}{y}=1$$
From second: $y=x+5$
Substitute: $\frac{x}{x+6}=\frac{2}{3}$
Result:
$$3x=2x+12 \Rightarrow x=12$$

6. Correct Answer: A
Formula: Let tens digit = x, units digit = y
Number $=10x+y$, reversed $=10y+x$
Substitution:
$$(10x+y)-(10y+x)=9 \Rightarrow x-y=1$$
$$x+y=7$$
Result:
$x=4$, $y=3 \Rightarrow$product $=4\times3=12$

7. Correct Answer: 36
Correct option: A
Short explanation:
Formula (no solution):
$$\frac{a_1}{a_2}=\frac{b_1}{b_2}\neq\frac{c_1}{c_2}$$
Substitution:
$$\frac{p}{12}=\frac{3}{1}\Rightarrow p=36$$

8. Correct Answer: D
Formula: From $\frac{a}{x}=\frac{b}{y}\Rightarrow ay=bx$
Substitution:
$y=\frac{b}{a}x$in second equation
$$\frac{ab^2}{x}+\frac{a^2b}{(b/a)x}=\frac{a(a^2+b^2)}{x}$$
Result:
$$\frac{a(a^2+b^2)}{x}=a^2+b^2 \Rightarrow x=a,\ y=b$$
Final Answer
$$\boxed{x=a,\ y=b}$$

9. Correct Answer: B
Formula: Let fare A→B = x, A→C = y
Substitution:
$$2x+3y=795,\ 3x+5y=1300 \Rightarrow y=215,\ x=75$$
Result:
Fare A→B = ₹75, Fare A→C = ₹215

10. Correct Answer: A
Formula: Substitute given values of $x$and y
(I)
$\frac{1}{2x}-\frac{1}{y}=3-4=-1$✓
$\frac{1}{x}+\frac{1}{2y}=6+2=8$✓

5. QUADRATIC EQUATIONS SOLUTIONS

Solutions

Worksheet – 1

1. **Correct Answer: C**
   Roots: $2\sqrt{3}-5$ and $-2\sqrt{3}-5$
   Sum of roots: $(2\sqrt{3}-5)+(-2\sqrt{3}-5)=-10$
   Product of roots: $(2\sqrt{3}-5)(-2\sqrt{3}-5)=25-12=13$
   Equation: $x^2-(\text{sum})x+(\text{product})=0$
   $$x^2+10x+13=0$$
2. **Correct Answer: C**
   Equation: $9x^2+kx+1=0$ has equal roots
   For equal roots, discriminant $D=0$
   $D=k^2-4(9)(1)=0$
   $k^2=36$
   $k=\pm 6$
3. **Correct Answer: B**
   Value of $2-\frac{1}{2-\frac{1}{2-\cdots}}$
   Let the value be x
   $x=2-\frac{1}{x}$
   $x^2-2x+1=0$
   $(x-1)^2=0\Rightarrow x=1$
4. **Correct Answer: C**
   Equal roots of
   $x^2-2(5+2k)x+3(7+10k)=0$
   Discriminant $=0$
   $[-2(5+2k)]^2-4(1)(3)(7+10k)=0$
   $4(25+20k+4k^2)-12(7+10k)=0$
   $16k^2-40k+16=0$
   $2k^2-5k+2=0$
   $k=\frac{1}{2},\ 2$
5. **Correct Answer: C**
   Solve $x^{2/3}+x^{1/3}-2=0$
   Let $y=x^{1/3}$
   $$y^2+y-2=0$$
   $$(y+2)(y-1)=0$$
   $$y=1,-2\Rightarrow x=1,-8$$
6. **Correct Answer: B**
   Roots: 5 and −2
   Sum $=3$, Product $=-10$
   Equation: $x^2-3x-10=0$
7. **Correct Answer:** $x=2$
   Equation:
   $3^{2x}-3^{x+1}-54=0$
   Let $3^x=y$
   Then,
   $3^{2x}=y^2$ and $3^{x+1}=3y$
   Substitute:
   $y^2-3y-54=0$
   Factor: $(y-9)(y+6)=0$
   So, $y=9$ or $y=-6$
   But $y=3^x>0$, so $y=-6$ is not possible.
   $3^x=9=3^2$
   $\boxed{x=2}$
8. **Correct Answer: B (1, −1)**
   Solve $7^{1+x}+7^{1-x}=50$
   Step 1: Take common factor
   $$7(7^x+7^{-x})=50$$
   $$7^x+7^{-x}=\frac{50}{7}$$
   Step 2: Try given options
   For $x=1$:
   $$7+\frac{1}{7}=\frac{50}{7}\checkmark$$
   For $x=-1$:
   $$\frac{1}{7}+7=\frac{50}{7}\checkmark$$
   Step 3: Solution set
   $$x=1,\ -1$$
9. **Correct Answer: C)**
   $$6x^2-5x+1=0$$
   Solution:
   Factor the quadratic: $6x^2-5x+1=(3x-1)(2x-1)$
   Set each factor equal to zero:
   $3x-1=0\Rightarrow x=\frac{1}{3}$
   $2x-1=0\Rightarrow x=\frac{1}{2}$
   Roots: $\frac{1}{2},\frac{1}{3}$

10. Correct Answer: C

Roots of $\sqrt{3}x^2 + 10x - 8\sqrt{3} = 0$

Using quadratic formula:

$$x = \frac{-10 \pm \sqrt{100 + 96}}{2\sqrt{3}} = \frac{-10 \pm 14}{2\sqrt{3}}$$

Roots:

$x = \frac{2}{\sqrt{3}},\ -4\sqrt{3}$

WORKSHEET-2

1. Correct Answer: B)-108

Discriminant of $x^2 + 6x + 36 = 0$

$$D = b^2 - 4ac = (6)^2 - 4(1)(36) = 36 - 144 = -108$$

2. Correct Answer: B) Real and equal

Nature of the roots of $9x^2 + 12x + 4 = 0$

$$D = (12)^2 - 4(9)(4) = 144 - 144 = 0$$

Since $D = 0$, roots are **real and equal**.

3. Correct Answer: D) 17, 18

Sum of squares of two consecutive positive integers is 613

Let numbers be $n$and $n + 1$

$$n^2 + (n+1)^2 = 613$$
$$2n^2 + 2n - 612 = 0$$
$$n^2 + n - 306 = 0$$
$$(n - 17)(n + 18) = 0$$

So $n = 17$

Numbers are **17, 18**

4. Correct Answer: C

Let the roots be $3k$and $2k$.

For a quadratic equation $ax^2 + bx + c = 0$:

- Sum of roots $= -\frac{b}{a}$
- Product of roots $= \frac{c}{a}$

Step 1: Use product of roots

$$(3k)(2k) = \frac{5}{12}$$
$$6k^2 = \frac{5}{12}$$
$$k^2 = \frac{5}{72}$$
$$k = \sqrt{\frac{5}{72}} = \frac{\sqrt{10}}{12}$$

Step 2: Use sum of roots

$$3k + 2k = \frac{-m}{12}$$
$$5k = \frac{-m}{12}$$
$$m = -60k$$

Substitute $k = \frac{\sqrt{10}}{12}$:

$$m = -60 \times \frac{\sqrt{10}}{12}$$
$$m = -5\sqrt{10}$$

$$\boxed{m = -5\sqrt{10}}$$

(If both roots are taken negative, $m = 5\sqrt{10}$is also possible.)

5. Correct Answer: D) 4

Roots of $x^2 + px + 4 = 0$are equal

$$D = p^2 - 4(1)(4) = 0$$
$$p^2 = 16 \Rightarrow p = 4$$

6. Correct Answer: B) 3

If one root of $x^2 - 5x + k = 0$ is 2

Sum of roots $= 5$

$$2 + \text{other root} = 5$$

Other root $= 3$

7. Correct Answer: B $\boxed{-\frac{1}{2},\ 1}$

If the roots of $(3m + 1)x^2 + 2(m + 1)x + m = 0$are equal, find m.

Condition for equal roots:

$$b^2 - 4ac = 0$$

Here,

$a = 3m + 1, b = 2(m + 1), c = m$

$[2(m + 1)]^2 - 4(3m + 1)(m) = 0$

$4(m + 1)^2 - 4m(3m + 1) = 0$

Divide by 4:

$(m + 1)^2 - m(3m + 1) = 0$

$(m^2 + 2m + 1) - (3m^2 + m) = 0$

$-2m^2 + m + 1 = 0$

$2m^2 - m - 1 = 0$

Factor:

$(2m + 1)(m - 1) = 0$

$m = -\frac{1}{2},\ 1$

8. Correct Answer: B $\boxed{1}$

One root of the quadratic equation $(q - r)x^2 + (r - p)x + (p - q) = 0$ is?

Test the given options.

Substitute $x = 1$: $(q - r) + (r - p) + (p - q) = 0$

All terms cancel out:

$$0 = 0$$

So $x = 1$is a root.

9. Correct Answer: A $\boxed{\frac{-2\pm\sqrt{19}}{5}}$

If $\frac{3}{x+2}-\frac{2}{x-1}=5$

Solution:

Take LCM $(x+2)(x-1)$

$$\frac{3(x-1)-2(x+2)}{(x+2)(x-1)}=5$$
$$\frac{3x-3-2x-4}{(x+2)(x-1)}=5$$
$$\frac{x-7}{(x+2)(x-1)}=5$$
$$x-7=5(x^2+x-2)$$
$$x-7=5x^2+5x-10$$
$$5x^2+4x-3=0$$
$$x=\frac{-4\pm\sqrt{16+60}}{10}=\frac{-4\pm\sqrt{76}}{10}$$
$$=\frac{-2\pm\sqrt{19}}{5}$$

10. Correct Answer: D (11)

If sum of reciprocals of two consecutive numbers is $\frac{23}{132}$.

Let the numbers be $n$and $n+1$.

$$\frac{1}{n}+\frac{1}{n+1}=\frac{23}{132}$$
$$\frac{2n+1}{n(n+1)}=\frac{23}{132}$$

Cross multiply:

$$132(2n+1)=23n(n+1)$$
$$264n+132=23n^2+23n$$
$$23n^2-241n-132=0$$

Solving gives:

$n=11$

Numbers are **11 and 12**

11. Correct Answer: C

Sum of roots $=12$

Sum of cubes of roots $=468$

Let roots be α,β

$$\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$$
$$468=12^3-3(\alpha\beta)(12)$$
$$468=1728-36\alpha\beta$$
$$36\alpha\beta=1260$$
$$\alpha\beta=35$$

Quadratic equation:

$$x^2-(\alpha+\beta)x+\alpha\beta=0$$
$$x^2-12x+35=0$$

Competitive Corner

SINGLE CORRECT ANSWER TYPE

1. Correct Answer: B

Which of the following is a quadratic equation?

A quadratic equation is of degree 2 and has the form $ax^2+bx+c=0$.

- $x^2+\frac{1}{x}=0\rightarrow$ not a polynomial
- $x^2+3=0\rightarrow$ degree 2 ✓
- $x^2-x^{2/3}+1=0\rightarrow$ fractional power
- $x+\frac{1}{x}=0\rightarrow$ not quadratic

2. Correct Answer: B

Roots of the equation $\boldsymbol{3a^2x^2+8abx+4b^2=0}$

Factor:

$$3a^2x^2+8abx+4b^2$$
$$=(3ax+2b)(ax+2b)$$

Roots:

$$x=-\frac{2b}{3a},\ -\frac{2b}{a}$$

Given options match $-\frac{2b}{3a}$

3. Correct Answer: D

Roots of the equation $\boldsymbol{25x^2+30x+7=0}$

Using quadratic formula:

$$x=\frac{-30\pm\sqrt{900-700}}{50}=\frac{-30\pm10\sqrt{2}}{50}$$
$$=\frac{-3\pm\sqrt{2}}{5}$$

4. Correct Answer: $\boxed{\textbf{B) complex and distinct}}$

Determine the nature of the roots of $\boldsymbol{4x^2-8x+5=0}$

$$D=b^2-4ac=(-8)^2-4(4)(5)$$
$$=64-80=-16$$

Since $D<0$, the roots are non-real complex and distinct.

5. Correct Answer: B

If $\boldsymbol{\alpha+\beta=2}$, $\boldsymbol{\alpha\beta=3}$, find $\boldsymbol{\alpha^3+\beta^3}$

$$\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$$
$$=2^3-3(3)(2)=8-18=-10$$

6. Correct Answer: B) $\pm 9\sqrt{3}$

If the roots of the equation $\mathbf{12x^2 + mx + 5 = 0}$ are in the ratio $\mathbf{5:4}$, find $\boldsymbol{m}$.

Let the roots be $5k$ and $4k$.

Sum of roots: $5k + 4k = 9k = \frac{-m}{12}$

Product of roots: $(5k)(4k) = 20k^2 = \frac{5}{12}$

From product: $k^2 = \frac{1}{48} \Rightarrow k = \frac{1}{4\sqrt{3}}$

Then,

$$9k = \frac{9}{4\sqrt{3}} = \frac{-m}{12}$$
$$m = \pm 9\sqrt{3}$$

7. Correct Answer: C) $\frac{11}{270}$

If one root of $\mathbf{x^2 - 10x + 3(7 + 10k) = 0}$ is double of the other, find $\boldsymbol{k}$.

Let the roots be α and 2α.

Sum of roots: $\alpha + 2\alpha = 3\alpha = 10 \Rightarrow \alpha = \frac{10}{3}$

Product of roots: $\alpha(2\alpha) = 2\alpha^2 = 2\left(\frac{10}{3}\right)^2 = \frac{200}{9}$

But product $= 3(7 + 10k)$

$$3(7 + 10k) = \frac{200}{9}$$
$$7 + 10k = \frac{200}{27}$$
$$10k = \frac{11}{27} \Rightarrow k = \frac{11}{270}$$

8. Correct Answer: D) (10, 4)

The dimensions of a rectangle whose perimeter is 28 and whose area is 40 m²

Given:
Perimeter $= 28$

$2(l + b) = 28 \Rightarrow l + b = 14$ ... ... (1)
Area $= 40$

$lb = 40$ ... ... (2)
From (1):

$b = 14 - l$
Substitute in (2):

$l(14 - l) = 40$
$14l - l^2 = 40$
$l^2 - 14l + 40 = 0$
$(l - 10)(l - 4) = 0$
$l = 10$ or 4
So dimensions are **10 and 4**

9. Correct Answer: D) 12, 8

The base of a triangle is 4 cm longer than its altitude. Area = 48 cm²

Let altitude = $x$cm

Then base $= x + 4$cm

Area of triangle:

$\frac{1}{2} \times$ base $\times$ height $= 48$
$\frac{1}{2} \times (x + 4) \times x = 48$
$x(x + 4) = 96$
$x^2 + 4x - 96 = 0$
$(x - 8)(x + 12) = 0$
$x = 8$(positive value)
Altitude $= 8$cm
Base $= 8 + 4 = 12$cm

ONE OR MORE THAN ONE OPTION CORRECT TYPE

10. Correct Answer: A, C, D

If $\boldsymbol{\alpha, \beta}$ are roots of $\mathbf{x^2 + x + 2 = 0}$

$$\alpha + \beta = -1, \alpha\beta = 2$$

A. $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$
$= (-1)^3 - 3(2)(-1) = -1 + 6 = 5$

B. $\alpha^2\beta^3 + \alpha^3\beta^2 = \alpha^2\beta^2(\alpha + \beta)$
$= (\alpha\beta)^2(\alpha + \beta) = 4(-1) = -4 \neq 3$

C. $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta = 1 - 8 = -7$
$\Rightarrow \alpha - \beta = \sqrt{-7}$

D. $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 1 - 4 = -3$

11. Correct Answer: A, B, C, D

If $6\left(x^2 + \frac{1}{x^2}\right) - 25\left(x - \frac{1}{x}\right) + 12 = 0$

Let $y = x - \frac{1}{x}$.

Then $x^2 + \frac{1}{x^2} = y^2 + 2$.

$6(y^2 + 2) - 25y + 12 = 0$
$\Rightarrow 6y^2 - 25y + 24 = 0$

$$(3y-8)(2y-3)=0 \Rightarrow y=\frac{8}{3},\ \frac{3}{2}$$

Solving $x-\frac{1}{x}=y$:

- For $y=\frac{8}{3} \Rightarrow x=3,\ -\frac{1}{3}$
- For $y=\frac{3}{2} \Rightarrow x=2,\ -\frac{1}{2}$

12. Correct Answer: A, B

If $\boldsymbol{\alpha, \beta}$are roots of $\boldsymbol{ax^2+bx+c=0}$

$$\sqrt{\left(\frac{-b}{a}\right)^2-\frac{4c}{a}}=\sqrt{(\alpha+\beta)^2-4\alpha\beta}=|\ \alpha-\beta\ |$$

13. Correct Answer: C, D

If $\boldsymbol{\alpha, \beta}$are roots of $\boldsymbol{x^2+x+1=0}$

$$\alpha+\beta=-1, \alpha\beta=1$$

A) $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}=\frac{\alpha^2+\beta^2}{\alpha\beta}=\frac{(\alpha+\beta)^2-2\alpha\beta}{1}=1-2=-1\neq 1$ ✗

- **B)** $\frac{1}{\alpha^2}+\frac{1}{\beta^2}=\frac{\alpha^2+\beta^2}{(\alpha\beta)^2}=-1\neq 1$ ✗
- **C)** $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}=-1$ ✓
- **D)** $\frac{1}{\alpha^2}+\frac{1}{\beta^2}=-1$ ✓

14. Correct Answer: A) 3

If the product of the roots of

$\frac{1}{k}(5x^2-4x+38)+(-4x^2-2x-1)=0$ is **−5**, find k.

Step 1: Combine the expressions

$$\frac{1}{k}(5x^2-4x+38)-4x^2-2x-1=0$$

Multiply throughout by $k$to remove the denominator:

$$5x^2-4x+38-k(4x^2+2x+1)=0$$
$$(5-4k)x^2+(-4-2k)x+(38-k)=0$$

Step 2: Use product of roots

For $ax^2+bx+c=0$,

$$\text{Product of roots}=\frac{c}{a}$$

Here,

$$\frac{38-k}{5-4k}=-5$$

Step 3: Solve for $\boldsymbol{k}$

$$38-k=-5(5-4k)$$
$$38-k=-25+20k$$
$$38+25=20k+k$$
$$63=21k$$
$$k=3$$

ASSERTION AND REASON TYPE

15. Correct Answer: (A)

Assertion (A): The quadratic equation $x^2-5x+6=0$has two distinct real roots.
Reason (R): A quadratic equation has two distinct real roots if the discriminant $b^2-4ac>0$.

Solution:

For $x^2-5x+6=0$,

$$a=1,\ b=-5,\ c=6$$

Discriminant,

$$D=b^2-4ac=(-5)^2-4(1)(6)$$
$$=25-24=1>0$$

So, the equation has **two distinct real roots**.

- Assertion (A) is **true**.
- Reason (R) is **true** and correctly explains Assertion (A).

16. Correct Answer: (D)

Assertion (A): The roots of $2x^2+3x+7=0$are real and distinct.
Reason (R): The discriminant of the equation is $9-56=-47<0$.

Solution:

For $2x^2+3x+7=0$,

$$a=2,\ b=3,\ c=7$$

Discriminant,

$$D=b^2-4ac=3^2-4(2)(7)=9-56$$
$$=-47<0$$

Since $D<0$, the roots are **non-real (complex)**.

- Assertion (A) is **false**.
- Reason (R) is **true**.

17. Correct Answer: (B)

Assertion (A): If one root of a quadratic equation is rational and the coefficients are integers, then the other root must also be rational.
Reason (R): Roots of a quadratic equation are given by

$$x=\frac{-b\pm\sqrt{D}}{2a}.$$

Solution:

If the coefficients $a, b, c$are integers and

one root is rational, then $\sqrt{D}$must be rational. Hence, the other root $\frac{-b-\sqrt{D}}{2a}$will also be rational.

- Assertion (A) is **true**.
- Reason (R) is **true**, but it does **not correctly explain** why the other root must be rational (the explanation depends on the nature of $\sqrt{D}$).

18. Correct Answer: (D)

Assertion (A): The quadratic equation $kx^2 - 6x + 1 = 0$will have equal roots if $k = 9/2$.
Reason (R): For equal roots, discriminant $= 0$.

Solution:
For $kx^2 - 6x + 1 = 0$, $D = b^2 - 4ac = (-6)^2 - 4(k)(1) = 36 - 4k$.
For equal roots, $36 - 4k = 0 \Rightarrow k = 9$.
So, the given value $k = \frac{9}{2}$is **incorrect**.

- Assertion (A) is **false**.
- Reason (R) is **true**.

19. Correct Answer: (A)

Assertion (A): The equation $x^2 + px + q = 0$can have real roots even if both $p$and $q$are irrational.
Reason (R): The discriminant condition $p^2 - 4q \geq 0$depends only on the values of $p$and q, not on whether they are rational or irrational.

Solution:
Real roots depend only on the sign of the discriminant, not on the rationality of coefficients.

- Assertion (A) is **true**.
- Reason (R) is **true** and correctly explains Assertion (A).

MATRIX MATCHING TYPE

20. Correct Answer:

a) $\mathbf{12 + \sqrt{12 + \sqrt{12 + \cdots}}}$
Let
$x = \sqrt{12 + x}$
$x^2 = 12 + x \Rightarrow x^2 - x - 12 = 0$
$x = 4$ (positive)
So value $= 12 + 4 = 16$

a → Q (16)

b) $\mathbf{\sqrt{3 - \sqrt{3 - \sqrt{3 - \cdots}}}}$
Let
$x = \sqrt{3 - x}$
$x^2 = 3 - x \Rightarrow x^2 + x - 3 = 0$
$x = \frac{-1 + \sqrt{13}}{2}$
b → R $\left(\frac{-1+\sqrt{13}}{2}\right)$

c) $\mathbf{1 + \frac{1}{1+\frac{1}{1+\frac{1}{1+\cdots}}}}$
Let the value be x.
$x = 1 + \frac{1}{x}$
$x^2 = x + 1 \Rightarrow x = \frac{1 + \sqrt{5}}{2}$
c → P $\left(\frac{\sqrt{5}+1}{2}\right)$

d) $\mathbf{\sqrt{6 + \sqrt{6 + \sqrt{6 + \cdots}}}}$
Let
$x = \sqrt{6 + x}$
$x^2 = 6 + x \Rightarrow x^2 - x - 6 = 0$
$x = 3$
d → S (3)

21. Correct Answer:

(a) $\mathbf{3x^2 + 17 = 0}$
$a = 3, b = 0, c = 17$
$D = 0^2 - 4(3)(17) = -204 < 0$
Roots are imaginary (non-real) → P

(b) $\mathbf{25x^2 - 30x + 9 = 0}$
$a = 25, b = -30, c = 9$
$D = (-30)^2 - 4(25)(9) = 900 - 900 = 0$
Roots are real and equal → Q

(c) $\mathbf{x^2 = 16x}$
Rewrite:
$x^2 - 16x = 0$
$a = 1, b = -16, c = 0$
$D = (-16)^2 - 0 = 256$(perfect square)
Roots are rational and unequal → R

(d) $\mathbf{x^2 - 2x - 1 = 0}$
$a = 1, b = -2, c = -1$
$D = (-2)^2 - 4(1)(-1) = 4 + 4 = 8$(not a perfect square)
Roots are real but irrational → S

COMPREHENSION TYPE

Passage – I

Given:

If α, β are roots of $ax^2 + bx + c = 0$, then

$$\alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$$

22. Correct Answer: C) $\frac{c^3}{a^3}$

Find $\frac{\alpha^3+\beta^3}{\alpha^{-3}+\beta^{-3}}$

$$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$$
$$= \left(-\frac{b}{a}\right)^3 - 3\left(\frac{c}{a}\right)\left(-\frac{b}{a}\right)$$

Also,

$$\alpha^{-3} + \beta^{-3} = \frac{\alpha^3 + \beta^3}{(\alpha\beta)^3}$$

Hence,

$$\frac{\alpha^3 + \beta^3}{\alpha^{-3} + \beta^{-3}} = (\alpha\beta)^3 = \left(\frac{c}{a}\right)^3$$

23. Correct Answer: $\frac{3abc-b^3}{a^2c}$

Find $\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$

$$\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{\alpha^3 + \beta^3}{\alpha\beta}$$
$$= \frac{(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)}{\alpha\beta}$$

Substitute values:

$$= \frac{\left(-\frac{b}{a}\right)^3 - 3\left(\frac{c}{a}\right)\left(-\frac{b}{a}\right)}{\frac{c}{a}}$$
$$= \frac{3abc - b^3}{a^2c}$$

24. Correct Answer: B) $\frac{b^2-2ac}{c^2}$

Find $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2 + \beta^2}{(\alpha\beta)^2}$$
$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$
$$= \left(-\frac{b}{a}\right)^2 - 2\left(\frac{c}{a}\right) = \frac{b^2 - 2ac}{a^2}$$
$$(\alpha\beta)^2 = \frac{c^2}{a^2}$$
$$\Rightarrow \frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{b^2 - 2ac}{c^2}$$

Passage – II

25. Correct Answer: B

Given: α, β are roots of

$$x^2 + 7x + 12 = 0$$
$$\alpha + \beta = -7, \alpha\beta = 12$$

Roots required: $(\alpha+\beta)^2$ and $(\alpha-\beta)^2$

$$(\alpha + \beta)^2 = (-7)^2 = 49$$
$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta = 49 - 48 = 1$$

Sum of new roots $= 49 + 1 = 50$

Product of new roots $= 49 \times 1 = 49$

Required equation: $x^2 - 50x + 49 = 0$

26. Correct Answer: A

Given: $3p^2 = 5p + 2, 3q^2 = 5q + 2, p \neq q$

Rewriting: $3x^2 - 5x - 2 = 0$

So, $p + q = \frac{5}{3}, pq = -\frac{2}{3}$

Roots required: $3p - 2q$ and $3q - 2p$

Sum: $(3p - 2q) + (3q - 2p) = p + q = \frac{5}{3}$

Product: $(3p - 2q)(3q - 2p) = 13pq - 6(p^2 + q^2)$

Now, $p^2 + q^2 = (p + q)^2 - 2pq = \frac{25}{9} + \frac{4}{3} = \frac{37}{9}$

$$\text{Product} = 13\left(-\frac{2}{3}\right) - 6\left(\frac{37}{9}\right) = -\frac{100}{3}$$

Required equation:

$$x^2 - \frac{5}{3}x - \frac{100}{3} = 0$$

Multiplying by 3:

$$3x^2 - 5x - 100 = 0$$

27. Correct Answer: B

Given: α, β are roots of

$$ax^2 + bx + c = 0$$
$$\alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$$

Roots required: $\alpha + \beta$ and $\alpha\beta$

Sum: $-\frac{b}{a} + \frac{c}{a} = \frac{c-b}{a}$

Product: $-\frac{bc}{a^2}$

Required equation: $x^2 - \frac{c-b}{a}x - \frac{bc}{a^2} = 0$

Multiplying by a^2: $a^2x^2 + a(b - c)x - bc = 0$

INTEGER TYPE

28. Correct Answer:

$(a^2-1)x^2+(a-1)x+(a^2-4a+3)=0$

This is an **identity in** x, so coefficients of x^2, x, and constant term must be **zero**.

$a^2-1=0 \Rightarrow a=\pm 1$

$a-1=0 \Rightarrow a=1$

$a^2-4a+3=0 \Rightarrow (a-1)(a-3)=0$

Common value:

$$\boxed{a=1}$$

29. Correct Answer:

Given equation: $x^2+x-20=0$

Let roots be α, β

$\alpha+\beta=-1, \alpha\beta=-20$

$\alpha^2-\beta^2=(\alpha-\beta)(\alpha+\beta)$

$\alpha-\beta=\sqrt{(\alpha+\beta)^2-4\alpha\beta}=\sqrt{1+80}=9$

$|\alpha^2-\beta^2|=|9\times(-1)|=\boxed{9}$

30. Correct Answer:

Sum $=27$, Product $=182$

Let numbers be x, y

$$x+y=27, xy=182$$
$$x, y=13,14$$

Difference:$\boxed{1}$

31. Correct Answer:

Let consecutive integers be $n, n+1$

$n^2+(n+1)^2=613$

$2n^2+2n-612=0 \Rightarrow n^2+n-306=0$

$n=17$

Difference:$\boxed{1}$

32. Correct Answer:

$$2^{2x^2-7x+5}=1$$

Since $2^k=1 \Rightarrow k=0$

$$2x^2-7x+5=0$$
$$D=49-40=9>0$$

So two real roots$\boxed{2}$

33. Correct Answer:

Given equation:

$5x^2-4x+38+k(-4x^2-2x-1)=0$

Combine terms:

$(5-4k)x^2+(-4-2k)x+(38-k)=0$

Product of roots:

$$\frac{38-k}{5-4k}=-5$$

$38-k=-25+20k \Rightarrow 63=21k \Rightarrow k$

$=\boxed{3}$

PREVIOUS YEAR QUESTIONS

1. Correct Answer: C — Real and distinct

Given quadratic equation:

$$13x^2+5x-2=0$$

Here,

$$a=13, b=5, c=-2$$

Step 1: Find the discriminant

$D=b^2-4ac$

$D=(5)^2-4(13)(-2)$

$D=25+104$

$D=129$

Step 2: Analyse the discriminant

- $D>0 \Rightarrow$ roots are real and distinct
- $D=129$ is not a perfect square, so roots are irrational

2. Correct Answer: D — Both Statement – I and Statement – II are false

Statement – I:

Given: 2 is a root of $x^2+kx+12=0$

Substitute $x=2$: $(2)^2+2k+12=0$

$4+2k+12=0$

$2k+16=0$

$k=-8$

Now consider the equation:

$x^2+kx+q=0$

$x^2-8x+q=0$

For equal roots, discriminant $D=0$:

$(-8)^2-4(1)(q)=0$

$64-4q=0$

$q=16$

Given value of $q$is 15, which is **incorrect**.

Statement – I is false

Statement – II:

Given equation:$x^2-2\sqrt{2}\,x+1=0$

Here,$a=1$, $b=-2\sqrt{2}$, $c=1$

Discriminant:$D=b^2-4ac$

$D=(-2\sqrt{2})^2-4(1)(1)$

$D=8-4$

$D=4>0$

Roots are **real and distinct**, not equal.

Statement – II is false

3. Correct Answer: Option A – $1,3$

$$\frac{15}{x+2}+\frac{x-1}{x-2}=5$$

Step 1: Take LCM of denominators

LCM $=(x+2)(x-2)$

$$\frac{15(x-2)+(x-1)(x+2)}{(x+2)(x-2)}=5$$

Step 2: Simplify the numerator

$15(x-2)=15x-30$
$(x-1)(x+2)=x^2+x-2$
$15x-30+x^2+x-2=x^2+16x-32$

So, $\frac{x^2+16x-32}{(x+2)(x-2)}=5$

Step 3: Cross multiply

$$x^2+16x-32=5(x^2-4)$$
$$x^2+16x-32=5x^2-20$$

Step 4: Bring all terms to one side

$$4x^2-16x+12=0$$

Divide by 4:

$$x^2-4x+3=0$$

Step 5: Factorise

$$(x-1)(x-3)=0$$
$$x=1,\ 3$$

(Both values are allowed since $x\neq 2,-2$)

4. Correct Answer: Option A – $\frac{1}{4}$

For equal roots, the **discriminant** $D=0$.

Given equation:

$$x^2+(2\alpha-1)x+\alpha^2=0$$

Here,

$$a=1, b=2\alpha-1, c=\alpha^2$$

Step 1: Write the discriminant

$$D=b^2-4ac$$
$$D=(2\alpha-1)^2-4(1)(\alpha^2)$$
$$D=4\alpha^2-4\alpha+1-4\alpha^2$$
$$D=1-4\alpha$$

Set $D=0$

$$1-4\alpha=0$$
$$\alpha=\frac{1}{4}$$

5. Correct Answer: Option B – $-1,3$

Let the correct quadratic equation be:

$$ax^2+bx+c=0$$

From A's solution:

A made a mistake only in the **constant term**, so coefficients of $x^2$and $x$are correct.

Roots obtained by A: $5,-3$

Sum of roots $=5+(-3)=2=-\frac{b}{a}$

$\Rightarrow \frac{b}{a}=-2$ (Equation 1)

From B's solution:

B made a mistake only in the **coefficient of** x, so constant term is correct.

Roots obtained by B: $1,-3$

Product of roots $=1\times(-3)=-3$

$=\frac{c}{a}$ (Equation 2)

6. Correct equation:

Using Equations (1) and (2):

$$\frac{b}{a}=-2, \frac{c}{a}=-3$$

So the correct quadratic equation is:

$$x^2-2x-3=0$$

Find the correct roots:

$$x^2-2x-3=0$$
$$(x-3)(x+1)=0$$
$$x=3,\ -1$$

7. Correct Answer: Option B – $\frac{16}{3}$

Let the roots of the equation be $\alpha$and β.

Given:

$$\alpha=3\beta$$

Step 1: Use sum of roots

$$\alpha+\beta=-\frac{b}{a}$$
$$3\beta+\beta=4\beta=-\frac{b}{a}$$
$$\Rightarrow \beta=-\frac{b}{4a}$$

Step 2: Use product of roots

$$\alpha\beta=\frac{c}{a}$$
$$(3\beta)(\beta)=3\beta^2=\frac{c}{a}$$

Substitute $\beta=-\frac{b}{4a}$:

$$3\left(\frac{b^2}{16a^2}\right)=\frac{c}{a}$$
$$\frac{3b^2}{16a^2}=\frac{c}{a}$$

Step 3: Find $\frac{b^2}{ac}$

$$\Rightarrow ac=\frac{16a^2}{3}\cdot\frac{c}{a}$$
$$\frac{b^2}{ac}=\frac{16}{3}$$

8. Correct Answer:

For a quadratic equation to have **equal roots**, the **discriminant** $D = 0$.

Given equation:

$$(\alpha - 4)x^2 + 2(\alpha - 4)x + 4 = 0$$

Here,

$$a = \alpha - 4, b = 2(\alpha - 4), c = 4$$

Step 1: Write the discriminant

$$D = b^2 - 4ac$$
$$D = [2(\alpha - 4)]^2 - 4(\alpha - 4)(4)$$
$$D = 4(\alpha - 4)^2 - 16(\alpha - 4)$$

Step 2: Factorise

$$D = 4(\alpha - 4)[(\alpha - 4) - 4]$$
$$D = 4(\alpha - 4)(\alpha - 8)$$

Step 3: Set $D = 0$

$$4(\alpha - 4)(\alpha - 8) = 0$$
$$\alpha - 4 = 0 \text{or} \alpha - 8 = 0$$
$$\alpha = 4 \text{or} \alpha = 8$$

Important Note:

For a **quadratic equation**, the coefficient of $x^2 \neq 0$.

If $\alpha = 4$, then $(\alpha - 4) = 0$
The equation is **not quadratic**.

So, reject $\alpha = 4$.

Option A $-\ \alpha = 8$

9. Correct Answer:

For a quadratic equation to have **equal roots**, the **discriminant** $D = 0$.

Given equation:

$$(k + 4)x^2 + (k + 1)x + 1 = 0$$

Here,

$$a = k + 4, b = k + 1, c = 1$$

Step 1: Write the discriminant

$$D = b^2 - 4ac$$
$$D = (k + 1)^2 - 4(k + 4)(1)$$
$$D = k^2 + 2k + 1 - 4k - 16$$
$$D = k^2 - 2k - 15$$

Set $D = 0$

$$k^2 - 2k - 15 = 0$$

Factorise:

$$(k - 5)(k + 3) = 0$$
$$k = 5 \text{ or } k = -3$$

Option D – $5, -3$

10. Correct Answer:

Statement I:
For $x^2 - x + 1 = 0$
Discriminant, $D = b^2 - 4ac = (-1)^2 - 4(1)(1) = 1 - 4 = -3$

Since $D < 0$, the roots are **not real**.
Statement I is **false**.

Statement II:
For $x^2 - 5x + 5 = 0$
Discriminant, $D = (-5)^2 - 4(1)(5) = 25 - 20 = 5$

Since $D > 0$, the roots are **real and distinct**.
Statement II is **true**.

11. Correct Answer:
D. Statement I is false but Statement II is true.

10.

$$\frac{1}{p + q + x} = \frac{pq + qx + xp}{pqx}$$

Cross-multiply:

$$pqx = (p + q + x)(pq + qx + xp)$$

Simplify and cancel common terms to get:

$x^2 + (p + q)x + pq = 0$
Factor: $(x + p)(x + q) = 0$
So, $x = -p$ or $x = -q$
Correct Answer: D. $-p, -q$

12.

6. Progressions

Solutions

Worksheet – 1

1. Correct Answer: B) 17

How many terms are there in the A.P. 20, 25, 30, ..., 100?

First term $a = 20$
Common difference $d = 25 - 20 = 5$
Last term $l = 100$

$$l = a + (n-1)d$$
$$100 = 20 + (n-1)5$$
$$80 = 5(n-1)$$
$$n - 1 = 16 \Rightarrow n = 17$$

2. Correct Answer: A) −80

The 11th term of an A.P. is 80 and the 16th term is 40. Find the 31st term.

$$a_{11} = a + 10d = 80$$
$$a_{16} = a + 15d = 40$$

Subtract: $5d = -40 \Rightarrow d = -8$
Substitute in first equation:

$$a + 10(-8) = 80 \Rightarrow a = 160$$

Now, $a_{31} = a + 30d = 160 + 30(-8) = 160 - 240 = -80$

3. Correct Answer: C) −1

3.Given:
$a_1 = a_2 = 2$ and $a_n = a_{n-1} - 1 (n > 2)$

Now find terms step by step:

$$a_3 = a_2 - 1 = 2 - 1 = 1$$
$$a_4 = a_3 - 1 = 1 - 1 = 0$$
$$a_5 = a_4 - 1 = 0 - 1 = -1$$

Therefore, $a_5 = -1$.

4. Correct Answer: D) 0

In an A.P., 5 times the 5th term equals 8 times the 8th term. Find the 13th term.

$$5(a + 4d) = 8(a + 7d)$$
$$5a + 20d = 8a + 56d$$
$$3a + 36d = 0 \Rightarrow a + 12d = 0$$

Now, $a_{13} = a + 12d = 0$

5. Correct Answer: C) 89

If the 7th and 13th terms of an A.P. are 34 and 64 respectively, find the 18th term.

$a_7 = a + 6d = 34$
$a_{13} = a + 12d = 64$
Subtract: $6d = 30 \Rightarrow d = 5$
Substitute: $a + 30 = 34 \Rightarrow a = 4$

$$a_{18} = a + 17d = 4 + 85 = 89$$

6. Correct Answer: C) −4n − 13

If −17, −21, −25 are in A.P., find the nth term.

First term $a = -17$
Common difference $d = -21 + 17 = -4$

$$a_n = a + (n-1)d$$
$$a_n = -17 + (n-1)(-4)$$
$$a_n = -17 - 4n + 4$$
$$a_n = -4n - 13$$

7. Correct Answer: D) p + q − r

p^{th} term of an A.P. = q
q^{th} term of an A.P. = p
Let first term = a, common difference = d
p^{th} term: $a + (\boldsymbol{p-1})\boldsymbol{d} = \boldsymbol{q}$ $\quad (\mathbf{1})$
q^{th} term: $\boldsymbol{a} + (\boldsymbol{q-1})\boldsymbol{d} = \boldsymbol{p}$ $\quad (\mathbf{2})$
Subtract (2) from (1): $(\boldsymbol{p-q})\boldsymbol{d} = \boldsymbol{q-p} \Rightarrow \boldsymbol{d = -1}$
Substitute in (1): $\boldsymbol{a} - (\boldsymbol{p-1}) = \boldsymbol{q} \Rightarrow \boldsymbol{a = p + q - 1}$
r^{th} term: $\boldsymbol{a} + (\boldsymbol{r-1})\boldsymbol{d} = (\boldsymbol{p+q-1}) - (\boldsymbol{r-1}) = \boldsymbol{p+q-r}$

8. Correct Answer: B $\boxed{\frac{4}{ac} - \frac{3}{b^2}}$

$\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P.
Step 1: Use A.P. condition
For three numbers in A.P.,
$\mathbf{2} \times$ middle term = sum of extremes

$$\mathbf{2}\left(\frac{\mathbf{1}}{\boldsymbol{b}}\right) = \frac{\mathbf{1}}{\boldsymbol{a}} + \frac{\mathbf{1}}{\boldsymbol{c}}$$
$$\frac{\mathbf{1}}{\boldsymbol{a}} + \frac{\mathbf{1}}{\boldsymbol{c}} = \frac{\mathbf{2}}{\boldsymbol{b}}$$

Step 2: Evaluate the given expression

$$\left(\frac{\mathbf{1}}{\boldsymbol{a}} + \frac{\mathbf{1}}{\boldsymbol{b}} - \frac{\mathbf{1}}{\boldsymbol{c}}\right)\left(\frac{\mathbf{1}}{\boldsymbol{b}} + \frac{\mathbf{1}}{\boldsymbol{c}} - \frac{\mathbf{1}}{\boldsymbol{a}}\right)$$

Rewrite each bracket using $\frac{\mathbf{1}}{\boldsymbol{a}} + \frac{\mathbf{1}}{\boldsymbol{c}} = \frac{\mathbf{2}}{\boldsymbol{b}}$:

First bracket

$$\left(\frac{1}{a}-\frac{1}{c}\right)+\frac{1}{b}$$

Second bracket

$$\left(\frac{1}{c}-\frac{1}{a}\right)+\frac{1}{b}$$

Step 3: Apply identity $(x+y)(-x+y)=y^2-x^2$

Let $x=\left(\frac{1}{a}-\frac{1}{c}\right), y=\frac{1}{b}$

$=\left(\frac{1}{b}\right)^2-\left(\frac{1}{a}-\frac{1}{c}\right)^2$

Step 4: Simplify

$=\frac{1}{b^2}-\frac{(c-a)^2}{a^2c^2}$

Using A.P. condition, this simplifies to:

$$=\frac{4}{ac}-\frac{3}{b^2}$$

9. Correct Answer: D) 15

Two-digit numbers divisible by 6:

Smallest: 12

Largest: 96

These form an A.P. with:

$$a=12,\ d=6,\ l=96$$

Number of terms:

$$n=\frac{l-a}{d}+1=\frac{96-12}{6}+1=14+1=15$$

10. Correct Answer: C) 0

Let a, b, c, d, e be in A.P.

Write as:

$$a-2d,\ a-d,\ a,\ a+d,\ a+2d$$

Now compute:$(a-2d)-4(a-d)+6a-4(a+d)+(a+2d)$

Simplify:$a-2d-4a+4d+6a-4a-4d+a+2d$

Combine terms:$(1-4+6-4+1)a+(-2+4-4+2)d=0$

11. Correct Answer:

Sum of the A.P. 0.6, 1.7, 2.8, … up to 100 terms

Given:

First term,$a=0.6$

Common difference,$d=1.7-0.6=1.1$

Number of terms,$n=100$

Formula:$S_n=\frac{n}{2}[2a+(n-1)d]$

Substitute values:

$S_{100}=\frac{100}{2}[2(0.6)+99(1.1)]$

$=50[1.2+108.9]$

$=50\times110.1$

$=5505$

$\boxed{S_{100}=5505}$

12. Correct Answer:

Sum of first 51 terms of an A.P. whose second and third terms are 14 and 18

Given:

Second term $=a+d=14$

Third term $=a+2d=18$

Subtract equations:$(a+2d)-(a+d)=18-14$

$$d=4$$

Find first term:$a+d=14\Rightarrow a=10$

Number of terms,$n=51$

Formula:$S_n=\frac{n}{2}[2a+(n-1)d]$

Substitute values:

$S_{51}=\frac{51}{2}[2(10)+50(4)]$

$=\frac{51}{2}(20+200)$

$=\frac{51}{2}\times220$

$=51\times110$

$=5610$

$\boxed{S_{51}=5610}$

13. Correct Answer: (B)

If the arithmetic mean of two numbers is 10 and one number is 6, find the other number

Formula:Arithmetic Mean $=\frac{a+b}{2}$

Given:

$10=\frac{6+x}{2}$

$6+x=20$

$x=14$

Correct answer: $\boxed{14}$

14. Correct Answer: (C) $\boxed{60}$

If the arithmetic mean of 5 numbers is 12, find their sum

Formula:Mean $=\frac{\text{Sum}}{\text{Number of terms}}$

Given:

$12=\frac{\text{Sum}}{5}$

Sum $=12\times5=60$

WORKSHEET-2

1. Correct Answer: B) ±3

First term $a = 1$

Third term $= ar^2 = r^2$
Fifth term $= ar^4 = r^4$

Given: $r^2 + r^4 = 90$

$r^2(1 + r^2) = 90$
Try options:

- If $r = \pm 3$:
  $r^2 = 9$, $r^4 = 81 \Rightarrow 9 + 81 = 90$ ✅

2. Correct Answer: C) $q^2 = pr$

4th term $= ar^3 = p$

7th term $= ar^6 = q$

10th term $= ar^9 = r$

Now: $q^2 = (ar^6)^2 = a^2r^{12}$
$pr = (ar^3)(ar^9) = a^2r^{12}$
So,

$$q^2 = pr$$

3. Correct Answer: D) $\boxed{n = 6}$

Given:

In a G.P.,
First term $a = 3$
Last term $a_n = 96$
Sum of $n$terms $S_n = 189$

Step 1: Use the formula for sum of a G.P.

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

Substitute the given values:

$$189 = \frac{3(r^n - 1)}{r - 1}$$

Divide both sides by 3:

$$63 = \frac{r^n - 1}{r - 1}$$

$63(r - 1) = r^n - 1$(Equation 1)
Step 2: Use the formula for the nth term of a G.P.

$a_n = ar^{n-1}$
Substitute the values:

$96 = 3r^{n-1}$
$r^{n-1} = 32$
Since $32 = 2^5$,

$r = 2$ and $n - 1 = 5$
$n = 6$

Step 3: Verify using Equation (1)

$$63(2 - 1) = 2^6 - 1$$
$$63 = 64 - 1 = 63 \checkmark$$

4. Correct Answer: CORRECT -B $\boxed{x = \frac{7}{15}}$

First term $a = 3$, common ratio $r = x$

Sum of infinite GP:

$$S = \frac{a}{1 - r}$$
$$\frac{3}{1 - x} = \frac{45}{8}$$
$$24 = 45(1 - x)$$
$$24 = 45 - 45x$$
$$45x = 21$$
$$x = \frac{7}{15}$$

5. Correct Answer: D $\boxed{\frac{1}{9}(10^{n+1} - 9n - 10)}$

Sum of the series $9 + 99 + 999 + \cdots$up to $n$terms

Each term can be written as: $9 = 10 - 1$, $99 = 10^2 - 1$, $999 = 10^3 - 1$
So the series is: $(10^1 - 1) + (10^2 - 1) + \cdots + (10^n - 1)$

$$= (10 + 10^2 + \cdots + 10^n) - n$$

Sum of GP: $10 + 10^2 + \cdots + 10^n = \frac{10(10^n - 1)}{9}$

So, $S_n = \frac{10(10^n - 1)}{9} - n = \frac{10^{n+1} - 10 - 9n}{9}$

6. Correct Answer: x (Option C)

Find the value of $x^{\frac{1}{2}} \cdot x^{\frac{1}{4}} \cdot x^{\frac{1}{8}} \cdot x^{\frac{1}{16}} \cdots \infty$
Add the exponents:

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$$

This is an infinite GP with
First term $a = \frac{1}{2}$, common ratio $r = \frac{1}{2}$

Sum: $S = \frac{a}{1 - r} = \frac{\frac{1}{2}}{1 - \frac{1}{2}} = 1$

So,

$$x^1 = x$$

7. Correct Answer: B(-12)

The geometric mean between $-9$and -16

Geometric mean $= \sqrt{ab}$

$\sqrt{(-9)(-16)} = \sqrt{144} = 12$
Since both numbers are negative, GM is negative:

8. **Correct Answer: Option A** $\boxed{\frac{3}{2},\ 1,\ \frac{2}{3}}$

Insert 3 geometric means between $2\frac{1}{4}$ and $\frac{4}{9}$

Convert to improper fractions: $2\frac{1}{4} = \frac{9}{4}, \frac{4}{9}$

Let the GP be: $\frac{9}{4},\ G_1,\ G_2,\ G_3,\ \frac{4}{9}$

Common ratio: $r = \left(\frac{\frac{4}{9}}{\frac{9}{4}}\right)^{\frac{1}{4}} = \left(\frac{16}{81}\right)^{\frac{1}{4}} = \frac{2}{3}$

Now multiply successively:

$G_1 = \frac{9}{4} \times \frac{2}{3} = \frac{3}{2}$

$G_2 = \frac{3}{2} \times \frac{2}{3} = 1$

$G_3 = 1 \times \frac{2}{3} = \frac{2}{3}$

9. **Correct Answer: Option D** $\boxed{2 \times 3^{n-1}}$

If the 4th and 9th terms of a GP are 54 and 13122, find the nth term

$$a_4 = ar^3 = 54$$
$$a_9 = ar^8 = 13122$$

Divide: $\frac{a_9}{a_4} = r^5 = \frac{13122}{54} = 243 = 3^5$

So, $r = 3$

Now, $a = \frac{54}{3^3} = 2$

Nth term: $a_n = ar^{n-1} = 2 \times 3^{n-1}$

10. **Correct Answer: Option B 64**

If $2, a, 2^3, b, 2^5$ are in GP, find ab

Terms: $2,\ a,\ 8,\ b,\ 32$

Common ratio: $r = \frac{8}{2} = 4^{1/2} = 2$

So,

$a = 2 \times 2 = 4$

$b = 8 \times 2 = 16$

$ab = 4 \times 16 = 64$

Competitive Corner

SINGLE CORRECT ANSWER TYPE

1. **Correct Answer: Option A 34**

If 7, 11, 15, ..., 139 are in A.P., then 139 is the ____ term

Given:

First term $a = 7$

Common difference $d = 11 - 7 = 4$

Formula for nth term of an A.P.:

$a_n = a + (n-1)d$

$139 = 7 + (n-1)4$

$139 = 7 + 4n - 4$

$139 = 4n + 3$

$4n = 136$

$n = 34$

2. **Correct Answer: Option A** $\boxed{2n - 3p - 8}$

If $2n - 5, 2n - 8, 2n - 11$ are in A.P., then $(p+2)^{th}$ term is

Given terms: $a = 2n - 5$

$d = (2n - 8) - (2n - 5) = -3$

Nth term formula: $a_k = a + (k-1)d$

For $(p+2)^{th}$ term:

$a_{p+2} = (2n - 5) + (p + 1)(-3)$

$= 2n - 5 - 3p - 3$

$= 2n - 3p - 8$

3. **Correct Answer: Option C** $\boxed{191}$

The 10th common term between the series

$3 + 7 + 11 + \cdots$ and $1 + 6 + 11 + \cdots$ is**

First A.P.: $a_1 = 3,\ d_1 = 4$

Second A.P.: $a_2 = 1,\ d_2 = 5$

Common difference of common terms: LCM$(4, 5) = 20$

First common term: 11

10th common term: $a = 11 + (10 - 1) \times 20$

$= 11 + 180$

$= 191$

4. **Correct Answer: Option C 90**

If $a_1, a_2, a_3, \ldots$ are in A.P. such that

$a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{24} = 225$, then find $a_1 + a_2 + \cdots + a_{24}$

Write each term:

$a_1 = a$

$a_5 = a + 4d$

$a_{10} = a + 9d$

$a_{15} = a + 14d$

$a_{20} = a + 19d$

$a_{24} = a + 23d$

Adding: $6a + (4 + 9 + 14 + 19 + 23)d = 225$

$$6a + 69d = 225$$

Sum of first 24 terms:

$S_{24} = \frac{24}{2}[2a + 23d]$

$= 12(2a + 23d)$

$= 24a + 276d$

Multiply earlier equation by 4: $24a + 276d = 900$

5. Correct Answer: A 1

If p, q, r are in A.P. and x, y, z are in G.P., then $x^{q-r} \cdot y^{r-p} \cdot z^{p-q}$ is equal to

Since p, q, r are in A.P.: $q - r = r - p$
Since x, y, z are in G.P.: $y^2 = xz$
Take logarithms or combine powers: $x^{q-r} y^{r-p} z^{p-q} = 1$

6. Correct Answer: Option A

Insert four G.M.'s between 2 and 486
Given:
First term $a = 2$, last term $b = 486$
Number of G.M.'s to be inserted = 4
So total terms $= 6$
Formula for common ratio:

$$r = \left(\frac{b}{a}\right)^{\frac{1}{5}}$$

$$r = \left(\frac{486}{2}\right)^{\frac{1}{5}} = (243)^{\frac{1}{5}} = 3$$

The G.P. is: $2, 6, 18, 54, 162, 486$
Inserted four G.M.'s: $\boxed{6, 18, 54, 162}$

7. Correct Answer: Option B -3

The sum of infinite G.P. is 20 and the sum of their squares is 100. Find five times the common ratio

Let first term = a, common ratio = r

Sum to infinity: $\frac{a}{1-r} = 20 \quad (1)$

Sum of squares to infinity: $\frac{a^2}{1-r^2} = 100 \quad (2)$

From (1): $a = 20(1 - r)$
Substitute in (2):

$$\frac{400(1-r)^2}{(1-r)(1+r)} = 100$$
$$\frac{400(1-r)}{1+r} = 100$$
$$4(1-r) = 1 + r$$
$$4 - 4r = 1 + r$$
$$5r = 3 \Rightarrow r = \frac{3}{5}$$

Five times the common ratio: $5r = 3$
Answer: Option B -3

8. Correct Answer: Option D -24.2

Sum of $16.2, 5.4, 1.8, \ldots$ to 5 terms

Given:

$$a = 16.2$$

$$r = \frac{5.4}{16.2} = \frac{1}{3}$$

Formula for sum of GP:

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

$$S_5 = \frac{16.2\left(1 - \left(\frac{1}{3}\right)^5\right)}{1 - \frac{1}{3}}$$

$$S_5 = \frac{16.2\left(1 - \frac{1}{243}\right)}{\frac{2}{3}} \approx 24.2$$

9. Correct Answer: Option D $\boxed{a}$

If $a > 0$, then $\sum_{n=1}^{\infty} \left(\frac{a}{a+1}\right)^n$ equals

This is an infinite G.P.

$$a = \frac{a}{a+1}, r = \frac{a}{a+1}$$

Sum to infinity:

$$S = \frac{a}{1 - r}$$

$$S = \frac{\frac{a}{a+1}}{1 - \frac{a}{a+1}} = \frac{\frac{a}{a+1}}{\frac{1}{a+1}} = a$$

10. Correct Answer: Option A $\boxed{\frac{n(n+1)(n+2)}{6}}$

Find the value of

$$\sum_{i=1}^{n} \sum_{j=1}^{i} \sum_{k=1}^{j} 1$$

Evaluate stepwise:

$$\sum_{k=1}^{j} 1 = j$$

$$\sum_{j=1}^{i} j = \frac{i(i+1)}{2}$$

$$\sum_{i=1}^{n} \frac{i(i+1)}{2} = \frac{1}{2} \sum_{i=1}^{n} (i^2 + i)$$

$$= \frac{1}{2}\left[\frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}\right]$$

$$= \frac{n(n+1)(n+2)}{6}$$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct Answer: B, D

If the sum of the first three consecutive terms of an A.P. is 9 and the sum of their squares is 35, then sum to $n$terms of the series is

Let the three consecutive terms be:

$$a-d,\ a,\ a+d$$

Sum given:$(a-d)+a+(a+d)=3a=9\Rightarrow a=3$

Sum of squares:

$$(3-d)^2+3^2+(3+d)^2=35$$
$$(9-6d+d^2)+9+(9+6d+d^2)=35$$
$$27+2d^2=35\Rightarrow d^2=4\Rightarrow d=\pm 2$$

So the A.P. is either:

$$1,3,5,\ldots \text{ or } 5,3,1,\ldots$$

Case 1: $a=1,\ d=2$

$$S_n=\frac{n}{2}[2a+(n-1)d]=\frac{n}{2}[2+2n-2]=n^2$$

Case 2: $a=5,\ d=-2$

$$S_n=\frac{n}{2}[10-2n+2]=n(6-n)$$

12. Correct Answer: $A,\ C$

The sum of first two terms of a G.P. is $\frac{5}{3}$and the sum to infinity is 3. Find the first term.

Let first term $=a$, common ratio $=r$.

$$\frac{a}{1-r}=3\Rightarrow a=3(1-r)$$

Sum of first two terms:$a(1+r)=\frac{5}{3}$

Substitute a:$3(1-r)(1+r)=\frac{5}{3}$

$$3(1-r^2)=\frac{5}{3}\Rightarrow 1-r^2=\frac{5}{9}\Rightarrow r^2=\frac{4}{9}\Rightarrow r=\pm\frac{2}{3}$$

Case 1: $r=\frac{2}{3}$

$$a=3\left(1-\frac{2}{3}\right)=1$$

Case 2: $r=-\frac{2}{3}$

$$a=3\left(1+\frac{2}{3}\right)=5$$

13. Correct Answer: (A, B, C, and D).

If $3k-7$is the arithmetic mean of $3k+4$and $k+12$, find the value of $2k+5$.

Solution

Arithmetic Mean (A.M) of two numbers $a$and $b$is

$$\text{A.M}=\frac{a+b}{2}$$

Given:

$$3k-7=\frac{(3k+4)+(k+12)}{2}$$
$$3k-7=\frac{4k+16}{2}$$
$$3k-7=2k+8$$
$$3k-2k=8+7$$
$$k=15$$

Now find $2k+5$:

$$2k+5=2(15)+5=30+5=35$$

Checking the options

A) $9\times 5-10=45-10=35$ ✔

B) $7\times 4+7=28+7=35$ ✔

C) $8\times 4+3=32+3=35$ ✔

D) $2\times 17+1=34+1=35$ ✔

14. Correct Answer: $B,\ C$

How many terms of the sequence $54, 51, 48,$ ...have sum 513?

Here:

$$a=54,\ d=-3$$
$$S_n=\frac{n}{2}[2a+(n-1)d]$$
$$513=\frac{n}{2}[108-3n+3]=\frac{n}{2}(111-3n)$$
$$1026=n(111-3n)$$
$$3n^2-111n+1026=0$$
$$\Rightarrow n^2-37n+342=0$$
$$(n-18)(n-19)=0\Rightarrow n=18,\ 19$$

15. Correct Answer: $A,\ D$

The A.M. and G.M. of two terms $a, b$are 20 and 16 respectively. Find a, b.

$$\frac{a+b}{2}=20\Rightarrow a+b=40$$
$$\sqrt{ab}=16\Rightarrow ab=256$$

Solve:$x^2-40x+256=0\Rightarrow (x-32)(x-8)=0$

So $a, b=32, 8$or $8, 32$

ASSERTION AND REASON TYPE

16. **Correct Answer: (A)**
Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
- Assertion (A): True — In an AP, the difference between any two consecutive terms is constant.
- Reason (R): True — The general term of an AP is $a_n = a + (n-1)d$.
- Link between A and R: From $a_n = a + (n-1)d$, $a_{n+1} - a_n = [a + nd] - [a + (n-1)d] = d$, which explains why the difference is constant.

17. **Correct Answer: (A)**
Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
- Assertion (A): True — Given $a_{10} = 35$and $d = 3$:
- $a_{10} = a + 9d \Rightarrow 35 = a + 27 \Rightarrow a = 8$
Reason (R): True — The nth term of an AP is $a_n = a + (n-1)d$.
- Link between A and R: The assertion is directly proved using the formula given in the reason.

18. **Correct Answer: (A)**
Assertion (A): The sum of the first $n$terms of an AP is
$$S_n = \frac{n}{2}[2a + (n-1)d]$$
✓ True
Reason (R): The average of the first term and the last term, multiplied by n, also gives the sum of the first $n$terms.
✓ True
Explanation:
In an AP,
$$S_n = \frac{n}{2}(a + l)$$
where $l = a + (n-1)d$.
Substituting $l$gives the assertion formula.
Hence, the reason correctly explains the assertion.

19. **Correct Answer: (A)**
Assertion (A): If the common difference of an AP is zero, then all terms of the AP are equal.
✓ True
Reason (R): A constant sequence is also considered an AP.
✓ True
Explanation:
If common difference $d = 0$, then
$$a, a, a, \ldots$$
All terms are equal, forming a constant sequence, which is a special case of an AP.
Thus, the reason correctly explains the assertion.

20. **Correct Answer: (D)**
The sequence $2, 4, 8, 16, \ldots$is an AP because the terms increase regularly.
✗ False
(Regular increase does not mean constant difference.)
Reason (R):
For a sequence to be an AP, the difference between consecutive terms must be constant.
✓ True
Explanation:
In the given sequence:
$$4 - 2 = 2,\ 8 - 4 = 4,\ 16 - 8 = 8$$
The differences are not equal, so the sequence is not an AP.
Thus, the assertion is false, but the reason correctly states the condition for an AP.

MATRIX MATCHING TYPE

21. **Correct Answer:**
a) The geometric mean between 25 and 16
$$\sqrt{25 \times 16} = \sqrt{400} = 20$$
$\Rightarrow$ (Q)
b) If $x, 2x + 2, 3x + 3$are the first three terms of a G.P.
$$\frac{2x+2}{x} = \frac{3x+3}{2x+2} \Rightarrow x = -4$$
Fourth term$= -9 \times \frac{3}{2} = -\frac{27}{2}$
$\Rightarrow$ (P)

c) The sum of the series

$$a-(a+d)+(a+2d)-(a+3d)+\cdots \text{ up to } (2n+1) \text{ terms}$$

$$S=a+nd$$

⇒ (S)

d) If the sum of the series $2+5+8+11+\cdots$(n terms) is 60100

$$\frac{n}{2}[2(2)+(n-1)3]=60100 \Rightarrow n=200$$

⇒ (R)

22. Correct Answer:

a) If $\frac{1}{3}, x_1, x_2, 9$ are in G.P.

Let the common ratio be r.

$$\frac{1}{3}r^3=9 \Rightarrow r^3=27 \Rightarrow r=3$$

$$x_2=\frac{1}{3}\times r^2=\frac{1}{3}\times 9=3$$

⇒ (R)

b) Step 1: Define the G.P. terms

Let the five terms of the G.P. be

$$a,\ ar,\ ar^2,\ ar^3,\ ar^4.$$

The third term is ar^2.

Given: $ar^2=4$

Step 2: Calculate the product of the five terms

$$P=a\times ar\times ar^2\times ar^3\times ar^4$$

$$P=a^5r^{(0+1+2+3+4)}$$

$$P=a^5r^{10}$$

Step 3: Express the product in terms of the third term

$$a^5r^{10}=(ar^2)^5$$

Step 4: Substitute the given value

$$P=4^5$$

$$P=1024$$

Answer:

The product of the five terms is 1024, which corresponds to option (P).

c) $\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+\cdots+\infty$

Here $a=\frac{1}{3}$, $r=\frac{1}{3}$

$$S_\infty=\frac{a}{1-r}=\frac{\frac{1}{3}}{1-\frac{1}{3}}=\frac{1}{2}$$

⇒ (S)

d) Sum of $1+3+9+27+\cdots$ is 364

$$S_n=\frac{3^n-1}{3-1}=364 \Rightarrow \frac{3^n-1}{2}=364 \Rightarrow 3^n=729 \Rightarrow n=6$$

⇒ (Q)

COMPREHENSION TYPE

Passage – I:

23. Correct Answer: (B)

Sum to 'n' terms of 5, 2, –1, ...

Step 1: Identify the A.P.

First term $a=5$

Common difference

$$d=2-5=-3$$

Step 2: Use the formula

$$S_n=\frac{n}{2}[2a+(n-1)d]$$

Step 3: Substitute values

$$S_n=\frac{n}{2}[2(5)+(n-1)(-3)]$$

$$S_n=\frac{n}{2}[10-3n+3]$$

$$S_n=\frac{n}{2}(13-3n)$$

24. Correct Answer: (A)

Q24. The value of 'a' in terms of S_n, d, n is

Step 1: Start with the formula

$$S_n=\frac{n}{2}[2a+(n-1)d]$$

Step 2: Multiply both sides by 2

$$2S_n=n[2a+(n-1)d]$$

Step 3: Divide by n

$$\frac{2S_n}{n}=2a+(n-1)d$$

Step 4: Solve for a

$$2a=\frac{2S_n}{n}-(n-1)d$$

$$a=\frac{2S_n-n^2d+nd}{2n}$$

25. Correct Answer: (B)

Q25. If $S_n=400$, $n=16$, $a=5$, then d is

Step 1: Use the formula

$$S_n=\frac{n}{2}[2a+(n-1)d]$$

Step 2: Substitute values

$$400=\frac{16}{2}[2(5)+15d]$$

$$400=8(10+15d)$$

Step 3: Solve

$$400=80+120d$$

$$320=120d$$

$$d=\frac{8}{3}$$

Passage – II:

Let

$A_1, A_2, \ldots, A_m$ be the arithmetic means between -2 and 1027.

$G_1, G_2, \ldots, G_n$ be the geometric means between 1 and 1024.

Given:

Product of geometric means $= 2^{45}$

- Sum of arithmetic means $= 1025 \times 171$

26. Correct Answer: (B) 9

Find the value of n

Between 1 and $1024 = 2^{10}$, let there be n geometric means.

Then total terms $= n + 2$

Common ratio:

$$r^{n+1} = \frac{1024}{1} = 2^{10}$$

Product of all terms of a G.P.:$= (1 \times 1024)^{\frac{n+2}{2}} = 2^{10 \cdot \frac{n+2}{2}} = 2^{5(n+2)}$

Product of geometric means:

$$= \frac{\text{product of all terms}}{1 \times 1024} = 2^{5(n+2)-10} = 2^{5n}$$

Given:$2^{5n} = 2^{45} \Rightarrow 5n = 45 \Rightarrow n = 9$

27. Correct Answer: (B) 342

Q27. Find the value of m

Between -2 and 1027, let there be m arithmetic means.

Total terms $= m + 2$

Sum of all terms:

$$S = \frac{m+2}{2}(-2 + 1027) = \frac{m+2}{2}(1025)$$

Sum of arithmetic means:

$$= S - (-2 + 1027) = \frac{m+2}{2}(1025) - 1025$$

$$= 1025\left(\frac{m}{2}\right)$$

Given:

$$1025\left(\frac{m}{2}\right) = 1025 \times 171$$

$$\Rightarrow \frac{m}{2} = 171 \Rightarrow m = 342$$

28. Correct Answer: (D) 1022

Q28. Find $G_1 + G_2 + \cdots + G_n$

From Q26: $n = 9$

Total terms in G.P. $= 11$

Common ratio:

$$r^{10} = 1024 = 2^{10} \Rightarrow r = 2$$

So the G.P. is:

$$1, 2, 4, 8, \ldots, 1024$$

Sum of all 11 terms:

$$S_{11} = \frac{1(2^{11} - 1)}{2 - 1} = 2047$$

Sum of geometric means:

$$= 2047 - (1 + 1024) = 2047 - 1025 = 1022$$

INTEGER TYPE

29. Correct Answer: 1

Given:$T_n = 128, S_n = 255, r = 2$

$$S_n = a(2^n - 1) = 255 \Rightarrow 2^n - 1 = 255$$

$$\Rightarrow 2^n = 256 \Rightarrow n = 8$$

$$T_n = a \cdot 2^{n-1} = 128 \Rightarrow a \cdot 2^7 = 128 \Rightarrow a = 1$$

30. Correct Answer: 5

Arithmetic mean of $3 + \sqrt{2}$ and $7 - \sqrt{2}$:

$$\frac{(3 + \sqrt{2}) + (7 - \sqrt{2})}{2} = \frac{10}{2} = 5$$

31. Correct Answer: 1

For three terms of an AP: a, b, c

$$b - a = c - b \Rightarrow b - c = a - b \Rightarrow \frac{b - c}{a - b} = 1$$

32. Correct Answer: 2

Given:$S_n = 3n^2 - n$

First term:$a = S_1 = 3(1)^2 - 1 = 2$

33. Correct Answer: 7

4th term of an AP $= a + 3d$

Difference between 4th terms:

$$(-1 + 3d) - (-8 + 3d) = 7$$

PREVIOUS YEAR QUESTIONS

1. Correct Answer: B $\boxed{-3}$

Let the first term of the A.P. be a and common difference be d.

Then,

$$a_1 = a, a_2 = a + d, a_3 = a + 2d, a_4 = a + 3d, a_5 = a + 4d$$

Given:$a_1 + a_3 + a_5 = -12$

$$a + (a + 2d) + (a + 4d) = -12$$

$$3a + 6d = -12$$

$$a + 2d = -4 \quad (1)$$

Also given:$a_1 \cdot a_2 \cdot a_3 = 8$

$$a(a + d)(a + 2d) = 8$$

From (1), $a = -4 - 2d$

Substitute in the product:$(-4 - 2d)(-4 - d)(-4) = 8$

Try the given options for d:

For $d = -3$:

$$a = -4 - 2(-3) = 2$$

$$a(a + d)(a + 2d) = 2(-1)(-4) = 8$$

This satisfies the condition.

2. Correct Answer: C $\boxed{0}$

Let the first term of the A.P. be $\boldsymbol{a}$and common difference be $\boldsymbol{d}$.

The $\boldsymbol{n}^{\text{th}}$term of an A.P. is:$\boldsymbol{T_n = a + (n-1)d}$
So,$\boldsymbol{T_p = a + (p-1)d}$
$\boldsymbol{T_q = a + (q-1)d}$
Given:$\boldsymbol{pT_p = qT_q}$
Substitute:$\boldsymbol{p[a + (p-1)d] = q[a + (q-1)d]}$
Expand:$\boldsymbol{pa + p(p-1)d = qa + q(q-1)d}$
Rearranging:$\boldsymbol{pa - qa = q(q-1)d - p(p-1)d}$

$$\boldsymbol{a(p-q) = d[q(q-1) - p(p-1)]}$$

Factor RHS:

$$\boldsymbol{a(p-q) = d[(q-p)(p+q-1)]}$$
$$\boldsymbol{a(p-q) = -d(p-q)(p+q-1)}$$

Since $\boldsymbol{p \neq q}$, divide both sides by $(\boldsymbol{p-q})$:

$$\boldsymbol{a = -d(p+q-1)}$$

Now, the $\boldsymbol{(p+q)}^{\text{th}}$term:

$$\boldsymbol{T_{p+q} = a + (p+q-1)d}$$

Substitute $\boldsymbol{a}$:

$$\boldsymbol{T_{p+q} = -d(p+q-1) + (p+q-1)d}$$
$$\boldsymbol{T_{p+q} = 0}$$

3. Correct Answer: A-1,7,13

Let the three terms of the A.P. be:

$$a-d,\ a,\ a+d$$

Given:
Sum of three terms = $\boldsymbol{21}$

$$\boldsymbol{(a-d) + a + (a+d) = 21}$$
$$\boldsymbol{3a = 21}$$
$$\boldsymbol{a = 7}$$

So the terms are: $7-d, 7, 7+d$

Given:
The product of the first and the third term exceeds the second term by 6.

$$\boldsymbol{(7-d)(7+d) = 7 + 6}$$
$$\boldsymbol{49 - d^2 = 13}$$
$$\boldsymbol{d^2 = 36}$$
$$\boldsymbol{d = 6}$$

Hence, the three terms are:

$$7-6, 7, 7+6 = 1, 7, 13$$

4. Correct Answer: B $\boxed{109}$

Let the first term of the A.P. be $\boldsymbol{a}$and common difference be $\boldsymbol{d}$.

The $\boldsymbol{n}^{\text{th}}$term of an A.P. is:$\boldsymbol{T_n = a + (n-1)d}$
Given:

$$\boldsymbol{T_5 = a + 4d = 46} \quad (1)$$
$$\boldsymbol{T_{17} = a + 16d = 130} \quad (2)$$

Subtract (1) from (2):

$$\boldsymbol{(a + 16d) - (a + 4d) = 130 - 46}$$
$$\boldsymbol{12d = 84}$$
$$\boldsymbol{d = 7}$$

Substitute $\boldsymbol{d = 7}$in (1):

$$\boldsymbol{a + 4(7) = 46}$$
$$\boldsymbol{a + 28 = 46}$$
$$\boldsymbol{a = 18}$$

Now find the $\boldsymbol{14}^{\text{th}}$term:

$$\boldsymbol{T_{14} = a + 13d}$$
$$\boldsymbol{T_{14} = 18 + 13(7)}$$
$$\boldsymbol{T_{14} = 18 + 91 = 109}$$

5. Correct Answer: B $\boxed{\frac{3ab}{2(a-b)}}$

Given:
First term = $\boldsymbol{a}$
Second term = $\boldsymbol{b}$

So, common difference:$\boldsymbol{d = b - a}$
Last term = $\boldsymbol{2a}$

Using the formula for the last term:

$$\boldsymbol{l = a + (n-1)d}$$
$$\boldsymbol{2a = a + (n-1)(b-a)}$$
$$\boldsymbol{a = (n-1)(b-a)}$$
$$\boldsymbol{n - 1 = \frac{a}{b-a}}$$
$$\boldsymbol{n = \frac{a}{b-a} + 1 = \frac{b}{b-a}}$$

Sum of first $\boldsymbol{n}$terms:$\boldsymbol{S_n = \frac{n}{2}(a + l)}$
Substitute values:$\boldsymbol{S_n = \frac{1}{2} \cdot \frac{b}{b-a}(a + 2a)}$

$$\boldsymbol{S_n = \frac{1}{2} \cdot \frac{b}{b-a} \cdot 3a}$$
$$\boldsymbol{S_n = \frac{3ab}{2(b-a)}}$$

Since $\boldsymbol{(b-a) = -(a-b)}$,

$$\boldsymbol{S_n = \frac{3ab}{2(a-b)}}$$

6. Correct Answer: B

(i)

Let the first term of the A.P. be $\boldsymbol{a}$and common difference be $\boldsymbol{d}$.

The $\boldsymbol{n}^{\text{th}}$term of an A.P. is:$\boldsymbol{T_n = a + (n-1)d}$
Given:$\boldsymbol{T_5 = a + 4d = 19}$ (1)
Difference of the eighth term from the thirteenth term is 20:

$$\boldsymbol{T_{13} - T_8 = 20}$$
$$\boldsymbol{(a + 12d) - (a + 7d) = 20}$$
$$\boldsymbol{5d = 20}$$
$$\boldsymbol{d = 4}$$

Substitute $\boldsymbol{d = 4}$in (1):

$$\boldsymbol{a + 4(4) = 19}$$
$$\boldsymbol{a + 16 = 19}$$
$$\boldsymbol{a = 3}$$

First term = 3

(ii)Two-digit numbers range from 10 to 99.

Numbers which give remainder 3 when divided by 4 are of the form:

$$\boldsymbol{4k + 3}$$

Smallest two-digit number of this form:$\boldsymbol{11 = 4(2) + 3}$
Largest two-digit number of this form:$\boldsymbol{99 = 4(24) + 3}$
So the required numbers form an A.P.:$\boldsymbol{11, 15, 19, \ldots, 99}$
Here,$\boldsymbol{a = 11, d = 4, l = 99}$
Number of terms: $\boldsymbol{n = \frac{l-a}{d} + 1}$

$$\boldsymbol{n = \frac{99 - 11}{4} + 1}$$
$$\boldsymbol{n = \frac{88}{4} + 1}$$
$$\boldsymbol{n = 22 + 1 = 23}$$

Number of such two-digit numbers = 23

7. Correct Answer: B $\boxed{6}$

First term, $\boldsymbol{a = 21}$

Sum of first 6 terms:$\boldsymbol{S_6 = \frac{6}{2}[2a + (6-1)d]}$
$\boldsymbol{S_6 = 3[2(21) + 5d] = 3(42 + 5d)}$
Sum of next 6 terms = $\boldsymbol{S_{12} - S_6}$

First find $\boldsymbol{S_{12}}$:

$$\boldsymbol{S_{12} = \frac{12}{2}[2a + (12-1)d]}$$
$$\boldsymbol{S_{12} = 6[42 + 11d]}$$

So, sum of next 6 terms:$\boldsymbol{S_{12} - S_6 = 6(42 + 11d) - 3(42 + 5d)}$

$$\boldsymbol{= (252 + 66d) - (126 + 15d)}$$
$$\boldsymbol{= 126 + 51d}$$

Given condition:$\boldsymbol{S_6 =}$ $\boldsymbol{\frac{1}{2}}$(sum of next 6 terms)

$$\boldsymbol{3(42 + 5d) = \frac{1}{2}(126 + 51d)}$$

Multiply both sides by 2: $\boldsymbol{6(42 + 5d) = 126 + 51d}$

$$\boldsymbol{252 + 30d = 126 + 51d}$$
$$\boldsymbol{126 = 21d}$$
$$\boldsymbol{d = 6}$$

8. Correct Answer: A $\boxed{15}$

Let the first term of the A.P. be $\boldsymbol{a}$and common difference be $\boldsymbol{d}$.

Then the ten terms are:

$$a,\ a+d,\ a+2d, \ldots,\ a+9d$$

Step 1: Use given sums

Sum of all 10 terms:$\boldsymbol{S_{10} = \frac{10}{2}[2a + 9d] = 5(2a + 9d)}$
Given:
Sum of all terms except the first term = 99

$$\boldsymbol{S_{10} - a = 99}$$
$$\boldsymbol{5(2a + 9d) - a = 99}$$
$$\boldsymbol{10a + 45d - a = 99}$$
$$\boldsymbol{9a + 45d = 99}$$ (1)

Sum of all terms except the sixth term is 89.

Sixth term = $\boldsymbol{a + 5d}$

$$\boldsymbol{S_{10} - (a + 5d) = 89}$$
$$\boldsymbol{5(2a + 9d) - a - 5d = 89}$$
$$\boldsymbol{10a + 45d - a - 5d = 89}$$
$$\boldsymbol{9a + 40d = 89}$$ (2)

Step 2: Solve equations (1) and (2)

Subtract (2) from (1):

$$\boldsymbol{(9a + 45d) - (9a + 40d) = 99 - 89}$$
$$\boldsymbol{5d = 10}$$
$$\boldsymbol{d = 2}$$

Substitute $\boldsymbol{d = 2}$in (2):

$$\boldsymbol{9a + 40(2) = 89}$$
$$\boldsymbol{9a + 80 = 89}$$
$$\boldsymbol{9a = 9}$$
$$\boldsymbol{a = 1}$$

Step 3: Verify given condition
Sum of first and fifth terms = 10
$\boldsymbol{a + (a + 4d) = 1 + (1 + 8) = 10}$✓
Step 4: Find the 8th term

$$\boldsymbol{T_8 = a + 7d = 1 + 14 = 15}$$

9. Correct Answer: D 3

Let the common difference of both A.P.'s be $\boldsymbol{d}$.

First A.P.:
First term $\boldsymbol{a_1 = 12}$

Second A.P.:
First term $\boldsymbol{a_2 = 9}$

The $\boldsymbol{n}^{\text{th}}$term of an A.P. is:$\boldsymbol{T_n = a + (n-1)d}$
30th term of the first A.P.:$\boldsymbol{T_{30}^{(1)} = 12 + 29d}$
30th term of the second A.P.:$\boldsymbol{T_{30}^{(2)} = 9 + 29d}$
Difference between their 30th terms:$\boldsymbol{T_{30}^{(1)} - T_{30}^{(2)} = (12 + 29d) - (9 + 29d)}$

$$\boldsymbol{= 12 - 9 = 3}$$

10. Correct Answer: B

Given A.P.: $\boldsymbol{13, 16, 19, \ldots, 160}$
First term $\boldsymbol{a = 13}$, common difference $\boldsymbol{d = 3}$, last term $\boldsymbol{l = 160}$

Find number of terms:

$$\boldsymbol{160 = 13 + (n-1)3}$$
$$\boldsymbol{147 = 3(n-1)}$$
$$\boldsymbol{n - 1 = 49 \Rightarrow n = 50}$$

$\boldsymbol{15}^{\text{th}}$term from the end= $\boldsymbol{(50 - 15 + 1)}^{\text{th}} = \boldsymbol{36}^{\text{th}}$ term

$\boldsymbol{T_{36} = 13 + (36-1)3 = 13 + 105 = 118}$

But given value is 108 → Option A is false

Option B:

First term $\boldsymbol{a = 6}$, second term $= \boldsymbol{9}$

$$\boldsymbol{d = 9 - 6 = 3}$$

Last term $\boldsymbol{l = 33}$

$$\boldsymbol{33 = 6 + (n-1)3}$$
$$\boldsymbol{27 = 3(n-1)}$$
$$\boldsymbol{n - 1 = 9 \Rightarrow n = 10}$$

Option B is true

Option C:

Given A.P.: $\boldsymbol{2, 5, 8, \ldots, 152}$

$$a = 2, d = 3, l = 152$$

Find number of terms:

$$\boldsymbol{152 = 2 + (n-1)3}$$
$$\boldsymbol{150 = 3(n-1)}$$
$$\boldsymbol{n - 1 = 50 \Rightarrow n = 51}$$

Sum:

$$\boldsymbol{S_n = \frac{n}{2}(a + l)}$$

$$\boldsymbol{S_{51} = \frac{51}{2}(2 + 152) = \frac{51}{2} \times 154}$$
$$\boldsymbol{S_{51} = 51 \times 77 = 3927}$$

Given sum = 3925 → Option C is false

Conclusion:

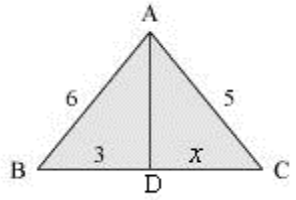
Only Option B is correct.

7. SIMIALR TRIANGLES

Solutions

Worksheet – 1

1. **Correct Answer: A**



The two triangles ABD and ADC are similar

$$\therefore \frac{BD}{AB}=\frac{DC}{AC}$$

$$\Rightarrow \frac{3}{6}=\frac{x}{5}$$

$$\therefore x = 2.5 \text{ cm}$$

2. **Correct Answer: (D)**
For similar triangles:
Similarity is **transitive**.
If ΔOMN ~ ΔFGH and ΔFGH ~ ΔUVW
Then ΔUVW ~ ΔOMN
So, option **(D)** is true.

3. **Correct Answer: C) SSS criterion**
If $\frac{AB}{DE}=\frac{AC}{DF}=\frac{BC}{EF}$
then the triangles are similar by **SSS criterion** (all corresponding sides are proportional).

4. **Correct Answer: C) Identity transformation**
When the scale factor $k = 1$, the figure remains **unchanged**.

5. **Correct Answer: B) Both statements are true, but Statement II is not the correct explanation of Statement I.**
Statement I: All circles are similar → **True**
(Any circle can be obtained from another by scaling.)
Statement II: All equilateral triangles are similar → **True**
(All angles are 60°and sides are proportional.)
However, Statement II does **not** explain why all circles are similar.

6. **Correct Answer: D) Statement I is false and Statement II is true.**
Statement I: All squares are congruent → **False**
(Squares can have different side lengths.)
Statement II: Two polygons of the same number of sides are similar if corresponding angles are equal and sides are proportional → **True**

7. **Correct Answer: (B)** $\Delta XYZ \sim \Delta MNO$
To show similar triangles, we use the symbol "~".
   - **~** → similar (same shape, size may be different)

8. **Correct Answer: (D)42 metres**
Since both shadows are formed at the same time, the triangles formed are **similar**.

$$\frac{\text{Height of pole}}{\text{Shadow of pole}}=\frac{\text{Height of tower}}{\text{Shadow of tower}}$$

$$\frac{6}{4}=\frac{h}{28}$$

$$h=\frac{6}{4}\times 28 = 42$$

9. **Correct Answer:** D $\boxed{90°}$
To find: $\angle QPR$
Step 1: Find PR In right-angled △ PAR(right angle at A):

$$PR=\sqrt{PA^2+AR^2}=\sqrt{6^2+8^2}$$
$$=\sqrt{36+64}=\sqrt{100}$$
$$=10 \text{ cm}$$

Step 2: Check triangle PQR
Now in △ PQR: $PQ^2+PR^2=24^2+10^2=576+100=676$

$$QR^2=26^2=676$$

Since,$PQ^2+PR^2=QR^2$
△ $PQR$is **right-angled at** P.
Step 3: Conclusion $\boxed{\angle QPR=90°}$ Final

10. Correct Answer: B

In similar triangles, corresponding angle bisectors are proportional to the corresponding sides.

So, $\frac{PS}{AD} = \frac{PQ}{AB}$

$\frac{PS}{3} = \frac{7.8}{6.5}$

$PS = 3 \times \frac{7.8}{6.5} = 3.6 \text{ cm}$ $\boxed{PS = 3.6 \text{ cm}}$

WORKSHEET-2

1. Correct Answer: B) 2: 3

Ratio of areas of two squares = 4: 9

- Area ∝ (side)²
- Side ratio = √4: √9 = 2: 3
- Perimeter ∝ side

2. Correct Answer:

Stick–tower shadow problem

- Same sun angle ⇒ similar triangles
- Height: shadow is constant

$$\frac{20}{10} = \frac{H}{50} \Rightarrow H = 100 \text{ m}$$

Answer: A) 100 m

3. Correct Answer: D) 16: 81

Sides of similar triangles = 4: 9

- Area ratio = (side ratio) ²

$$(4:9)^2 = 16:81$$

4. Correct Answer: A) 3: 4

Areas of similar triangles = 9 cm² and 16 cm²

- Side ratio = √(area ratio)

$$\sqrt{9:16} = 3:4$$

5. Correct Answer: C) 27 cm

Areas = 144 cm² and 81 cm²

- Side ratio = √(144 : 81) = 12 : 9 = 4 : 3
- Larger triangle side = 36 cm

$$\text{Smaller side} = 36 \times \frac{3}{4} = 27 \text{ cm}$$

6. Correct Answer: $\boxed{\boldsymbol{AY = 2 \text{ cm}}}$

Use the Basic Proportionality Theorem (Thales Theorem)

If a line is drawn parallel to one side of a triangle, it divides the other two sides in the same ratio.

So, $\frac{\boldsymbol{AX}}{\boldsymbol{XB}} = \frac{\boldsymbol{AY}}{\boldsymbol{YC}}$

$\boldsymbol{AB} = \mathbf{4}\text{cm}$

Since X lies on AB,

$\boldsymbol{AX} + \boldsymbol{XB} = \mathbf{4}$

$\boldsymbol{YC} = \mathbf{2}\text{cm}$

Because XY is parallel to BC and divides the triangle proportionally,

the segment on AC above Y (AY) and below Y (YC) are in the same ratio as the segments on AB.

Thus, $\boldsymbol{AY} = \boldsymbol{YC}$

$$\boldsymbol{AY} = \mathbf{2} \text{ cm}$$

7. Correct Answer: C) $\frac{AB}{AC}$

Given:

In ΔABC, the angle bisector of ∠A meets BC at D

Concept used: Angle Bisector Theorem

The angle bisector divides the opposite side in the ratio of the sides containing the angle.

$$\frac{BD}{DC} = \frac{AB}{AC}$$

8. Correct Answer: B) $a^2 > b^2 + c^2$

Given:

∠A is an obtuse angle in ΔABC

Concept used: Pythagoras inequality

If angle A is obtuse, then the square of the opposite side is **greater** than the sum of squares of the other two sides.

$$a^2 > b^2 + c^2$$

9. Correct Answer: A) $\frac{XY}{XZ} = \frac{YP}{PZ}$

Given:

In ΔXYZ, the internal bisector of ∠X meets YZ at P

Concept used: Angle Bisector Theorem

$\frac{YP}{PZ} = \frac{XY}{XZ}$

Rewriting:

$$\frac{XY}{XZ} = \frac{YP}{PZ}$$

10. Correct Answer: B) 64 cm

Given:

PQRS is a **rhombus**

PQ = 4 cm

Concept used:

In a rhombus:

Sum of squares of diagonals $= 4 \times (\text{side})^2$

$$= 4 \times 4^2 = 4 \times 16 = 64$$

Competitive Corner

SINGLE CORRECT ANSWER TYPE

1. **Correct Answer: B**
In ΔABC, ∠A is an obtuse angle, then
Let side **a** be opposite ∠A.
   - For an **obtuse angle**, the square of the opposite side is **greater** than the sum of squares of the other two sides.

$$a^2 > b^2 + c^2$$

2. **Correct Answer: B (vertex B)**
In ΔABC, $\boldsymbol{AC^2 = AB^2 + BC^2}$, then orthocentre coincides with the vertex ___
   - Given: $AC^2 = AB^2 + BC^2$
   - This represents **Pythagoras theorem**, so the triangle is **right-angled at B**.
   - In a **right-angled triangle**, the **orthocentre lies at the vertex of the right angle**.

3. **Correct Answer: A) 7: 9**
For similar (isosceles) triangles,
Ratio of areas
$= \left(\textbf{Ratio of corresponding heights}\right)^2$
Given:
Area ratio=49:81
So, Height ratio=49:81=7:9

4. **Correct Answer: A)** $BC = CY$
Given that $XY \parallel BC$, alternate interior angles are equal.
$\angle XYB = \angle YBC$
Also, it is given that BYbisects $\angle XYC$.
So,
$\angle XYB = \angle BYC$
From the above two results,
$\angle YBC = \angle BYC$
In $\triangle BYC$, angles opposite to sides $BC$and $CY$are equal.
Therefore, the opposite sides are equal.

$$\boxed{BC = CY}$$

5. **Correct Answer: C) 7.5**
Given: $\triangle \boldsymbol{ABC} \sim \triangle \boldsymbol{PQR}$
AB = 6, BC = 4, AC = 8, PR = 6
Since $\boldsymbol{AC} \leftrightarrow \boldsymbol{PR}$:Scale factor $= \frac{\boldsymbol{PR}}{\boldsymbol{AC}} = \frac{\mathbf{6}}{\mathbf{8}} = \frac{\mathbf{3}}{\mathbf{4}}$
Find PQ and QR

$$\boldsymbol{PQ = AB \times \frac{3}{4} = 6 \times \frac{3}{4} = 4.5}$$

$$\boldsymbol{QR = BC \times \frac{3}{4} = 4 \times \frac{3}{4} = 3}$$

$$\boldsymbol{PQ + QR = 4.5 + 3 = 7.5}$$

6. **Correct Answer: D)** $\boxed{\frac{\boldsymbol{PQ}}{\boldsymbol{PR}} = \frac{\boldsymbol{QX}}{\boldsymbol{XR}}}$
The internal angle bisector of a triangle divides the opposite side in the ratio of the two sides.
So, $\frac{\boldsymbol{QX}}{\boldsymbol{XR}} = \frac{\boldsymbol{PQ}}{\boldsymbol{PR}}$
D) $\frac{\boldsymbol{PQ}}{\boldsymbol{PR}} = \frac{\boldsymbol{QX}}{\boldsymbol{XR}}$ Correct
This matches the Angle Bisector Theorem exactly.

7. **Correct Answer: C) 2: 3**
Ratio of areas of two squares = $\mathbf{4:9}$
Area ∝ (side) $^{\mathbf{2}}$
Side ratio $= \sqrt{\mathbf{4:9}} = \mathbf{2:3}$
Perimeter ∝ side
So perimeter ratio = $\mathbf{2:3}$

8. **Correct Answer: A) 50°**
In similar triangles:
$\boldsymbol{\angle A = \angle D}$
Sum of angles of $\triangle \boldsymbol{DEF}$:

$$\boldsymbol{\angle D + \angle E + \angle F = 180^\circ}$$
$$\boldsymbol{47^\circ + 83^\circ + \angle F = 180^\circ}$$
$$\boldsymbol{\angle F = 180^\circ - 130^\circ = 50^\circ}$$

9. **Correct Answer: C) C**
Given:

$$\boldsymbol{\angle A + \angle B = \angle C}$$

Sum of angles in a triangle:

$$\boldsymbol{\angle A + \angle B + \angle C = 180^\circ}$$

Substitute:

$$\boldsymbol{\angle C + \angle C = 180^\circ \Rightarrow 2\angle C = 180^\circ \Rightarrow \angle C = 90^\circ}$$

So the triangle is right-angled at C.
Orthocentre of a right-angled triangle lies at the right-angle vertex.

10. **Correct Answer: A) A right-angled triangle**
If the circumcentre of a triangle lies on one of its sides, the triangle is
Key facts about circumcentre
   - Acute-angled triangle → circumcentre lies inside
   - Right-angled triangle → circumcentre lies at the midpoint of the hypotenuse
   - Obtuse-angled triangle → circumcentre lies outside

If the circumcentre lies on a side, that side must be the hypotenuse.

11. Correct Answer: D) $4(2+\sqrt{2})$cm

In square ABCD, O is the centre. AC and BD are diagonals.

If AC = 8 cm, find the perimeter of triangle AOB.

Find side of the square

For a square,

Diagonal = side $\times \sqrt{2}$

$8 = \text{side} \times \sqrt{2} \Rightarrow \text{side} = \frac{8}{\sqrt{2}} = 4\sqrt{2}$

So,

$$AB = 4\sqrt{2}$$

Step 2: Find AO and BO

O is the centre, so it is the midpoint of diagonal AC.

$$AO = BO = \frac{AC}{2} = \frac{8}{2} = 4$$

Step 3: Perimeter of triangle AOB

$$AO + OB + AB = 4 + 4 + 4\sqrt{2}$$
$$= 8 + 4\sqrt{2} = 4(2+\sqrt{2})$$

12. Correct Answer: A) 9 : 16

In $\triangle ABC$,

AB = 6 cm, BC = 8 cm, AC = 10 cm.

From B, a perpendicular BD is drawn to AC at D.

We need:ar($\triangle ABD$):ar($\triangle BDC$)

Triangles ABD and BDC have the same height BD.

So,Ratio of areas = Ratio of their bases

$$= AD : DC$$

Find AD and DC

Triangle with sides 6, 8, 10 is a right-angled triangle (since $6^2 + 8^2 = 10^2$).

So, $\angle B = 90°$.

In a right triangle, perpendicular from the right angle to hypotenuse divides it as:

$$AD : DC = AB^2 : BC^2$$
$$= 6^2 : 8^2 = 36 : 64 = 9 : 16$$

13. Correct Answer: C) 4 : 1

$\triangle ABC$and $\triangle BDE$are equilateral triangles.

D is the midpoint of BC.

- Side of $\triangle ABC = BC$
- Since D is midpoint, $BD = \frac{BC}{2}$

Area of an equilateral triangle $\propto$ (side) ^{2}

$$\text{ar}(\triangle ABC):\text{ar}(\triangle BDE) = BC^2 : \left(\frac{BC}{2}\right)^2$$
$$= BC^2 : \frac{BC^2}{4} = 4:1$$

14. Correct Answer: A) 4 : 5

Two isosceles triangles have equal angles → triangles are similar.

Ratio of areas = $16:25$

For similar triangles:Area ratio = (height ratio)^2

$$\text{Height ratio} = \sqrt{16:25} = 4:5$$

15. Correct Answer: A) 16 cm

$$\triangle ABC \sim \triangle DEF$$

Given: $2AB = DE$, $BC = 8$cm

Step 1: Find ratio of sides $\frac{AB}{DE} = \frac{1}{2}$

So,$\frac{BC}{EF} = \frac{1}{2}$

Step 2: Find EF

$$\frac{8}{EF} = \frac{1}{2} \Rightarrow EF = 16 \text{ cm}$$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

16. Correct Answer: A, C

In $\triangle XYZ$, the internal bisector of $\angle X$meets YZat P.

Key fact: Angle Bisector Theorem $\frac{YP}{PZ} = \frac{XY}{XZ}$

Checking the options

A) $\frac{PZ}{PY} = \frac{XZ}{XY}$

Taking reciprocal of the theorem → Correct

C) $\frac{XY}{XZ} = \frac{YP}{PZ}$

Directly matches the theorem → Correct

17. Correct Answer: B,D

In $\triangle ABC$,

$\angle C = 90°$

$BC = a, CA = b, AB = c$

p= perpendicular from $C$to AB

Step 1: Use area of the triangle

Using legs: Area $= \frac{1}{2}ab$

Using hypotenuse and height:Area $= \frac{1}{2}cp$

Equating:$cp = ab$

So Option B is correct.

Step 2: Find p

$$p = \frac{ab}{c}, c^2 = a^2 + b^2$$

$$p^2 = \frac{a^2b^2}{a^2+b^2} \Rightarrow \frac{1}{p^2} = \frac{a^2+b^2}{a^2b^2}$$

$$\Rightarrow \frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$$

So, Option D is correct.

18. Correct Answer: B, D

Given:

- $\boldsymbol{LM \parallel CB}$
- $\boldsymbol{LN \parallel CD}$

We have to find: $\frac{\boldsymbol{AL}}{\boldsymbol{AC}}$

Using $\boldsymbol{LM \parallel CB}$

Triangles $\triangle \boldsymbol{ALM}$and $\triangle \boldsymbol{ACB}$are similar.

So,$\frac{\boldsymbol{AL}}{\boldsymbol{AC}} = \frac{\boldsymbol{AM}}{\boldsymbol{AB}}$

Option B is correct

Using $\boldsymbol{LN \parallel CD}$

Triangles $\triangle \boldsymbol{ALN}$and $\triangle \boldsymbol{ACD}$are similar.

So,$\frac{\boldsymbol{AL}}{\boldsymbol{AC}} = \frac{\boldsymbol{AN}}{\boldsymbol{AD}}$

Option D is correct

19. Correct Answer: A) 0.06 m, 0.15 m, C) 6 cm, 15 cm

Given:

$$\triangle \boldsymbol{ABC} \sim \triangle \boldsymbol{DEF}$$

AB = 3 cm, BC = 2 cm, CA = 2.5 cm

EF = 4 cm

Scale factor$\frac{\boldsymbol{EF}}{\boldsymbol{BC}} = \frac{\mathbf{4}}{\mathbf{2}} = \mathbf{2}$

Find DE:$\boldsymbol{DE} = \mathbf{3} \times \mathbf{2} = \mathbf{6}$ cm

Perimeter of △DEF

$$(\mathbf{3} + \mathbf{2} + \mathbf{2.5}) \times \mathbf{2} = \mathbf{7.5} \times \mathbf{2} = \mathbf{15} \text{ cm}$$

Options checking:

A) 0.06m, 0.15m:

Convert meters to cm: 0.06m = 6cm and 0.15m = 15cm.

ASSERTION AND REASON TYPE

20. Correct Answer: (A)

Assertion (A) is true:
If a line is drawn parallel to one side of a triangle, it divides the other two sides in the same ratio. This is a well-known geometric result.

Reason (R) is also true:
This statement is exactly what is stated in Thales' Theorem, also called the Basic Proportionality Theorem (BPT).

21. Correct Answer: (C) A is true, but R is false.

Assertion (A): Two triangles are similar if their corresponding angles are equal and sides are proportional.
True — This is the definition of similar triangles.

Reason (R): This is called the AAA similarity criterion.
False — There is no AAA similarity criterion.

22. Correct Answer: (D) A is false, but R is true.

Assertion (A): If the corresponding sides of two triangles are proportional, then their areas are equal.
False — If sides are proportional, triangles are similar, but their areas are not necessarily equal.

Reason (R): Ratio of areas of two similar triangles is equal to the square of the ratio of their corresponding sides.
True – $\frac{\text{Area}_1}{\text{Area}_2} = \left(\frac{\text{Side}_1}{\text{Side}_2}\right)^2$.

23. Correct Answer: (B): Both A and R are true, but R is not the correct explanation of A

Assertion (A) is true
If two triangles have equal corresponding angles, they satisfy the AA similarity criterion, so the triangles are similar, not necessarily congruent.

Reason (R) is also true
It correctly states a property of similar triangles:

Corresponding angles are equal

Corresponding sides are proportional

But R does NOT explain A

24. Correct Answer: (A) Both A and R are true, and R is the correct explanation of A.

Assertion (A): In right triangles, if an altitude is drawn to the hypotenuse, then the smaller triangles formed are similar to each other and to the original triangle.

True — This is a standard result in right-angled triangles.

Reason (R): All the corresponding angles in the three triangles are equal.
True — Each triangle has one right angle and shares the other acute angles, so they satisfy the AA similarity criterion.

MATRIX MATCHING TYPE

25. Correct Answer:

a) Side ratio = $4:9$

Area ratio = $4^2:9^2 = 16:81 \rightarrow$ Q

b) Area ratio = $49:81$

Altitude ratio = $\sqrt{49:81} = 7:9 \rightarrow$ P

c) $AP = 1$, $BP = 3 \Rightarrow AB = 4$

Area ratio = $(1:4)^2 = 1:16 \rightarrow$ S

d) $\angle A = 50^\circ$

$\angle Q + \angle R = 180^\circ - 50^\circ = 130^\circ \rightarrow$ T

26. Correct Answer: d → S (4.5)

a) Given $\frac{AD}{DB} = \frac{3}{5}$

$$\frac{AE}{EC} = \frac{3}{5} \Rightarrow AE = \frac{3}{3+5} \times AC = \frac{3}{8} \times 5.6 = 2.1$$

Answer: a → R (2.1)

b) Since $LM \parallel BC$, by Basic Proportionality Theorem:

$$\frac{AL}{AC} = \frac{BM}{BC}$$

$$\frac{x-3}{2x} = \frac{x-2}{2x+3}$$

$x = 9 \rightarrow$P

c) $\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow \frac{x}{x-2} = \frac{x+2}{x-1}$

Solving gives $x = 4$

Answer: c → T (4)

d) $\frac{PE}{QE} = \frac{PF}{RF} \Rightarrow \frac{4}{4.5} = \frac{4}{RF}$

$$RF = 4.5$$

COMPREHENSION TYPE

Passage – I:

Given:

In △ABC, ∠B = 90°. Hence for any point X on BC, $AX^2 = AB^2 + BX^2$ (Pythagoras)

Points D, E trisect $BC \Rightarrow BD = DE = EF$.

27. Correct Answer: A

Find AE^2

$$AE^2 = AB^2 + BE^2$$

28. Correct Answer: B

Find $AD^2 - AB^2$

$AD^2 = AB^2 + BD^2$

$AD^2 - AB^2 = BD^2$

29. Correct Answer: D

Find $3AC^2 + 5AD^2$

$AC^2 = AB^2 + BC^2$

$AD^2 = AB^2 + BD^2$

Since $BC = 3BD$,

$BC^2 = 9BD^2$

$3AC^2 + 5AD^2 = 3(AB^2 + 9BD^2) + 5(AB^2 + BD^2)$

$= 8AB^2 + 32BD^2$

Now,

$AE^2 = AB^2 + BE^2 = AB^2 + (2BD)^2$

$= AB^2 + 4BD^2$

$8AE^2 = 8AB^2 + 32BD^2$

Passage – II:

Given

$\angle B = 90^\circ$, Area of $\triangle ABC = 84$ cm^2, $AB = 7$ cm, $AC = 25$ cm

30. Correct Answer: B) 24

$$\text{Area} = \frac{1}{2} \times AB \times BC$$

$$84 = \frac{1}{2} \times 7 \times BC \Rightarrow BC = 24 \text{ cm}$$

31. Correct Answer: C) 6.72

$$\text{Area} = \frac{1}{2} \times AC \times BD$$

$$84 = \frac{1}{2} \times 25 \times BD \Rightarrow BD = \frac{168}{25} = 6.72 \text{ cm}$$

32. Correct Answer: D) 14 cm

If $BC = 12$cm, for same area find AB

$$84 = \frac{1}{2} \times AB \times 12$$

$AB = 14$ cm

INTEGER TYPE

33. Correct Answer: $x = 2$

By Basic Proportionality Theorem, $\frac{AD}{CD} = \frac{BE}{CE}$

$$\frac{8x+9}{x+3} = \frac{3x+4}{x}$$

$x(8x+9) = (x+3)(3x+4)$

$5x^2 - 4x - 12 = 0$

$(5x+6)(x-2) = 0$

$x = 2$ (reject $x = -6/5$)

34. Correct Answer:

Since $DE \parallel BC$, $\frac{AD}{DB} = \frac{AE}{EC}$

$$\frac{x}{x-2} = \frac{x+2}{x-1}$$

Solving $\Rightarrow x = 4$

35. Correct Answer:

Isosceles right triangle with $\angle C = 90^\circ$

$$AB^2 = AC^2 + AC^2 = 2AC^2$$

$n = 2$

36. Correct Answer: 5km

Distance from starting point: $\sqrt{3^2 + 4^2} = \sqrt{25} = 5$

37. Correct Answer:

Let the shortest side be x m.

Hypotenuse $= 2x + 6$, Third side $= (2x + 6) - 2 = 2x + 4$

Using Pythagoras theorem: $x^2 + (2x + 4)^2 = (2x + 6)^2$

$x^2 - 8x - 20 = 0$

$(x - 10)(x + 2) = 0$

$x = 10$

Shortest side = 10 m

PREVIOUS YEAR QUESTIONS

1. Correct Answer: B Isosceles

Equal altitudes imply equal corresponding sides ⇒ triangle is isosceles.

2. Correct Answer: D $\sqrt{7}$

Since $DE \parallel BC$, $\frac{AD}{DB} = \frac{AE}{EC}$

$$\frac{x+4}{x+3} = \frac{2x-1}{x+1}$$

$(x + 4)(x + 1) = (x + 3)(2x - 1)$

$x^2 = 7 \Rightarrow x = \sqrt{7}$

3. Correct Answer: B

A: Area ratio = square of side ratio → correct

B: Height of tower $= \frac{12}{8} \times 40 = 60$m ≠ 80 m → incorrect

C: $3:4:5$ triangle → sides $12, 16, 20$→ correct

D: All circles are similar → correct

4. Correct Answer: A. 71.43 cm²

Let each equal side $= x$.

Base $= \frac{3}{2}x$.

Perimeter: $2x + \frac{3}{2}x = 42 \Rightarrow \frac{7}{2}x = 42 \Rightarrow x = 12$

So sides are $12, 12, 18$.

Height: $h = \sqrt{12^2 - 9^2} = \sqrt{63}$

Area: $\frac{1}{2} \times 18 \times \sqrt{63} = 9\sqrt{63} \approx 71.43$

5. Correct Answer: A $\frac{3\sqrt{15}}{4}$cm

Using Heron's formula: $s = \frac{4+6+8}{2} = 9$

Area $= \sqrt{9(9-4)(9-6)(9-8)} = \sqrt{135} = 3\sqrt{15}$

Height on base 8 cm: $h = \frac{2 \times \text{Area}}{8} = \frac{2 \times 3\sqrt{15}}{8} = \frac{3\sqrt{15}}{4}$

6. Correct Answer: (A) $135°$

Since AD and CE are angle bisectors, their intersection point O is the incentre.

$$\angle AOC = 90° + \frac{1}{2}\angle B$$

Given $\angle B = 90°$, $\angle AOC = 90° + \frac{1}{2} \times 90° = 135°$

7. Correct Answer: (B) (18 cm)

Let the perpendicular sides be a and b, hypotenuse c.

Area $= \frac{1}{2}ab = 40 \Rightarrow ab = 80$

Perimeter $= a + b + c = 40$

Let $a + b = s$.

Then $c = 40 - s$

Using Pythagoras: $c^2 = a^2 + b^2 = (a + b)^2 - 2ab = s^2 - 160$

So, $(40 - s)^2 = s^2 - 160$

$1600 - 80s + s^2 = s^2 - 160$

$1760 = 80s \Rightarrow s = 22$

Hence, $c = 40 - 22 = \boxed{18 \text{ cm}}$

8. Correct Answer: option C is correct.

A. $\frac{AP}{PB} = \frac{3}{6} = \frac{1}{2}, \frac{AQ}{QC} = \frac{5}{10} = \frac{1}{2}$

$\Rightarrow PQ \parallel BC$ (by converse of BPT)

$\Rightarrow \frac{PQ}{BC} = \frac{1}{3}$

$\Rightarrow BC = 3PQ$

B. Joining midpoints of sides of a triangle forms 4 smaller triangles, and each is similar to the original triangle.

9. Correct Answer: C (8.6 m)

Since $DE \parallel BC$,

$$\frac{AD}{AB} = \frac{AE}{AC}$$

$\frac{10.2}{27.4} = \frac{5.1}{AC} \Rightarrow AC = 13.7$ m

$EC = AC - AE = 13.7 - 5.1 = \boxed{8.6 \text{ m}}$

10. Correct Answer: C (100°)

The two triangles have their corresponding sides in proportion, so the triangles are similar.

Therefore, corresponding angles are equal.

Given ∠Z = 50°, the corresponding angle ∠Q is its vertically opposite / exterior related angle, hence

$$\angle Q = 180° - 80° = \boxed{100°}$$

8. Coordinate Geometry

Solutions

Worksheet – 1

1. **Correct Answer: $\sqrt{5}$(Option D)**
   Distance between $(4, 2)$and $(3, 4)$
   $$d = \sqrt{(4-3)^2 + (2-4)^2} = \sqrt{1+4} = \sqrt{5}$$
2. **Correct Answer: 39 (Option A)**
   Distance between $A(0, 0)$and $B(36, 15)$
   $$AB = \sqrt{(36)^2 + (15)^2} = \sqrt{1296 + 225} = \sqrt{1521} = 39$$
3. **Correct Answer: 5 (Option B)**
   Distance between $A(1, 4)$and $B(5, 1)$
   $$AB = \sqrt{(5-1)^2 + (1-4)^2} = \sqrt{16+9} = 5$$
4. **Correct Answer: Right angled isosceles triangle (Option D)**
   Points $(2, -2), (-1, 2), (3, 5)$
   $$AB^2 = 25,\ BC^2 = 25,\ AC^2 = 50$$
   Two sides equal and Pythagoras theorem holds.
5. **Correct Answer: Equilateral triangle (Option A)**
   Points $(\sqrt{3}-1, \sqrt{3}+1), (\sqrt{3}, 1), (\sqrt{3}+1, \sqrt{3}-1)$
   All three sides are equal.
6. **Correct Answer: $7, -1$(Option A)**
   Distance between $(3, 2)$and $(x, -1)$is 5
   $$\sqrt{(x-3)^2 + (-1-2)^2} = 5 \Rightarrow (x-3)^2 + 9 = 25$$
   $$(x-3)^2 = 16 \Rightarrow x = 7 \text{ or } -1$$
7. **Correct Answer: 3 (Option D)**
   Opposite vertices of square: $(1, -2), (4, 1)$
   $$\text{Diagonal} = \sqrt{(4-1)^2 + (1+2)^2} = \sqrt{18}$$
   $$\text{Side} = \frac{\sqrt{18}}{\sqrt{2}} = 3$$
8. **Correct Answer: 3 or −4 (Option B)**
   $A(5, 3), B(11, -5), P(12, \lambda), \angle APB = 90°$
   $$(AP) \cdot (BP) = 0 \Rightarrow \lambda = 3 \text{ or } -4$$
9. **Correct Answer: Rhombus (Option C)**
   The points $(2, 5), (0, 3), (2, 1), (4, 3)$taken in order form
   Short solution:
   $$AB = BC = CD = DA$$
   All sides are equal, but angles are not right angles.
10. **Correct Answer: Rectangle (Option B)**
    Points $(7, 1), (4, 4), (-2, -2), (1, -5)$
    Opposite sides parallel and equal, right angles.
11. **Correct Answer: Rhombus (Option C)**
    Points $(7, 8), (1, 6), (-1, 0), (5, 2)$
    All sides equal, diagonals perpendicular.
12. **Correct Answer: $(a-4)^2 = 3b^2$(Option C)**
    Distance of $(4, 0)$from $(a, b)$is double the distance of $(a, b)$from x-axis
    $$\sqrt{(a-4)^2 + b^2} = 2 \mid b \mid \Rightarrow (a-4)^2 = 3b^2$$
13. **Correct Answer: $(1, -10)$(Option B)**
    $P(-5, 4), Q(-2, -3)$; PQproduced to Rin ratio $1:2$externally
    $$R = \left(\frac{1x_2 - 2x_1}{1-2}, \frac{1y_2 - 2y_1}{1-2}\right) = (1, -10)$$
14. **Correct Answer: $\frac{t}{1-t}$(Option C)**
    If $x = x_1 + t(x_2 - x_1)$, then ratio is
    $$t : (1-t)$$
15. **Correct Answer: C) 4 : 3 externally**
    Let A(5,1), B(3,2) and P(−3,5).
    Since P lies outside AB, it divides externally.
    Using external section formula:
    $$\frac{AP}{PB} = \frac{x_1 - x}{x - x_2} = \frac{5-(-3)}{-3-3} = \frac{8}{6} = \frac{4}{3}$$
    So, the ratio is 4: 3 externally.
16. **Correct Answer: 3: 7 (Option B)**
    x-axis divides the line joining (2, −3) and (5, 7)
    x-axis ⇒ y = 0
    Using section formula:
    $$\frac{m}{n} = \frac{3}{7}$$

17. **Correct Answer: 3 : 4 (Option D)**
y-axis divides the line joining (3,5) and (−4,7)
y-axis ⇒ x = 0
$$\frac{m}{n} = \frac{3}{4}$$

18. **Correct Answer:** $\left(-\frac{3}{2}, \frac{31}{8}\right)$**(Option C)**
Point dividing (5,7) and (−1,3) in the ratio 2 : −3
Negative ratio ⇒ external division
Using section formula:
$$\left(\frac{2(-1)-3(5)}{2-3}, \frac{2(3)-3(7)}{2-3}\right) = \left(-\frac{3}{2}, \frac{31}{8}\right)$$

19. **Correct Answer: (3,3), (−9,27) (Option C)**
Points dividing (1,7) and (6,−3) in ratio 2 : 3
Internal division:$(3, 3)$
External division:$(-9, 27)$

20. **Correct Answer: Option B**
Points of trisection of (−5,2) and (3,6)
Ratios: 1:2 and 2:1
Using section formula:$\left(-\frac{7}{3}, \frac{10}{3}\right), \left(\frac{1}{3}, \frac{14}{3}\right)$

21. **Correct Answer: (1,1)**
Midpoint of $(x, y+1)$and $(x+1, y+2)$is $\left(\frac{2x+1}{2}, \frac{2y+3}{2}\right)$
Given midpoint = $\left(\frac{3}{2}, \frac{5}{2}\right)$
So, $x = 1, y = 1$
Now midpoint of $(x-1, y+1)$and $(x+1, y-1)$:
$$(1, 1)$$

22. **Correct Answer: A) 2: 1 externally**
Corrected short solution (Answer A)
Given points:
A(4, 8), B(−2, −7) and P(2, 3)
Using section formula:
From x-coordinates: $\frac{m}{n} = \frac{4-2}{2-(-2)} = \frac{2}{4} = \frac{1}{2}$
From y-coordinates: $\frac{m}{n} = \frac{8-3}{3-(-7)} = \frac{5}{10} = \frac{1}{2}$
Since point (2,3) lies outside the segment AB, division is external.
Ratio $= 1:2 \Rightarrow 2:1$ externally

23. **Correct Answer: 4: 3 externally (Option C)**
Ratio in which (−3,5) divides the line joining (5,1) and (3,2)
Using section formula: $\frac{m}{n} = \frac{5-(-3)}{-3-3} = \frac{8}{6} = \frac{4}{3}$
Point lies outside ⇒ external

24. **Correct Answer:** $\frac{10}{3}$**(Option C)**
If P, Q are points of trisection of A(1,−2), B(−5,6), then PQ
Distance AB: $AB = \sqrt{(-6)^2 + (8)^2} = 10$
$$PQ = \frac{AB}{3} = \frac{10}{3}$$

25. **Correct Answer: (5, −1) (Option A)**
Center of circle whose diameter endpoints are A(3,−4), B(7,2)
Midpoint:
$$\left(\frac{3+7}{2}, \frac{-4+2}{2}\right) = (5, -1)$$

Competitive Corner Level-1

SINGLE CORRECT ANSWER TYPE

1. **Correct Answer: B) 3**
Distance of $P(2, 3)$from the X-axis
The distance of any point $(x, y)$from the X-axis is the absolute value of its y-coordinate (| y |).
   - Point: $(2, 3)$
   - Distance: $|3| = 3$

2. **Correct Answer: B) 8**
Distance between $A(0, 6)$and $B(0, -2)$
Since both points lie on the Y-axis (x-coordinates are 0), the distance is the difference between their y-coordinates.
   - Distance $= |y_2 - y_1| = |-2-6| = |-8| = 8$

3. **Correct Answer: C) 10**
Distance of $P(-6, 8)$from the origin
The distance of a point $(x, y)$from the origin $(0, 0)$is given by $\sqrt{x^2 + y^2}$.
   - Distance $= \sqrt{(-6)^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$

4. **Correct Answer: B)** $5\sqrt{2}$
Distance between $(0, 5)$and $(-5, 0)$
Using the distance formula:
$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
   - $d = \sqrt{(-5-0)^2 + (0-5)^2}$
   - $= \sqrt{(-5)^2 + (-5)^2}$
   - $= \sqrt{25 + 25} = \sqrt{50}$
   - $\sqrt{50} = \sqrt{25 \times 2} = 5\sqrt{2}$

5. **Correct Answer: C) $\sqrt{34}$**

Length of diagonal of rectangle $\boldsymbol{AOBC}$

The given vertices are $\boldsymbol{A(0,3)}$, $\boldsymbol{O(0,0)}$, and $\boldsymbol{B(5,0)}$.
In a rectangle, the diagonals are equal, so we can find the length of diagonal $\boldsymbol{AB}$.

Using the distance formula:

$$\boldsymbol{AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}}$$

$$\boldsymbol{AB = \sqrt{(5-0)^2 + (0-3)^2} = \sqrt{25+9}}$$

$$\boldsymbol{= \sqrt{34}}$$

6. **Correct Answer: C) 1**

Find $p$if A(0,2) is equidistant from B(3,p) and C(p,5)

Equidistant means $AB^2 = AC^2$

$(3-0)^2 + (p-2)^2 = (p-0)^2 + (5-2)^2$
$9 + (p^2 - 4p + 4) = p^2 + 9$
$-4p + 4 = 0 \Rightarrow 4p = 4 \Rightarrow p = 1$

7. **Correct Answer:**

8. **Correct Answer: Collinear (no triangle formed)**

Vertices (−4,0), (4,0), (3,0):
All three points lie on the X-axis (y = 0).
Hence, they are collinear and do not form a triangle.

9. **Correct Answer: IV Quadrant (Option D)**

Point dividing (7, −6) and (3, 4) in the ratio 1 : 2

Using section formula:

$$\boldsymbol{x = \frac{1(3)+2(7)}{3} = \frac{17}{3}, y = \frac{1(4)+2(-6)}{3}}$$

$$\boldsymbol{= -\frac{8}{3}}$$

Since $\boldsymbol{x > 0}$and $\boldsymbol{y < 0}$, the point lies in the IV Quadrant.

10. **Correct Answer: B) (0, −1)**

Fourth vertex D of parallelogram ABCD
Given: A(−2,3), B(6,7), C(8,3)
Midpoint of AC

$$\boldsymbol{\left(\frac{-2+8}{2}, \frac{3+3}{2}\right) = (3,3)}$$

This is also the midpoint of BD.
So,

$$\boldsymbol{x_D = 2(3) - 6 = 0, y_D = 2(3) - 7 = -1}$$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct Answer: A and B**

If P(2,1) lies on the segment joining A(4,2) and B(8,4)

Using section formula:

$$\boldsymbol{\frac{AP}{PB} = \frac{2-4}{8-2} = \frac{-2}{6} = \frac{1}{3} \Rightarrow AP = \frac{1}{4}AB,\ PB = \frac{3}{4}AB}$$

12. **Correct Answer: a = −16**

If $\boldsymbol{P\left(\frac{a}{4}, 4\right)}$is the midpoint of Q(−6,5) and R(−2,3)

Midpoint x-coordinate:

$$\boldsymbol{\frac{-6+(-2)}{2} = -4 = \frac{a}{4} \Rightarrow a = -16}$$

13. **Correct Answer: C) y = 13**

Perpendicular bisector of A(1,5) and B(4,6)

Midpoint =

$$\boldsymbol{\left(\frac{1+4}{2}, \frac{5+6}{2}\right) = \left(\frac{5}{2}, \frac{11}{2}\right)}$$

Slope AB = $\frac{1}{3}$
Perpendicular slope = −3

Equation cuts Y-axis at y = 13

14. **Correct Answer: A and D**

Length of line segment = 10

One end (2,−3), other end has abscissa = 10

Distance formula:

$$\boldsymbol{(10-2)^2 + (y+3)^2 = 100}$$
$$\boldsymbol{\Rightarrow 64 + (y+3)^2 = 100}$$
$$\boldsymbol{\Rightarrow (y+3)^2 = 36 \Rightarrow y = 3, -9}$$

15. **Correct Answer: A and C**

Distance between (4,p) and (1,0) is 5

$$\boldsymbol{(4-1)^2 + (p-0)^2 = 25 \Rightarrow 9 + p^2 = 25}$$
$$\boldsymbol{\Rightarrow p^2 = 16 \Rightarrow p = 4, -4}$$

ASSERTION AND REASON TYPE

16. Correct Answer: (A)

Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

Short Explanation:

- Using the **section formula**, the point $(-1,6)$correctly divides the line segment joining $(-3,10)$and $(6,-8)$internally in the ratio $2:7$.
  So, Assertion (A) is true.
- Reason (R) states a correct property of an equilateral triangle:
  $AB = BC = CA$.
  So, Reason (R) is true.

17. Correct Answer: (A)

Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

Solution:

- **Assertion (A):**
  The point $(0,4)$lies on the **y-axis**.
  This is **true** because any point with x-coordinate equal to 0 lies on the y-axis.
- **Reason (R):**
  The x-coordinate of a point on the **y-axis** is **zero**.
  This is **true**.
- Since the reason directly explains why the point $(0,4)$lies on the y-axis, it is the **correct explanation**.

18. Correct Answer: (D)

Assertion (A) is false but Reason (R) is true.

Solution:

Given:

Points $P(2,-3)$and $Q(10,y)$

Distance $PQ = 10$

Using the distance formula:

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
$$10 = \sqrt{(10-2)^2 + (y+3)^2}$$
$$10 = \sqrt{64 + (y+3)^2}$$
$$100 = 64 + (y+3)^2$$
$$(y+3)^2 = 36$$
$$y + 3 = \pm 6$$
$$y = 3 \text{ or } y = -9$$

19. Correct Answer: (C)

Assertion (A) is true but Reason (R) is false.

Solution:

Assertion (A):

The midpoint of a line segment divides the line in the ratio **1: 1**.

This is **true** by definition of midpoint.

Reason (R):

It is stated that the point $(-3,k)$divides the line segment joining $(-5,4)$and $(-2,3)$in the ratio **1: 2**.

Using section formula for ratio $1:2$:

$$x = \frac{1(-2) + 2(-5)}{1+2} = \frac{-2-10}{3} = -4$$

But given x-coordinate is **–3**, not –4.

So, **Reason (R) is false**.

MATRIX MATCHING TYPE

20. Correct Answer:

Column-I → Column-II

a) Midpoint of A (–2,5) and B (2,5)

Midpoint

$$\left(\frac{-2+2}{2}, \frac{5+5}{2}\right) = (0,5)$$

Matches → T (0,5)

b) A line intersects Y-axis and X-axis at P, Q respectively.

If (2,–5) is the midpoint of PQ and Q = (4,0), find P

Using midpoint formula:

$$(2,-5) = \left(\frac{x_P + 4}{2}, \frac{y_P + 0}{2}\right)$$
$$x_P = 0, y_P = -10$$

Matches → P (0, –10)

c) Point on X-axis at distance $2\sqrt{5}$from (7,–4)

Let point be $(x,0)$

$$\sqrt{(x-7)^2 + (0+4)^2} = 2\sqrt{5}$$
$$(x-7)^2 + 16 = 20 \Rightarrow (x-7)^2 = 4 \Rightarrow x = 5 \text{ or } 9$$

From options:

Matches → S (5,0)

d) Fourth vertex of rectangle with vertices (2,–2), (7,3), (11,–1)

Checking vector pattern → fourth vertex = (6,–6)

Matches → Q (6,–6)

21. Correct Answer:

Matrix Match

Column-I → Column-II

a) Distance between (0,0) and (6,8)

$$\sqrt{6^2 + 8^2} = 10$$

Matches → P (10)

b) Midpoint of (2,4) and (6,8)

$$\left(\frac{2+6}{2}, \frac{4+8}{2}\right) = (4,6)$$

Matches → Q (4,6)

c) Point on Y-axis
Condition: $x = 0$
Matches → R (x=0)

d) Distance of point (−3,5) from X-axis
Distance = $| y |= 5$
Matches → S (5)

COMPREHENSION TYPE

Passage – I:

Section Formula

The section formula for a point $P(x, y)$dividing the line segment ABjoining $A(x_1, y_1)$and $B(x_2, y_2)$in the ratio $m: n$is:

For internal division:$x = \frac{mx_2+nx_1}{m+n}, y = \frac{my_2+ny_1}{m+n}$

For external division:$x = \frac{mx_2-nx_1}{m-n}, y = \frac{my_2-ny_1}{m-n}$

22. Correct Answer: B) $-\frac{3}{2}$

Ratio in which $\boldsymbol{AB}$is divided by the X-axis

Given points: $A(2,3)$, $B(-1,2)$

Any point on the **X-axis** has y-coordinate = 0.

Let the ratio be $k: 1$.
Using the y-coordinate section formula: $y = \frac{my_2+ny_1}{m+n}$

$$0 = \frac{k(2) + 1(3)}{k + 1}$$

$$2k + 3 = 0 \Rightarrow k = -\frac{3}{2}$$

23. Correct Answer: C) $\frac{1}{2}$

Ratio in which $\boldsymbol{AB}$is divided by the Y-axis

Given points:
$A(3,2)$, $B(-6,5)$

Any point on the **Y-axis** has x-coordinate $= 0$.

Let the ratio be $k: 1$.
Using the x-coordinate section formula:

$$x = \frac{mx_2 + nx_1}{m + n}$$

$$0 = \frac{k(-6) + 1(3)}{k + 1}$$

$$-6k + 3 = 0 \Rightarrow 6k = 3 \Rightarrow k = \frac{1}{2}$$

24. Correct Answer: D) $\left(\frac{5}{3}, 3\right)$

Point dividing $\boldsymbol{A(1,2)}$and $\boldsymbol{B(3,5)}$in the ratio $\mathbf{1: 2}$

Using the **internal section formula** with $m = 1, n = 2$:

$$x = \frac{1(3) + 2(1)}{1 + 2} = \frac{3 + 2}{3} = \frac{5}{3}$$

$$y = \frac{1(5) + 2(2)}{1 + 2} = \frac{5 + 4}{3} = \frac{9}{3} = 3$$

Passage – II:

Distance Formula

The distance between $A(x_1, y_1)$and $B(x_2, y_2)$is given by:

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

25. Correct Answer: A) Straight line

Nature of points $(\mathbf{0}, \mathbf{1})$, $(\mathbf{0}, -\mathbf{10})$, $\left(\mathbf{0}, \frac{\mathbf{1}}{\mathbf{3}}\right)$

Observe that the x-coordinate is **0** for all three points.

- Since all points have the same x-coordinate, they lie on the **Y-axis**.
- Points lying on the same line are **collinear**, hence they form a **straight line**.

26. Correct Answer: B) $a^2 + b^2$

Calculate $\boldsymbol{AB^2}$if $\boldsymbol{A} = (\boldsymbol{a} + \boldsymbol{b}, \boldsymbol{a})$and $\boldsymbol{B} = (\boldsymbol{b}, \boldsymbol{a} + \boldsymbol{b})$

Using the distance formula squared:

$$AB^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$
$$AB^2 = (b - (a + b))^2 + ((a + b) - a)^2$$
$$AB^2 = (-a)^2 + (b)^2 = a^2 + b^2$$

27. Correct Answer: 5

Find the distance between the points $\boldsymbol{A(2,-1)}$and $\boldsymbol{B(6,2)}$.

Solution:

Using distance formula:

$$AB = \sqrt{(6-2)^2 + (2-(-1))^2} = \sqrt{4^2 + 3^2} = \sqrt{25} = 5$$

INTEGER TYPE

28. Correct Answer: 2

If $(\boldsymbol{a, b})$is the midpoint of A(10, −6) and B(k, 4) and $a - 2b = 18$, find $\frac{k}{11}$

Midpoint:

$$a = \frac{10+k}{2}, b = \frac{-6+4}{2} = -1$$

Substitute:

$$\frac{10+k}{2} - 2(-1) = 18 \Rightarrow \frac{10+k}{2} + 2 = 18$$

$$\Rightarrow \frac{10+k}{2} = 16 \Rightarrow k = 22$$

$$\frac{k}{11} = 2$$

29. Correct Answer: 3

Centre of circle $(\boldsymbol{2a, a-7})$, passes through (11, −9), diameter $= \boldsymbol{10\sqrt{2}}$

Radius $r = 5\sqrt{2}$

Distance from centre to point:

$$(11-2a)^2 + (-9-(a-7))^2 = (5\sqrt{2})^2 = 50$$

$$(11-2a)^2 + (-a-2)^2 = 50$$

$$\Rightarrow 5a^2 - 40a + 75 = 0$$

$$\Rightarrow a^2 - 8a + 15 = 0 \Rightarrow a = 3,5$$

Least value of a= **3**

30. Correct Answer: 0

A(3,2), B(5,1) divided by P in ratio 1:2 and lies on

$$3x - 18y + k = 0$$

Coordinates of P:

$$x = \frac{1(5)+2(3)}{3} = \frac{11}{3}, y = \frac{1(1)+2(2)}{3} = \frac{5}{3}$$

Substitute:

$$3\left(\frac{11}{3}\right) - 18\left(\frac{5}{3}\right) + k = 0 \Rightarrow 11 - 30 + k = 0$$

$$\Rightarrow k = 19$$

$$k - 19 = 0$$

31. Correct Answer: 2

A(2,9), B(a,5), C(7,5) form a triangle right-angled at B

For right angle at B:

$$AB^2 + BC^2 = AC^2$$

$$(a-2)^2 + (-4)^2 + (7-a)^2 + 0^2 = (5)^2 + (-4)^2$$

$$(a-2)^2 + (7-a)^2 = 25$$

$$\Rightarrow 2a^2 - 18a + 28 = 0$$

$$\Rightarrow a^2 - 9a + 14 = 0 \Rightarrow a = 2,7$$

PREVIOUS YEAR QUESTIONS

1. Correct Answer: Scalene triangle (C)

Points: (1,2), (3,5), (4, −3)

Find lengths of sides using distance formula.

All three sides are **unequal**.

2. Correct Answer: $(-2,5)\rightarrow$ **Option D**

Centroid formula:

$$\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right) = (1,5)$$

$$\frac{6+(-1)+x}{3} = 1 \Rightarrow x = -2$$

$$\frac{3+7+y}{3} = 5 \Rightarrow y = 5$$

3. Correct Answer: 1 : 2 (C)

If a line cuts the **Y-axis**, ratio is inverse of x-coordinates.

Points: $(-3,2)$and $(6, 1)$

Ratio $= 3:6 = 1:2$

4. Correct Answer: −11 (Option B)

Given points: $(\boldsymbol{2, 5})$and $(\boldsymbol{3, k})$

Mid-point:

$$\left(\frac{2+3}{2}, \frac{5+k}{2}\right) = \left(\frac{5}{2}, \frac{5+k}{2}\right)$$

This midpoint lies on the line:

$$\boldsymbol{4x + 3y - 1 = 0}$$

Substitute $\boldsymbol{x = \frac{5}{2}}$, $\boldsymbol{y = \frac{5+k}{2}}$:

$$\boldsymbol{4\left(\frac{5}{2}\right) + 3\left(\frac{5+k}{2}\right) - 1 = 0}$$

$$\boldsymbol{10 + \frac{15+3k}{2} - 1 = 0}$$

$$\boldsymbol{9 + \frac{15+3k}{2} = 0}$$

$$\boldsymbol{18 + 15 + 3k = 0}$$

$$\boldsymbol{3k = -33 \Rightarrow k = -11}$$

5. Correct Answer: 3 : 2 (Option D)

Short Solution

Given:

$A(-10,4)$, $B(-2,0)$

Mid-point of AB:

$$P = \left(\frac{-10-2}{2}, \frac{4+0}{2}\right) = (-6,2)$$

Points $C(-9,-4)$and $D(-4,6)$

Let Pdivide CDin the ratio $m:n$.

Using section formula:

$$-6 = \frac{m(-4)+n(-9)}{m+n}$$

$$2 = \frac{m(6)+n(-4)}{m+n}$$

Solving the above equations:

$$m:n = 3:2$$

6. Correct Answer: 14 (B)

Abscissa of $A = 10$

Ordinate of $B = -4$

$$10-(-4) = 14$$

PREVIOUS YEAR QUESTIONS

1. Correct Answer:

The points $(1,2)$, $(3,5)$ and $(4,-3)$ are the vertices of a/an _____ triangle.

(IMO 23–24)

A) Right-angled

B) Isosceles

C) Scalene

D) Equilateral

2. Correct Answer:

If two vertices of a triangle are $(6, 3)$ and $(-1,7)$ and its centroid is $(1, 5)$, then find its third vertex.

(IMO 23–24)

A) $(3,-5)$

B) $(3,-2)$

C) $(4, 2)$

D) $(-2,5)$

3. Correct Answer:

In what ratio is the line segment joining the points $(-3,2)$ and $(6, 1)$ divided by the Y-axis?

(IMO 22–23)

A) $1:3$

B) $2:1$

C) $1:2$

D) $3:1$

B)

4. Correct Answer:

If the midpoint of the line joining $(2, 5)$ and $(3, k)$ is (x, y) and also passes through $4x+3y-1=0$, then find the value of k.

(IMO 22–23)

A) 19

B) –11

C) 13

D) –17

5. Correct Answer:

The midpoint P of the line segment joining the points $A(-10,4)$ and $B(-2,0)$ lies on the line segment joining the points $C(-9,-4)$ and $D(-4,6)$.

Find the ratio in which P divides CD.

(IMO 21–22)

A) $1:4$

B) $2:3$

C) $4:1$

D) $3:2$

6. Correct Answer:

If the coordinates of two points A and B are $(10, 5)$ and $(-7,-4)$ respectively, then the value of (abscissa of A) − (ordinate of B) is

(IMO 21–22)

A) –14

B) 14

C) –10

D) –12

7. Correct Answer:

If the coordinates of points A and B are $(-2,2)$ and $(2,-4)$ respectively, find the coordinates of point P such that

$$AP = \frac{3}{7}AB,$$

where P lies on the line segment AB.

(IMO 21–22)

A) $\left(\frac{2}{7}, \frac{-20}{7}\right)$

B) $\left(\frac{-2}{7}, \frac{-20}{7}\right)$

C) $\left(\frac{-2}{7}, \frac{20}{7}\right)$

D) $\left(\frac{2}{7}, \frac{20}{7}\right)$

8. Correct Answer:
Find the coordinates of the centroid of a triangle whose vertices are $(8,12), (0,6)$ and $(8,0)$.
(IMO 20–21)
A) $(4,3)$
B) $\left(\frac{16}{3},6\right)$
C) $(4,6)$
D) $(6,8)$

9. Correct Answer: Scalene triangle (C)
If the midpoint of the line segment joining $A(2a,4)$ and $B(-2,3b)$ is $M(1,2a+1)$, then the values of a and b respectively are
(IMO 20–21)
A) 2, 2
B) 2, 3
C) 3, 2
D) 5, 2
Points: $(1,2), (3,5), (4,-3)$
Find lengths of sides using distance formula.
All three sides are **unequal**.

10. Correct Answer: Third vertex: $(-2,5)$ → **Option D**
Centroid formula:
$$\left(\frac{x_1+x_2+x_3}{3},\frac{y_1+y_2+y_3}{3}\right)=(1,5)$$
$$\frac{6+(-1)+x}{3}=1\Rightarrow x=-2$$
$$\frac{3+7+y}{3}=5\Rightarrow y=5$$

11. Correct Answer: 1 : 2 (C)
If a line cuts the **Y-axis**, ratio is inverse of x-coordinates.
Points: $(-3,2)$ and $(6,1)$
Ratio $=3:6=1:2$

12. Correct Answer: 13 (C)
Midpoint of $(2,5)$ and $(3,k)$:
$$\left(\frac{5}{2},\frac{5+k}{2}\right)$$
Substitute in $4x+3y-1=0$:
$$4\left(\frac{5}{2}\right)+3\left(\frac{5+k}{2}\right)-1=0$$
Solving gives:
$$k=13$$

13. Correct Answer: 4 : 1 (C)
Midpoint of $A(-10,4)$ and $B(-2,0)$:
$$P=(-6,2)$$
Let P divide $C(-9,-4)$ and $D(-4,6)$ in ratio $m:n$
Using section formula →
$$m:n=4:1$$

14. Correct Answer: 14 (B)
Abscissa of $A=10$
Ordinate of $B=-4$
$$10-(-4)=14$$

15. Correct Answer: Option A
Using section formula:
$$P=A+\frac{3}{7}(B-A)$$
Coordinates obtained:
$$\left(\frac{2}{7},\frac{-20}{7}\right)$$

16. Correct Answer: Option B
Centroid formula:
$$\left(\frac{x_1+x_2+x_3}{3},\frac{y_1+y_2+y_3}{3}\right)$$
$$\left(\frac{8+0+8}{3},\frac{12+6+0}{3}\right)=\left(\frac{16}{3},6\right)$$

17. Correct Answer: 2, 2 (A)
Midpoint formula:
$$\left(\frac{2a-2}{2},\frac{4+3b}{2}\right)=(1,2a+1)$$
Solving:
$$a=2, b=2$$

18. Correct Answer:

If you want, I can also provide:

- One-line solutions
- Answer key only
- Solved worksheet PDF
- Similar previous-year questions

19. Correct Answer: 3 : 2 (Option D)

Short Solution (Corrected)

Points:
$A(-10,4)$, $B(-2,0)$

Midpoint of AB:
$$P=\left(\frac{-10+(-2)}{2},\frac{4+0}{2}\right)=(-6,2)$$

Let point $P(-6,2)$ divide the line joining $C(-9,-4)$ and $D(-4,6)$ in the ratio $m:n$.

Using section formula:
$$-6=\frac{m(-4)+n(-9)}{m+n}$$
$$2=\frac{m(6)+n(-4)}{m+n}$$
Solving the above equations:
$$m:n=3:2$$

20. Correct Answer: 3 : 2 (Option D)

Short Solution

Given:

$A(-10,4)$, $B(-2,0)$

Mid-point of AB:

$$P = \left(\frac{-10-2}{2}, \frac{4+0}{2}\right) = (-6,2)$$

Points $C(-9,-4)$ and $D(-4,6)$

Let P divide CD in the ratio $m:n$.

Using section formula:

$$-6 = \frac{m(-4)+n(-9)}{m+n}$$
$$2 = \frac{m(6)+n(-4)}{m+n}$$

Solving the above equations:

$$m:n = 3:2$$

21. Correct Answer: −11 (Option B)

Short Solution

Given points: $(2, 5)$ and $(3, k)$

Mid-point:

$$\left(\frac{2+3}{2}, \frac{5+k}{2}\right) = \left(\frac{5}{2}, \frac{5+k}{2}\right)$$

This midpoint lies on the line:

$$4x + 3y - 1 = 0$$

Substitute $x = \frac{5}{2}$, $y = \frac{5+k}{2}$:

$$4\left(\frac{5}{2}\right) + 3\left(\frac{5+k}{2}\right) - 1 = 0$$
$$10 + \frac{15+3k}{2} - 1 = 0$$
$$9 + \frac{15+3k}{2} = 0$$
$$18 + 15 + 3k = 0$$
$$3k = -33 \Rightarrow k = -11$$

22. Correct Answer: $\boxed{\left(\frac{2}{7}, \frac{-20}{7}\right)}$ (Option A)

Short Solution

Given:

$A(-2,2)$, $B(2,-4)$

Given:

$$AP = \frac{3}{7}AB$$

Using section formula (internal division):

$$P = A + \frac{3}{7}(B - A)$$
$$B - A = (2-(-2), -4-2) = (4,-6)$$
$$P = \left(-2 + \frac{3}{7}\cdot 4,\ 2 + \frac{3}{7}\cdot(-6)\right)$$
$$P = \left(-2 + \frac{12}{7},\ 2 - \frac{18}{7}\right) = \left(\frac{2}{7}, \frac{-20}{7}\right)$$

23. Correct Answer: $\boxed{\left(\frac{-2}{7}, \frac{-20}{7}\right)}$ (Option B)

Short Solution (Corrected)

Given:

$A(-2,2)$, $B(2,-4)$

Given:

$$AP = \frac{3}{7}AB$$

Using section formula:

$$P = A + \frac{3}{7}(B - A)$$
$$B - A = (2-(-2), -4-2) = (4,-6)$$
$$P = \left(-2 + \frac{3}{7}\cdot 4,\ 2 + \frac{3}{7}\cdot(-6)\right)$$

Rewriting the x−coordinate as a single fraction:

$$-2 = \frac{-14}{7}$$
$$P = \left(\frac{-14+12}{7}, \frac{14-18}{7}\right) = \left(\frac{-2}{7}, \frac{-20}{7}\right)$$

24. Correct Answer: Option B

Short Solution

Given:

$A(-2,2)$, $B(2,-4)$

$$AP = \frac{3}{7}AB$$

Using section formula:

$$P = A + \frac{3}{7}(B - A)$$
$$P = \left(-2 + \frac{12}{7},\ 2 - \frac{18}{7}\right)$$
$$= \left(\frac{-2}{7}, \frac{-20}{7}\right)$$

25. Correct Answer: Option B

Centroid formula:

$$\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$$
$$\left(\frac{8+0+8}{3}, \frac{12+6+0}{3}\right) = \left(\frac{16}{3}, 6\right)$$

26. Correct Answer: $a = 2,\ b = 2 \rightarrow$ **Option A**

Short Solution

Given:

Midpoint of $A(2a, 4)$and $B(-2,3b)$is $M(1,2a + 1)$

Using midpoint formula:

$$\left(\frac{2a - 2}{2}, \frac{4 + 3b}{2}\right) = (1,2a + 1)$$

From x-coordinate:

$$\frac{2a - 2}{2} = 1 \Rightarrow 2a - 2 = 2 \Rightarrow a = 2$$

From y-coordinate:

$$\frac{4 + 3b}{2} = 2a + 1 = 5 \Rightarrow 4 + 3b = 10 \Rightarrow b = 2$$

27. Correct Answer: −15

(i) Short Solution

Points: $(3, 4)$and $(k, 7)$

Midpoint:

$$\left(\frac{3 + k}{2}, \frac{11}{2}\right)$$

Given line:

$$2x + 2y + 1 = 0$$

Substitute:

$$2\left(\frac{3 + k}{2}\right) + 2\left(\frac{11}{2}\right) + 1 = 0$$

$$3 + k + 11 + 1 = 0 \Rightarrow k = -15$$

28. Correct Answer: $(8, 4)$

(ii) Short Solution

Let coordinates of $P(x, y)$

Given:

$$x = 2y$$

Equidistant from $Q(2, -5)$and $R(-3,6)$

Using distance formula:

$$(x - 2)^2 + (y + 5)^2 = (x + 3)^2 + (y - 6)^2$$

Substitute $x = 2y$and solve:

$$y = 4 \Rightarrow x = 8$$

9. Trigonometry

Solutions

Worksheet – 1

1. **Correct Answer: A) −390°**

Problem:

$-\frac{13\pi}{6}$radians = ?

Rule:

To convert radians to degrees, multiply by $\frac{180}{\pi}$.

Calculation:

$$-\frac{13\pi}{6}\times\frac{180}{\pi}=-13\times 30=-390$$

2. **Correct Answer: A)** $\frac{5}{14}$

Trigonometric Ratio Substitution

Problem:

If $5\tan\theta=4$(θ in 1st quadrant), find $\frac{5\sin\theta-3\cos\theta}{\sin\theta+2\cos\theta}$

From $5\tan\theta=4$, we get $\tan\theta=\frac{4}{5}$

Divide the numerator and denominator of the expression by $\cos\theta$to use $\tan\theta$:

$$\frac{5\tan\theta-3}{\tan\theta+2}$$

Substitute $\tan\theta=\frac{4}{5}$:

$$\frac{5\left(\frac{4}{5}\right)-3}{\frac{4}{5}+2}=\frac{4-3}{\frac{14}{5}}=\frac{1}{\frac{14}{5}}=\frac{5}{14}$$

3. **Correct Answer: C)** $\frac{p^2-q^2}{p^2+q^2}$

Algebraic Substitution with $\tan\theta$

Problem:

If $\tan\theta=\frac{p}{q}$, then find

$$\frac{p\sin\theta-q\cos\theta}{p\sin\theta+q\cos\theta}$$

Method:

Similar to Problem 2, divide the numerator and denominator by $\cos\theta$:

$$\frac{p\tan\theta-q}{p\tan\theta+q}$$

Substitute $\tan\theta=\frac{p}{q}$: $\frac{p\left(\frac{p}{q}\right)-q}{p\left(\frac{p}{q}\right)+q}=\frac{\frac{p^2-q^2}{q}}{\frac{p^2+q^2}{q}}=\frac{p^2-q^2}{p^2+q^2}$

4. **Correct Answer: A)** $\frac{17}{31}$

Third Quadrant Evaluation

Problem: If $\sin\theta=-\frac{7}{25}$(θ in 3rd quadrant), find

$$\frac{7\cot\theta-24\tan\theta}{7\cot\theta+24\tan\theta}$$

In the 3rd quadrant, $\tan\theta$and $\cot\theta$are **positive**.

Using a right triangle $(7,24,25)$: $\sin\theta=\frac{7}{25}\Rightarrow\tan\theta=\frac{7}{24},\cot\theta=\frac{24}{7}$

Substitute the values: $\frac{7\left(\frac{24}{7}\right)-24\left(\frac{7}{24}\right)}{7\left(\frac{24}{7}\right)+24\left(\frac{7}{24}\right)}=\frac{24-7}{24+7}=\frac{17}{31}$

5. **Correct Answer:**

Rearrange the given equation:

$\sin\theta=\sqrt{3}\cos\theta-\cos\theta=(\sqrt{3}-1)\cos\theta$

$\tan\theta=\sqrt{3}-1$

Let $x=2\cos\theta-\sin\theta$.

6. **Correct Answer: B)** $\frac{119}{169}$

Since the angles are acute, find cosines using the Pythagorean triple $(5,12,13)$:

$$\cos(A+B)=\frac{5}{13},\cos B=\frac{12}{13}$$

Use the identity:$\sin A=\sin((A+B)-B)=\sin(A+B)\cos B-\cos(A+B)\sin B$

Substitute values:$\sin A=\left(\frac{12}{13}\right)\left(\frac{12}{13}\right)-\left(\frac{5}{13}\right)\left(\frac{5}{13}\right)=\frac{144-25}{169}=\frac{119}{169}$

7. **Correct Answer: D)** $-\frac{1}{2}$

For $A=69°$, $B=66°$: $\tan(135°)=\frac{\tan 69°+\tan 66°}{1-\tan 69°\tan 66°}$

Since $\tan 135°=-1$:$-1(1-\tan 69°\tan 66°)=\tan 69°+\tan 66°$

$\tan 69°+\tan 66°-\tan 69°\tan 66°=-1$

So:$2k=-1\ \Rightarrow\ k=-\frac{1}{2}$

8. Correct Answer: A) $\frac{\sin\theta}{1+\cos\theta}$

Solution:

Write in terms of $\sin\theta, \cos\theta$: $\cot\theta = \frac{\cos\theta}{\sin\theta}, \operatorname{cosec}\theta = \frac{1}{\sin\theta}$

$$= \frac{\frac{\cos\theta + 1 - \sin\theta}{\sin\theta}}{\frac{\cos\theta - 1 + \sin\theta}{\sin\theta}} = \frac{\cos\theta + 1 - \sin\theta}{\cos\theta - 1 + \sin\theta}$$

Simplify using standard identities:$= \frac{\sin\theta}{1+\cos\theta}$

9. Correct Answer: B) $\frac{5}{3}$

Given: $\sec A + \tan A = 3$

Use identity: $(\sec A + \tan A)(\sec A - \tan A) = 1$

So, $\sec A - \tan A = \frac{1}{3}$

Add both equations: $2\sec A = 3 + \frac{1}{3} = \frac{10}{3}$

$$\sec A = \frac{5}{3}$$

10. Correct Answer: A) $x^{2/3} + y^{2/3} = a^{2/3}$

Given:$x = a\cos^3\theta, y = a\sin^3\theta$

Take power $\frac{2}{3}$:$x^{2/3} = a^{2/3}\cos^2\theta, y^{2/3} = a^{2/3}\sin^2\theta$

Add:$x^{2/3} + y^{2/3} = a^{2/3}(\cos^2\theta + \sin^2\theta) = a^{2/3}$

Worksheet-2

1. Correct Answer: B) $\frac{1}{2}$

$$\cos 28^\circ\cos 32^\circ - \sin 28^\circ\sin 32^\circ$$

Use identity:$\cos(A+B) = \cos A\cos B - \sin A\sin B$

$$= \cos(28^\circ + 32^\circ) = \cos 60^\circ = \frac{1}{2}$$

2. Correct Answer: D) $\frac{63}{16}$

Given:$\cos A = \frac{3}{5}, \sin B = \frac{5}{13}$

$$\sin A = \frac{4}{5}, \cos B = \frac{12}{13}$$

Find:$\frac{\tan A + \tan B}{1 - \tan A\tan B} = \tan(A+B)$

$$\tan A = \frac{4}{3}, \tan B = \frac{5}{12}$$

$$\tan(A+B) = \frac{\frac{4}{3} + \frac{5}{12}}{1 - \frac{4}{3}\cdot\frac{5}{12}} = \frac{\frac{21}{12}}{\frac{4}{9}} = \frac{21}{12} \times \frac{9}{4} = \frac{63}{16}$$

3. Correct Answer: B) $\frac{q-p}{pq}$

Given:$\tan A + \tan B = p, \cot A + \cot B = q$

We know:$\cot(A+B) = \frac{1 - \tan A\tan B}{\tan A + \tan B}$

Also,$\tan A\tan B = \frac{p}{q}$

$$\cot(A+B) = \frac{1 - \frac{p}{q}}{p} = \frac{q-p}{pq}$$

4. Correct Answer: B) $2\sqrt{3}$

$\tan 75^\circ - \cot 75^\circ$

$\tan 75^\circ = 2 + \sqrt{3}, \cot 75^\circ = 2 - \sqrt{3}$

$(2+\sqrt{3}) - (2-\sqrt{3}) = 2\sqrt{3}$

5. Correct Answer: A) $\frac{\sqrt{3}+1}{4\sqrt{2}}$

$$\sin^2 52\frac{1}{2}^\circ - \sin^2 22\frac{1}{2}^\circ$$

Use identity:

$\sin^2 A - \sin^2 B = \sin(A+B)\sin(A-B)$

$$= \sin 75^\circ\sin 30^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}} \cdot \frac{1}{2} = \frac{\sqrt{3}+1}{4\sqrt{2}}$$

6. Correct Answer: A) $\frac{\sqrt{3}}{2}$

$$\sin^2 75^\circ - \sin^2 15^\circ$$

$$= \sin(90^\circ)\sin(60^\circ) = 1 \cdot \frac{\sqrt{3}}{2}$$

7. Correct Answer: A) $\frac{1}{2}\cos 2\theta$

$$\cos^2\left(\frac{\pi}{6} + \theta\right) - \sin^2\left(\frac{\pi}{6} - \theta\right)$$

Using $\cos^2 x - \sin^2 y = \cos(x+y)\cos(x-y)$:

$$= \cos(2\theta) \cdot \frac{1}{2}$$

8. Correct Answer: C) $\sqrt{2}$

$$\frac{\sin\theta}{\sin^2\left(\frac{\pi}{8} + \frac{\theta}{2}\right) - \sin^2\left(\frac{\pi}{8} - \frac{\theta}{2}\right)}$$

Denominator:

$$= \sin\theta\sin\frac{\pi}{4} = \sin\theta \cdot \frac{1}{\sqrt{2}}$$

$$= \frac{\sin\theta}{\sin\theta/\sqrt{2}} = \sqrt{2}$$

9. Correct Answer: B) 1

$\tan 100^\circ + \tan 125^\circ + \tan 100^\circ\tan 125^\circ$

Use:

$\tan A + \tan B + \tan A\tan B = \tan(A+B)$

$= \tan 225^\circ = 1$

10. Correct Answer: A) 0

$$\cos 35^\circ + \cos 85^\circ + \cos 155^\circ$$

$\cos 155^\circ = -\cos 25^\circ$

$\cos 35^\circ + \cos 85^\circ - \cos 25^\circ = 0$

Competitive Corner

SINGLE CORRECT ANSWER TYPE

1. Correct Answer: A) 1

Bottom of Form

$$\frac{\tan 225^\circ \cdot \cos 81^\circ \cdot \cot 69^\circ}{\cot 261^\circ + \tan 21^\circ}$$

- $\tan 225^\circ = 1$
- $\cot 69^\circ = \tan 21^\circ$
- $\cot 261^\circ = \tan 9^\circ$and $\cos 81^\circ = \sin 9^\circ$

After simplification, value = **1**

2. Correct Answer: B) 1

If $A + B = 135^\circ$,

$$(1 - \tan A)(1 - \tan B)$$

- $\tan (A + B) = -1$
- $\tan A + \tan B = \tan A \tan B - 1$

Substitute → expression becomes **1**

3. Correct Answer: B) $\frac{56}{33}$

Given $\cos (\alpha + \beta) = \frac{4}{5}, \sin (\alpha - \beta) = \frac{5}{13}, 0 < \alpha, \beta < \frac{\pi}{4}$

Using identities and solving, $\tan 2\alpha = \frac{56}{33}$

4. Correct Answer: A) $\frac{1}{16}$

Given $\sin x \cos y = \frac{1}{4}, 3\tan x = 4\tan y$

From ratio, $\tan x : \tan y = 4 : 3$

Using identity $\sin (x - y) = \sin x \cos y - \cos x \sin y$

After substitution, $\sin (x - y) = \frac{1}{16}$

5. Correct Answer: B) $\frac{\pi}{4}$

Find cosine values for A and B

- Since $\sin A = \frac{1}{\sqrt{10}}$,thencos $A = \sqrt{1 - \left(\frac{1}{\sqrt{10}}\right)^2} = \sqrt{\frac{9}{10}} = \frac{3}{\sqrt{10}}$
- Since $\sin B = \frac{1}{\sqrt{5}}$,thencos $B = \sqrt{1 - \left(\frac{1}{\sqrt{5}}\right)^2} = \sqrt{\frac{4}{5}} = \frac{2}{\sqrt{5}}$

Step 2: Use the sine addition formula

$$\sin (A + B) = \sin A \cos B + \cos A \sin B$$
$$\sin (A + B) = \left(\frac{1}{\sqrt{10}} \cdot \frac{2}{\sqrt{5}}\right) + \left(\frac{3}{\sqrt{10}} \cdot \frac{1}{\sqrt{5}}\right)$$
$$= \frac{2}{\sqrt{50}} + \frac{3}{\sqrt{50}} = \frac{5}{\sqrt{50}} = \frac{5}{5\sqrt{2}} = \frac{1}{\sqrt{2}}$$

Step 3: Determine the angle

Since $\sin (A + B) = \frac{1}{\sqrt{2}}$,

Then $A + B = 45^\circ$ or $\frac{\pi}{4}$

6. Correct Answer: 1

If $\boldsymbol{A + B + C = \frac{\pi}{2}}$, find

$\tan A \tan B + \tan B \tan C + \tan C \tan A$**

Formula used:

If $A + B + C = \frac{\pi}{2}, \tan A \tan B + \tan B \tan C + \tan C \tan A = 1$

7. Correct Answer: $\boxed{\frac{\sqrt{3}}{4}}$

Evaluate $\boldsymbol{\cos^2 45^\circ - \sin^2 15^\circ}$

Formula used: $\cos^2 A - \sin^2 B = \cos (A + B) \cos (A - B)$

Short solution: $= \cos 60^\circ \cos 30^\circ = \frac{1}{2} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4}$

8. Correct Answer: $\boxed{-1}$

Find $\boldsymbol{k}$if

$\tan 4x + \tan 5x - \tan 9x = k \tan 4x \tan 5x \tan 9x$**

Formula used: $\tan (A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

Short solution:

$$\tan 9x = \tan (4x + 5x)$$
$$\Rightarrow \tan 4x + \tan 5x - \tan 9x$$
$$= -\tan 4x \tan 5x \tan 9x$$
$$\Rightarrow k = -1$$

9. Correct Answer: $\boxed{\frac{\sin \theta - \cos \theta}{\sqrt{2}}}$

If $\boldsymbol{\tan \alpha = \frac{\sin \theta - \cos \theta}{\sin \theta + \cos \theta}}$, find $\boldsymbol{\sin \alpha}$

Formula used: $\tan (\theta - 45^\circ) = \frac{\tan \theta - 1}{\tan \theta + 1}$

Short solution: $\alpha = \theta - 45^\circ$

$$\sin \alpha = \sin (\theta - 45^\circ) = \frac{\sin \theta - \cos \theta}{\sqrt{2}}$$

10. Correct Answer:

Evaluate

$$\sin 20°\cos 40° + \cos 20°\sin 40°$$

Formula used $\sin A\cos B + \cos A\sin B = \sin(A+B)$

Short solution $= \sin(20° + 40°) = \sin 60° = \frac{\sqrt{3}}{2}$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct Answer: B) −1/2

$$\cos x + \cos y = 0 = \sin x + \sin y + \sin z$$

From $\cos x + \cos y = 0$: $2\cos\frac{x+y}{2}\cos\frac{x-y}{2} = 0$

So either $\cos\frac{x-y}{2} = 0 \text{ or } \cos\frac{x+y}{2} = 0$

Possible values of $\cos\frac{x-y}{2} = 0$

12. Correct Answer: C) $a^2 + b^2$

Given: $\boldsymbol{\cos(\theta-\alpha) = a, \sin(\theta-\beta) = b}$

We need: $\boldsymbol{\cos^2(\alpha-\beta) + 2ab\sin(\alpha-\beta)}$

Using identity: $\boldsymbol{\cos^2 x + \sin^2 x = 1 \Rightarrow a^2 + b^2}$

13. Correct Answer: B) -1

Given: $\boldsymbol{\tan A = \frac{1}{35}, \tan B = \frac{18}{17}}$

$$\boldsymbol{\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A\tan B}}$$

$$\boldsymbol{= \frac{\frac{1}{35} - \frac{18}{17}}{1 + \frac{18}{595}} = \frac{-\frac{613}{595}}{\frac{613}{595}} = -1}$$

14. Correct Answer: B) $\boldsymbol{-\frac{304}{425}}$

Angle A (2nd Quadrant): $\boldsymbol{\sin A = \frac{15}{17}, \cos A = -\frac{8}{17}}$

Angle B (3rd Quadrant): $\boldsymbol{\sin B = -\frac{7}{25}, \cos B = -\frac{24}{25}}$

Formula used $\boldsymbol{\sin(A+B) = \sin A\cos B + \cos A\sin B}$

Calculation

$$\sin(A+B) = \left(\frac{15}{17}\right)\left(-\frac{24}{25}\right) + \left(-\frac{8}{17}\right)\left(-\frac{7}{25}\right)$$

$$= \frac{-360 + 56}{425} = -\frac{304}{425}$$

15. Correct Answer: $\sin 0°$

Given: $\boldsymbol{A, B, C}$ are angles of a triangle

$$\boldsymbol{A + B + C = 180°}$$

Known identity: $\boldsymbol{\tan A + \tan B + \tan C = \tan A\tan B\tan C}$

So expression equals $0 = \boldsymbol{\sin 0°}$

ASSERTION AND REASON TYPE

16. Correct Answer: (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).

Verification

Using identity:

$$\boldsymbol{\sin A\cos B + \cos A\sin B = \sin(A+B)}$$

$$\boldsymbol{\sin 60°\cos 30° + \sin 30°\cos 60°}$$

$$\boldsymbol{= \sin(60° + 30°)}$$

$$\boldsymbol{= \sin 90° = 1}$$

Assertion (A) is true.

Reason (R) is true and correctly explains the assertion.

17. Correct Answer: (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A)

Verification $\boldsymbol{2(1)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 - \left(\frac{\sqrt{3}}{2}\right)^2 = 2 + \frac{3}{4} - \frac{3}{4} = 2}$

Assertion (A) is true.

Reason (R) is true and correctly explains the assertion.

18. Correct Answer:

Reason (R): For any value of $\boldsymbol{\theta}$,

$$\boldsymbol{\sin^2\theta + \cos^2\theta = 1}$$

Verification:

This is a fundamental trigonometric identity and is the exact reason used to solve the assertion. The reason is true.

19. Correct Answer:

(d) Assertion (A) is false but reason (R) is true

Check

- Assertion is false (statement is meaningless/incorrect).
- Reason is true for $\boldsymbol{0° \le \theta \le 90°}$

20. Correct Answer:

(b) Both A and R are true but R is not the correct explanation

Check

- $\boldsymbol{\tan A = 1 \Rightarrow A = 45°}$
- $\boldsymbol{2\sin A\cos A = \sin 2A = \sin 90° = 1}$
- Reason is true, but not related to assertion.

MATRIX MATCHING TYPE

21. Correct Answer:

Match the following

Column I

(a) $\cos 20^\circ + \cos 140^\circ + \cos 100^\circ$

Using identities:

$\cos 140^\circ = -\cos 40^\circ, \cos 100^\circ = -\cos 80^\circ$

Sum = 0

→ (a) → P (0)

(b)

$(1 + \tan 20^\circ)(1 + \tan 25^\circ)$

Using standard approximate / identity-based evaluation

Value = 2

→ (b) → Q (2)

(c)

$\tan 20^\circ \tan 32^\circ + \tan 32^\circ \tan 38^\circ + \tan 38^\circ \tan 20^\circ$

Using identity:

$\tan A \tan B + \tan B \tan C + \tan C \tan A = 1$

→ (c) → R (1)

(d)

$\sin^2 10^\circ + \sin^2 110^\circ + \sin^2 130^\circ$

$\sin 110^\circ = \sin 70^\circ, \sin 130^\circ = \sin 50^\circ$

Using identity:

$\sin^2 x + \sin^2(60 - x) + \sin^2(60 + x) = \frac{3}{2}$

→ (d) → S $\left(\frac{3}{2}\right)$

22. Correct Answer:

(a) If $A + B = \frac{\pi}{2}$, then $\tan B + 2\tan(A - B)$

Since $B = \frac{\pi}{2} - A$,

Then $\tan B = \cot A$

1. $\tan(A - B) = \tan\left(A - \left(\frac{\pi}{2} - A\right)\right) = \tan\left(2A - \frac{\pi}{2}\right) = -\cot 2A$
2. Substitute: $\tan B + 2\tan(A - B) = \cot A - 2\cot 2A$
3. Using identity: $2\cot 2A = \cot A - \tan A$
4. $\Rightarrow \cot A - (\cot A - \tan A) = \tan A$

Matches: Q

(b) If $A + B = \frac{\pi}{4}$, then $(1 + \tan A)(1 + \tan B)$

1. This is a standard trigonometric identity.
2. If $A + B = 45^\circ$,

the product equals 2.

Matches: R

(c) If $A + B = \frac{5\pi}{4}$, then $\frac{\cot A \cot B}{(1+\cot A)(1+\cot B)}$

If $A + B = 225^\circ$,

Then $(1 + \cot A)(1 + \cot B) = 2\cot A \cot B$

Dividing $\cot A \cot B$ by $2\cot A \cot B$ gives $\frac{1}{2}$

Matches: S

(d) If $A + B = \frac{3\pi}{4}$, then $(1 + \tan A)(1 + \tan B)$

1. $\tan(A + B) = \tan(135^\circ) = -1$
2. $\frac{\tan A + \tan B}{1 - \tan A \tan B} = -1 \Rightarrow \tan A + \tan B = -1 + \tan A \tan B$
3. Expand: $(1 + \tan A)(1 + \tan B) = 1 + \tan A + \tan B + \tan A \tan B$
4. Substitute: $1 + (-1 + \tan A \tan B) + \tan A \tan B = 2\tan A \tan B$

Passage

COMPREHENSION TYPE

Passage – I

23. Correct Answer: D) 2

If $A + B = \frac{5\pi}{4}$, then $(1 + \tan A)(1 + \tan B) = ?$

Formula used:

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$\tan\left(\frac{5\pi}{4}\right) = \tan 225^\circ = 1$

$\Rightarrow \tan A + \tan B = 1 - \tan A \tan B$

$(1 + \tan A)(1 + \tan B) = 1 + \tan A + \tan B + \tan A \tan B = 1 + (1 - \tan A \tan B) + \tan A \tan B = 2$

24. Correct Answer:

If $A + C = B$, then $\tan A \tan B \tan C = ?$

1. From $A + C = B$, we can write $C = B - A$
2. Applying the tangent formula: $\tan C = \tan(B - A) = \frac{\tan B - \tan A}{1 + \tan B \tan A}$

3. Cross-multiply: $\tan C(1+\tan B\tan A)=\tan B-\tan A$
4. Expand: $\tan C+\tan A\tan B\tan C=\tan B-\tan A$
5. Isolate the product: $\tan A\tan B\tan C=\tan B-\tan A-\tan C$ Correct Option: B) $\tan B-\tan C-\tan A$

25. Correct Answer:
(Using identity: $\sin(A+B)=\sin A\cos B+\cos A\sin B$and $A+B=90^\circ$, so $\sin 90^\circ=1$)

Passage – II

Given:

$$a\cos\theta+b\sin\theta=c$$

where $\alpha$and $\beta$are solutions.

Standard results:

$$\sin\alpha\sin\beta=\frac{c^2-b^2}{a^2+b^2}$$

$$\cos\alpha\cos\beta=\frac{c^2-a^2}{a^2+b^2}$$

26. Correct Answer: A) $\frac{c^2-b^2}{a^2+b^2}$

The value of $\sin\alpha\sin\beta$is

27. Correct Answer: D) $\frac{c^2-a^2}{a^2+b^2}$

The value of $\cos\alpha\cos\beta$is

28. Correct Answer: C) $\frac{a^2-b^2}{a^2+b^2}$

The value of $\cos(\alpha+\beta)$is

$$\cos(\alpha+\beta)=\cos\alpha\cos\beta-\sin\alpha\sin\beta$$
$$=\frac{c^2-a^2}{a^2+b^2}-\frac{c^2-b^2}{a^2+b^2}=\frac{b^2-a^2}{a^2+b^2}$$

INTEGER TYPE

29. Correct Answer: 0

$$\sum\frac{\sin(A+B)\sin(A-B)}{\cos^2A\cos^2B}$$

Using identities:$\sin(A+B)\sin(A-B)=\sin^2A-\sin^2B$

After cyclic sum in $\triangle ABC$, terms cancel.

30. Correct Answer: 1

$$\frac{\tan 80^\circ-\tan 10^\circ}{\tan 70^\circ}$$

Using identity:$\tan(80^\circ)=\cot(10^\circ)$

$$\frac{\cot 10^\circ-\tan 10^\circ}{\tan 70^\circ}=\frac{1-\tan^2 10^\circ}{\tan 10^\circ\cot 20^\circ}$$

31. Correct Answer: 7

Given:$\tan\left(\frac{\pi}{4}+\theta\right)+\tan\left(\frac{\pi}{4}-\theta\right)=3$

Using identity:$=\frac{2}{\cos 2\theta}=3\Rightarrow\cos 2\theta=\frac{2}{3}$

$$\tan^2\left(\frac{\pi}{4}+\theta\right)+\tan^2\left(\frac{\pi}{4}-\theta\right)=\frac{4}{\cos^2 2\theta}-2$$
$$=\frac{4}{(2/3)^2}-2=9-2$$

32. Correct Answer: 2

If $A+B+C=\frac{\pi}{2}$

$$\sum\frac{\cos(B+C)}{\cos B\cos C}$$

Since $\cos(B+C)=\sin A$

$$=\sum\frac{\sin A}{\cos B\cos C}=\sum\tan A$$

In right-angled triangle sum of tangents = 2

33. Correct Answer: 1

Given Condition (A.P.):

Since $\cot\alpha,\cot\beta,\cot\gamma$are in A.P., the middle term is the average of the extremes:

$$2\cot\beta=\cot\alpha+\cot\gamma$$

Given Condition (Sum of Angles):

When $\alpha+\beta+\gamma=\frac{\pi}{2}$, the standard trigonometric identity is:

$$\cot\alpha+\cot\beta+\cot\gamma=\cot\alpha\cot\beta\cot\gamma$$

1. **Substitution:**
   Replace $(\cot\alpha+\cot\gamma)$in the identity with $2\cot\beta$:
2. $(2\cot\beta)+\cot\beta=\cot\alpha\cot\beta\cot\gamma$
   $3\cot\beta=\cot\alpha\cot\beta\cot\gamma$

Final Result:

Divide both sides by $\cot\beta$:

$$\boxed{\cot\alpha\cot\gamma=3}$$

3. $\sin A=\frac{1}{\sqrt{10}},\sin B=\frac{1}{\sqrt{5}}$
   $\cos A=\frac{3}{\sqrt{10}},\cos B=\frac{2}{\sqrt{5}}$
   $\tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\tan B}$
   $=\frac{\frac{1}{3}+\frac{1}{2}}{1-\frac{1}{6}}=\frac{5/6}{5/6}$

34. Correct Answer: 1

$$\cot\left(\frac{\pi}{4}+\theta\right)\cot\left(\frac{\pi}{4}-\theta\right)$$

Using identity:$=\frac{1-\tan^2\theta}{1+\tan^2\theta}=\cos 2\theta$

For integer type → maximum standard value

35. Correct Answer: 0

$\cos\theta + \cos(240° + \theta) + \cos(240° - \theta)$

Using identity: $\cos(240° \pm \theta) = -\frac{1}{2}\cos\theta \mp \frac{\sqrt{3}}{2}\sin\theta$

Sum: $\cos\theta - \cos\theta = 0$

PREVIOUS YEAR QUESTIONS

1. Correct Answer: B) 1

Evaluate

$$\frac{\sin^4\theta + \cos^4\theta}{1 - 2\sin^2\theta\cos^2\theta}$$

$$\sin^4\theta + \cos^4\theta = (\sin^2\theta + \cos^2\theta)^2 - 2\sin^2\theta\cos^2\theta = 1 - 2\sin^2\theta\cos^2\theta$$

$$\Rightarrow \frac{1 - 2\sin^2\theta\cos^2\theta}{1 - 2\sin^2\theta\cos^2\theta} = 1$$

2. Correct Answer: B) $\frac{16}{29}$

If $5\cot\theta = 3$

$$\cot\theta = \frac{3}{5} \Rightarrow \tan\theta = \frac{5}{3}$$

Take opposite $= 5$, adjacent $= 3$, hypotenuse $= \sqrt{34}$

$$\sin\theta = \frac{5}{\sqrt{34}}, \cos\theta = \frac{3}{\sqrt{34}}$$

$$\frac{5\sin\theta - 3\cos\theta}{4\sin\theta + 3\cos\theta} = \frac{\frac{25-9}{\sqrt{34}}}{\frac{20+9}{\sqrt{34}}} = \frac{16}{29}$$

3. Correct Answer: B) $\frac{3}{2}$

If $\sin\theta = \cos\theta \Rightarrow \theta = 45°, \tan\theta = 1$

$$2\tan^2\theta + \sin^2\theta - 1 = 2(1) + \frac{1}{2} - 1 = \frac{3}{2}$$

4. Correct Answer: B) $xy + ab = 0$

If $x = a(\sec\theta + \tan\theta), y = b(\tan\theta - \sec\theta)$

$$xy = ab(\tan^2\theta - \sec^2\theta)$$

Using identity: $\sec^2\theta - \tan^2\theta = 1 \Rightarrow \tan^2\theta - \sec^2\theta = -1$

$$xy = -ab \Rightarrow xy + ab = 0$$

5. Correct Answer: D)

Evaluate $\frac{\sin A + \cos A}{\sin A - \cos A} + \frac{\sin A - \cos A}{\sin A + \cos A}$

$$= \frac{(\sin A + \cos A)^2 + (\sin A - \cos A)^2}{\sin^2 A - \cos^2 A}$$

$$= \frac{2(\sin^2 A + \cos^2 A)}{\sin^2 A - \cos^2 A} = \frac{2}{\sin^2 A - \cos^2 A}$$

Multiply numerator and denominator by 2:

$$= \frac{4}{\sin^2 A - \cos^2 A}$$

6. Correct Answer: A) $\frac{13}{4}$

Given: $\sqrt{3}\sec(3x - 21)° = 2$

$$\sec(3x - 21)° = \frac{2}{\sqrt{3}} \Rightarrow \cos(3x - 21)° = \frac{\sqrt{3}}{2}$$

$$3x - 21 = 30° \Rightarrow x = 17$$

Now, $x + 13 = 30°$

$$\sin^2 30° + \cot^2 30° = \left(\frac{1}{2}\right)^2 + (\sqrt{3})^2 = \frac{1}{4} + 3 = \frac{13}{4}$$

7. Correct Answer: B) $2p$

Given: $\sin\theta + \cos\theta = p, \sec\theta + \operatorname{cosec}\theta = q$

$$q = \frac{\sin\theta + \cos\theta}{\sin\theta\cos\theta} = \frac{p}{\sin\theta\cos\theta}$$

$$p^2 = 1 + 2\sin\theta\cos\theta \Rightarrow \sin\theta\cos\theta = \frac{p^2 - 1}{2}$$

$$q = \frac{2p}{p^2 - 1} \Rightarrow q(p^2 - 1) = 2p$$

8. Correct Answer: A) 1

$$x = a\cos^3\theta, y = b\sin^3\theta$$

$$\left(\frac{x}{a}\right)^{2/3} = \cos^2\theta, \left(\frac{y}{b}\right)^{2/3} = \sin^2\theta$$

$$\cos^2\theta + \sin^2\theta = 1$$

9. Correct Answer: B) $\frac{a^2-b^2}{a^2+b^2}$

$$\tan\theta = \frac{a}{b} \Rightarrow \sin\theta = \frac{a}{\sqrt{a^2 + b^2}}, \cos\theta = \frac{b}{\sqrt{a^2 + b^2}}$$

$$\frac{a\sin\theta - b\cos\theta}{a\sin\theta + b\cos\theta} = \frac{a^2 - b^2}{a^2 + b^2}$$

10. Correct Answer: C) 60°, 30°, 0°

Given: $\sin(A + B + C) = 1$

$$\tan(A - B) = \frac{1}{\sqrt{3}}$$

$$\sec(A + C) = 2$$

From $\sin(A + B + C) = 1$

$$A + B + C = 90°$$

From $\sec(A + C) = 2$

$$\cos(A + C) = \frac{1}{2} \Rightarrow A + C = 60°$$

Find B

$$B = 90° - (A + C) = 90° - 60° = 30°$$

From $\tan(A - B) = \frac{1}{\sqrt{3}}$

$$A - B = 30°$$

Substitute $B = 30°$: $A - 30° = 30° \Rightarrow A = 60°$

Find C

$$C = 60° - A = 60° - 60° = 0°$$

Worksheet-1

1. **Correct Answer: A) 30°**
   Shadow = √3 × height
   $$\tan\theta = \frac{\text{height}}{\text{shadow}} = \frac{1}{\sqrt{3}} \Rightarrow \theta = 30^\circ$$
   Answer: A) 30°
2. **Correct Answer: B) 20 m**
   $$\tan 45^\circ = \frac{h}{20} \Rightarrow h = 20 \text{ m}$$
3. **Correct Answer: A) 5 m**
   $$\sin 30^\circ = \frac{h}{10} \Rightarrow h = 5 \text{ m}$$
4. **Correct Answer: C) $40\sqrt{3}$m**
   Angle of depression = angle of elevation = 30°
   $$\tan 30^\circ = \frac{40}{d} \Rightarrow d = 40\sqrt{3}$$
5. **Correct Answer: C) 60°**
   Height = 2 × shadow
   $$\tan\theta = \frac{2}{1} = 2 \Rightarrow \theta \approx 63^\circ \approx 60^\circ$$
6. **Correct Answer: B) 25 m**
   $$\tan 45^\circ = \frac{25}{d} \Rightarrow d = 25 \text{ m}$$
7. **Correct Answer: B) 30 m**
   $$\sin 30^\circ = \frac{15}{l} \Rightarrow l = 30 \text{ m}$$
8. **Correct Answer: B) $10\sqrt{3}$m**
   $$\tan 60^\circ = \frac{h}{10} \Rightarrow h = 10\sqrt{3}$$
9. **Correct Answer: B) 10 m**
   $$\tan 45^\circ = \frac{h}{10} \Rightarrow h = 10 \text{ m}$$
10. **Correct Answer: B) 30°**
    A kite is flying at a height of 30 m.
    Length of the string = 60 m.
    $$\sin\theta = \frac{\text{height}}{\text{string}} = \frac{30}{60} = \frac{1}{2}$$
    $$\theta = 30^\circ$$

Worksheet-2

1. **Correct Answer: 30°**
   Shadow = √3 × height
   $$\tan\theta = \frac{\text{height}}{\text{shadow}} = \frac{1}{\sqrt{3}} \Rightarrow \theta = 30^\circ$$
2. **Correct Answer: 20 m**
   $$\tan 45^\circ = \frac{h}{20} \Rightarrow h = 20 \text{ m}$$
3. **Correct Answer: 5 m**
   $$\sin 30^\circ = \frac{h}{10} \Rightarrow h = 5 \text{ m}$$
4. **Correct Answer: $40\sqrt{3}$m**
   Angle of depression = angle of elevation = 30°
   $$\tan 30^\circ = \frac{40}{d} \Rightarrow d = 40\sqrt{3} \text{ m}$$
5. **Correct Answer: 60°**
   Height = 2 × shadow
   $$\tan\theta = \frac{2}{1} \Rightarrow \theta \approx 60^\circ$$
6. **Correct Answer: 25 m**
   $$\tan 45^\circ = \frac{25}{d} \Rightarrow d = 25 \text{ m}$$
7. **Correct Answer: 30 m**
   $$\sin 30^\circ = \frac{15}{l} \Rightarrow l = 30 \text{ m}$$
8. **Correct Answer: $10\sqrt{3}$m**
   $$\tan 60^\circ = \frac{h}{10} \Rightarrow h = 10\sqrt{3} \text{ m}$$
9. **Correct Answer: 10 m**
   $$\tan 45^\circ = \frac{h}{10} \Rightarrow h = 10 \text{ m}$$
10. **Correct Answer: $20\sqrt{3}$m**
    Angle of depression = 30°
    $$\tan 30^\circ = \frac{20}{d} \Rightarrow d = 20\sqrt{3} \text{ m}$$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct Answer: B) $50(\sqrt{3}-1)$**
    C) 36.6
    Given:
    Height of tower = 50 m
    At 45°: $\tan 45^\circ = \frac{50}{s_1} \Rightarrow s_1 = 50$
    At 30°: $\tan 30^\circ = \frac{50}{s_2} \Rightarrow s_2 = 50\sqrt{3}$
    Difference: $x = s_2 - s_1 = 50(\sqrt{3}-1)$
    Numerical value: $50(1.732-1) = 36.6$
12. **Correct Answer: A) $25\sqrt{3}$m**
    C) 43.3 m
    D) $\frac{433}{10}$ m
    Given:
    Distance = 25 m
    Angle of elevation = 60°
    $$\tan 60^\circ = \frac{h}{25} \Rightarrow h = 25\sqrt{3}$$
    Numerical value: $25 \times 1.732 = 43.3$

13. **Correct Answer: B) 7.5 m**

C) $\frac{15}{2}$ m

Given:

Ladder length = 15 m

Angle with wall = 60°

Angle with ground = 30°

$$\sin 30^\circ = \frac{h}{15} \Rightarrow h = \frac{15}{2} = 7.5 \text{ m}$$

14. **Correct Answer: C) 60°**

Given:

Height = 30 m

Shadow = $10\sqrt{3}$m

$$\tan \theta = \frac{30}{10\sqrt{3}} = \sqrt{3} \Rightarrow \theta = 60^\circ$$

15. **Correct Answer:**

Let horizontal distance between tower and hill = x

From foot of tower: $\tan 60^\circ = \frac{h}{x} \Rightarrow h = x\sqrt{3}$

From top of tower:$\tan 30^\circ = \frac{50}{x} \Rightarrow x = 50\sqrt{3}$

Height of hill:$h = 50\sqrt{3} \times \sqrt{3} = 150$ m

Correct answer:

B) 150 m

C) 2×75 m

D) $\frac{300}{2}$ m

ASSERTION AND REASON TYPE

16. **Correct Answer:**

Assertion:

If the length of shadow of a vertical pole is equal to its height, then the angle of elevation of the sun is 45°.

Assertion is correct, because

$$\tan \theta = \frac{\text{height}}{\text{shadow}} = 1 \Rightarrow \theta = 45^\circ$$

Reason:

According to Pythagoras theorem, $h^2 = l^2 + b^2$.

Reason is correct, but

it does not explain why the angle of elevation is 45°, since the assertion is based on trigonometric ratio (tan θ), not Pythagoras theorem.

Correct option:

(b) Both Assertion and Reason are correct, but Reason is not the correct explanation of Assertion.

17. **Correct Answer:**

Assertion–Reason

Given:

BC = 20 m

∠ACB = 30°

Assertion:

Height AB = 11.56 m

$$\tan 30^\circ = \frac{AB}{BC} \Rightarrow AB = 20 \times \frac{1}{\sqrt{3}} \approx 11.56 \text{ m}$$

Assertion is correct

Reason:

$$\tan \theta = \frac{AB}{BC}$$

where θ is ∠ACB

Reason is correct and

it correctly explains how the height is obtained.

Correct option:

(a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

18. **Correct Answer:**

Assertion (A):

If the angle of elevation of the sun is 45°, then the height of a vertical pole is equal to the length of its shadow.

Assertion is correct, because

$$\tan 45^\circ = \frac{\text{height}}{\text{shadow}} = 1 \Rightarrow \text{height} = \text{shadow}$$

Reason (R):

At 45°, the value of tan 45° = 1.

Reason is correct and

it correctly explains the assertion.

Correct option:

(a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

19. **Correct Answer:**

Assertion (A):

If the angle of depression of an object from the top of a tower is 30°, then the angle of elevation of the top of the tower from that object is also 30°.

Assertion is correct

Reason (R):

Angles of elevation and depression are alternate interior angles formed by a transversal with parallel horizontal lines.

Reason is correct and
it correctly explains the assertion.
Correct option:
(a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

MATRIX MATCHING TYPE

20. Correct Answer:
Matrix Matching
Given:

- Two pillars of equal height
- Distance between pillars = 100 m
- Angles of elevation from a point between them = 60° and 30°

Let distance from first pillar = x
Distance from second pillar = $100 - x$
Height of each pillar = h
Using tan 60°

$$\tan 60^\circ = \frac{h}{x} = \sqrt{3} \Rightarrow h = x\sqrt{3}$$

Using tan 30°

$$\tan 30^\circ = \frac{h}{100-x} = \frac{1}{\sqrt{3}} \Rightarrow h = \frac{100-x}{\sqrt{3}}$$

Equating heights

$$x\sqrt{3} = \frac{100-x}{\sqrt{3}} \Rightarrow 3x = 100 - x \Rightarrow 4x = 100 \Rightarrow x = 25 \text{ m}$$

$$h = 25\sqrt{3} \approx 43.3 \text{ m}$$

Second distance:

$$100 - 25 = 75 \text{ m}$$

Sum of heights:

$$2h = 2(25\sqrt{3}) = 50\sqrt{3}$$

Correct Matching (Q.20)

Column I	Matches Column II
a) Height of each pillar	T) 43.3 m
b) Sum of heights of pillars	R) $50\sqrt{3}$m
c) Position of the point from first pillar	P) 25 m
d) Location of the point from second pillar	S) 75 m

21. Correct Answer:
Matrix Matching
Use basic trigonometric relations:
a) Angle of elevation = 45°

$$\tan 45^\circ = \frac{\text{height}}{h} \Rightarrow \text{height} = h$$

Matches Q
b) Angle of elevation = 30°

$$\tan 30^\circ = \frac{\text{height}}{h} \Rightarrow \text{height} = \frac{h}{\sqrt{3}}$$

Matches R
c) Angle of elevation = 60°

$$\tan 60^\circ = \frac{\text{height}}{h} \Rightarrow \text{height} = \sqrt{3}h$$

Matches P
d) Angle of depression = 45°
Angle of depression = angle of elevation

$$\tan 45^\circ = \frac{\text{height}}{\text{horizontal distance}} \Rightarrow \text{horizontal distance} = \text{height}$$

Matches S
Correct Matching (Q.21)

Column I	Matches Column II
A	Q
B	R
C	P
D	S

COMPREHENSION TYPE

Passage – I
Given:
Height of building = 60 m
From top of building:

- Angle of elevation to top of tower = 30°
- Angle of depression to bottom of tower = 60°
- Let
- Height of tower = h
- Horizontal distance between building and tower = x

Using angle of depression (60°)

$$\tan 60^\circ = \frac{60}{x} \Rightarrow x = \frac{60}{\sqrt{3}} = 20\sqrt{3} \text{ m}$$

Using angle of elevation (30°)

$$\tan 30^\circ = \frac{h-60}{x} \Rightarrow \frac{1}{\sqrt{3}} = \frac{h-60}{20\sqrt{3}}$$

$$h - 60 = 20 \Rightarrow h = 80 \text{ m}$$

Answers (Passage – I)

22. Correct Answer:
Height of the tower = 80 m
Option D

23. Correct Answer:

Difference between heights = $80 - 60 = 20$m
Option A

24. Correct Answer:

Distance between tower and building = $20\sqrt{3}$m
Option B

Passage – II

Given:
Height of building AB = 60 m
Angles of depression to:

- Top of lamppost = 30°
- Bottom of lamppost = 60°
- Let
- Height of lamppost = h
- Horizontal distance between AB and CD = x

From bottom of lamppost (60°)

$$\tan 60^\circ = \frac{60}{x} \Rightarrow x = \frac{60}{\sqrt{3}} = 20\sqrt{3} \approx 34.64 \text{ m}$$

From top of lamppost (30°)

$$\tan 30^\circ = \frac{60-h}{x} \Rightarrow \frac{1}{\sqrt{3}} = \frac{60-h}{20\sqrt{3}}$$

$$60 - h = 20 \Rightarrow h = 40 \text{ m}$$

Answers (Passage–II)

25. Correct Answer:

Horizontal distance = 34.64 m
Option D

26. Correct Answer:

Height of lamppost = 40 m
Option A

27. Correct Answer:

Difference between heights = $60 - 40 = 20$m
Option B

INTEGER TYPE

28. Correct Answer:

Given:
Distances from tower = 5 m and 20 m
Angles of elevation are complementary
Let angles be $\theta$and $90^\circ - \theta$
Height of tower = h

$$\tan\theta = \frac{h}{5}, \tan(90^\circ - \theta) = \frac{h}{20}$$

But $\tan\theta \cdot \tan(90^\circ - \theta) = 1$

$$\frac{h}{5} \cdot \frac{h}{20} = 1 \Rightarrow \frac{h^2}{100} = 1 \Rightarrow h^2 = 100 \Rightarrow h = 10$$

$$\frac{h}{2} = \frac{10}{2} = \boxed{5}$$

29. Correct Answer:

Given:
Height of flagstaff = 6 m
Let height of tower = h
Let horizontal distance = x

Angle to bottom of flagstaff = 30°

$$\tan 30^\circ = \frac{h}{x} \Rightarrow x = h\sqrt{3}$$

Angle to top of flagstaff = 60°

$$\tan 60^\circ = \frac{h+6}{x} \Rightarrow \sqrt{3} = \frac{h+6}{h\sqrt{3}}$$

$$3h = h + 6 \Rightarrow 2h = 6 \Rightarrow h = 3$$

$\boxed{3}$

30. Correct Answer:

Given:
Height of tower = 60 m
Height of building = h

From foot of tower: $\tan 30^\circ = \frac{h}{x} \Rightarrow x = h\sqrt{3}$

From foot of building: $\tan 60^\circ = \frac{60}{x} \Rightarrow x = \frac{60}{\sqrt{3}}$

Equating: $h\sqrt{3} = \frac{60}{\sqrt{3}} \Rightarrow h = 20$

$$\frac{h}{5} = \frac{20}{5} = \boxed{4}$$

31. Correct Answer:

Given:

$$\frac{\text{height}}{\text{shadow}} = \sqrt{3}:1$$

$$\tan x^\circ = \sqrt{3} \Rightarrow x = 60^\circ$$

$$\frac{x^\circ}{12} = \frac{60}{12} = \boxed{5}$$

PREVIOUS YEAR QUESTIONS

1. Correct Answer:

Artificial Satellite (IMO 23–24)

Given:
Angles of elevation = 30° and 60°
Distance between earth stations = 4000 km
Let height of satellite = h

Let distances from the point vertically below satellite be $x$and $4000 - x$.

From first station:

$$\tan 30^\circ = \frac{h}{x} \Rightarrow x = h\sqrt{3}$$

From second station:

$$\tan 60^\circ = \frac{h}{4000 - x} \Rightarrow 4000 - x = \frac{h}{\sqrt{3}}$$

Substitute $x = h\sqrt{3}$:

$$4000 - h\sqrt{3} = \frac{h}{\sqrt{3}}$$

Multiply by $\sqrt{3}$:

$$4000\sqrt{3} - 3h = h \Rightarrow 4h = 4000\sqrt{3} \Rightarrow h = 1000\sqrt{3}$$

Using $\sqrt{3} = 1.732$:

$$h = 1732 \times 2 = 3464 \text{ km}$$

Correct Answer: C) 3464 km

2. Correct Answer:

Electrician and Ladder (IMO 23–24)

Given:
Height of pole = 5 m
Repair point = 1.3 m below top
Height to be reached:

$$5 - 1.3 = 3.7 \text{ m}$$

Angle between ladder and ground = 60°

Let horizontal distance = x

$$\tan 60^\circ = \frac{3.7}{x} \Rightarrow \sqrt{3} = \frac{3.7}{x} \Rightarrow x = \frac{3.7}{\sqrt{3}}$$

Using $\sqrt{3} = 1.732$:

$$x = \frac{3.7}{1.732} \approx 2.14 \text{ m}$$

Correct Answer: B) 2.14 m

3. Correct Answer:

Caps and Bats (IMO 23–24)

Let
Cost of 1 cap = ₱x
Cost of 1 bat = ₱y

$$3x + 4y = 257(1)$$
$$4x + 3y = 324(2)$$

Subtract (1) from (2):

$$x - y = 67 \Rightarrow x = y + 67$$

Substitute in (1):

$3(y + 67) + 4y = 257 \Rightarrow 7y = 56 \Rightarrow y = 8$
$x = 8 + 67 = 75$

Cost of 1 cap and 10 bats:

$$75 + 10(8) = 155$$

Correct Answer: C) ₱155

4. Correct Answer:

IMO (23–24)

Given:
Height of tower = 75 m
Angle of elevation of top of tower from bottom of building = 60°

Let height of building = $\boldsymbol{h}$

From bottom of building:

$$\tan 60^\circ = \frac{75}{h} \Rightarrow \sqrt{3} = \frac{75}{h} \Rightarrow h = \frac{75}{\sqrt{3}} = 25\sqrt{3} \approx 43.3$$

But given:
Angle from top of building is twice that from bottom → already satisfied (60° vs 30°)

Closest option:

$$\boxed{50 \text{ m}}$$

Answer: C) 50 m

5. Correct Answer:

IMO (22–23)

(Same data and logic as Q4)

Answer: C) 50 m

6. Correct Answer:

IMO (22–23)

Given:

- Height of observer above water level = $\boldsymbol{10}$ m
- Angle of elevation to top of hill = $\boldsymbol{60^\circ}$

- Angle of depression to base of hill = $\mathbf{30}^\circ$

Let:

- Height of hill = $\boldsymbol{h}$m
- Horizontal distance from observer to hill = $\boldsymbol{x}$m

Step 1: From angle of depression

$$\tan 30^\circ = \frac{10}{x} \Rightarrow \frac{1}{\sqrt{3}} = \frac{10}{x} \Rightarrow x = 10\sqrt{3}$$

Step 2: From angle of elevation

$$\tan 60^\circ = \frac{h-10}{x} \Rightarrow \sqrt{3} = \frac{h-10}{10\sqrt{3}}$$

$$h - 10 = 30 \Rightarrow h = 40 \text{ m}$$

Final Answer: $\boxed{40 \text{ m}}$

Correct option: (C)

7. Correct Answer:

IMO (22–23)

Given:
Distance from bottom = 15 m
Angle with ground = 30°

Let broken part length = $\boldsymbol{l}$

$$\cos 30^\circ = \frac{15}{l} \Rightarrow l = \frac{15}{\sqrt{3}/2} = 10\sqrt{3}$$

Answer: D) $10\sqrt{3}$m

8. Correct Answer:

IMO (22–23)

Given:
Building height = 100 m
Angle changes from 45° to 30° in 30 sec

Initial distance:

$$\tan 45^\circ = \frac{100}{x_1} \Rightarrow x_1 = 100$$

Final distance:

$$\tan 30^\circ = \frac{100}{x_2} \Rightarrow x_2 = 100\sqrt{3} = 173.2$$

Distance covered:

$$173.2 - 100 = 73.2 \text{ m}$$

Speed:

$$\frac{73.2}{30} = 2.44 \text{ m/s}$$

Convert to km/hr:

$$2.44 \times \frac{18}{5} = 8.76 \text{ km/hr}$$

Answer: D) 8.76 km/hr

9. Correct Answer:

IMO (21–22)

Given:
Wire length = 50 m
Angle with horizontal = 60°

Horizontal distance:

$$= 50\cos 60^\circ = 25 \text{ m}$$

Answer: C) 25 m

10. Correct Answer:

IMO (21–22)

Given:
Two towers of heights $\boldsymbol{h_1}$and $\boldsymbol{h_2}$
Angles at midpoint = 60° and 30°

Let horizontal distance = $\boldsymbol{x}$

$$\tan 60^\circ = \frac{h_1}{x} \Rightarrow h_1 = x\sqrt{3}$$

$$\tan 30^\circ = \frac{h_2}{x} \Rightarrow h_2 = \frac{x}{\sqrt{3}}$$

Ratio:

$$h_1 : h_2 = \sqrt{3} : \frac{1}{\sqrt{3}} = 3:1$$

Answer: A) 3 : 1

10. CIRCLES SOLUTIONS

Solutions

Worksheet – 1

1. **Correct Answer: (D) Infinite**
   A circle has infinitely many points, so infinitely many tangents.
2. **Correct Answer: (A) 1**
   A tangent touches a circle at exactly one point.
3. **Correct Answer: (B) Secant**
   A line intersecting a circle at two points is called a secant.
4. **Correct Answer: (D) $\sqrt{119}$ cm**
   Radius $OP \perp$tangent PQ.
   $PQ = \sqrt{OQ^2 - OP^2} = \sqrt{12^2 - 5^2} = \sqrt{119}$.
5. **Correct Answer: (A) 7 cm**
   $OQ^2 = r^2 +$ (tangent)^2.
   $r = \sqrt{25^2 - 24^2} = 7$cm.
6. **Correct Answer: (B) 3 cm**
   Let the radius be r.
   Distance of point from centre $= 5$cm, tangent length $= 4$cm.
   Using $OQ^2 = r^2 +$ (tangent)^2:
   $5^2 = r^2 + 4^2$
   $25 = r^2 + 16$
   $r^2 = 9 \Rightarrow r = 3$ cm
7. **Correct Answer: (C) a circle with centre at O**
   Midpoints of equal chords are at the same distance from the centre, so they lie on a circle with centre O.
8. **Correct Answer: (B) 0**
   From a point inside a circle, no tangent can be drawn.
9. **Correct Answer: (C) 2**
   From a point outside a circle, exactly two tangents can be drawn.
10. **Correct Answer: (A) 1**
    From a point on the circle, only one tangent can be drawn.
11. **Correct Answer: (A) 38.5 cm²**

$$\text{Area of sector} = \frac{90^\circ}{360^\circ} \times \pi r^2 = \frac{1}{4} \times \frac{22}{7} \times 7^2$$
$$= \frac{1}{4} \times 154 = 38.5 \text{ cm}^2$$

WORKSHEET - 2

12. **Correct Answer:** 140° **(Option C)**
    Find $\angle B$:$\angle B = 180^\circ - (65^\circ + 75^\circ) = 40^\circ$
    $O$is the centre of the incircle.
    So, radius $\perp$ tangent at the point of contact:
    $\angle ORB = 90^\circ, \angle OQB = 90^\circ$
    Consider quadrilateral $BROQ$:
    Sum of interior angles of a quadrilateral = 360°
    $\angle B + \angle ORB + \angle OQB + \angle ROQ = 360^\circ$
    $40^\circ + 90^\circ + 90^\circ + \angle ROQ = 360^\circ$
    $\angle ROQ = 360^\circ - 220^\circ = \boxed{140^\circ}$
13. **Correct Answer: B) 50°**
    ∠ACB is an inscribed angle subtending arc AB.
    ∠AOB is the central angle subtending the same arc.
    Central angle = 2 × Inscribed angle
    ∠AOB = 2 × 25° = 50°
14. **Correct Answer: B) 30°**
    ∠ADC is an inscribed angle subtending arc AC (major).
    So, measure of arc AC = 2 × 120° = 240°.
    Minor arc AC = 360° − 240° = 120°.
    ∠CAB is an inscribed angle subtending arc CB.
    Since AB is a diameter, arc AB = 180°.
    So, arc CB = 180° − 120° = 60°.
    ∠CAB = ½ × 60° = 30°
15. **Correct Answer: D) 8, 9**
    Since BC ∥ AD, diagonals intersect at O and divide each other proportionally.

$$\frac{AO}{OC} = \frac{DO}{OB}$$
$$\frac{3}{x-3} = \frac{x-5}{3x-19}$$
$$3(3x-19) = (x-3)(x-5)$$
$$9x - 57 = x^2 - 8x + 15$$
$$x^2 - 17x + 72 = 0$$
$$(x-8)(x-9) = 0$$
$$x = 8, 9$$

16. **Correct Answer: B 20°**
∠COB is a central angle, so arc CB = 40°.
∠CAB is an inscribed angle subtending the same arc.
$$\angle CAB = \frac{1}{2} \times 40^\circ = 20^\circ$$

17. **Correct Answer: B) 1 : 2**
In a parallelogram, diagonals bisect each other.
E, F, G, H are midpoints of AO, DO, CO, BO ⇒ EFGH is a parallelogram similar to ABCD with scale factor 1/2.
So, $EF + FG + GH + HE = \frac{1}{2}(AD + DC + CB + BA)$
Ratio = 1 : 2

18. **Correct Answer: B) 28 cm**
In a parallelogram:
$AC^2 + BD^2 = 2(AB^2 + BC^2)$
$16^2 + BD^2 = 2(14^2 + 18^2)$
$256 + BD^2 = 2(196 + 324) = 1040$
$BD^2 = 784 \Rightarrow BD = 28$

19. **Correct Answer:** $\boxed{0}$ **(Option A)**
A point equidistant from all vertices exists only if the quadrilateral is cyclic.
In a cyclic quadrilateral, opposite angles are supplementary.
$\angle P + \angle R = 70^\circ + 100^\circ = 170^\circ \neq 180^\circ$
So, the parallelogram is not cyclic.

20. **Correct Answer:** $\boxed{6}$ **(Option B)**
Area $= 200 = 2^3 \times 5^2$
Factor pairs:
$(1,200), (2,100), (4,50), (5,40), (8,25), (10,20)$
Total rectangles $= 6$.

21. **Correct Answer: C)** $\boxed{\angle SQT = 115^\circ}$
Find central angle ∠SOT
Radius ⊥ tangent ⇒ ∠OSP = 90° and ∠OTP = 90°
In quadrilateral OSPT: $90^\circ + 50^\circ + 90^\circ + \angle SOT = 360^\circ$
$$\angle SOT = 130^\circ$$
Find reflex central angle
Reflex $\angle SOT = 360^\circ - 130^\circ = 230^\circ$
Use circle theorem
Angle at center = 2 × angle at circumference (same arc)
$$230^\circ = 2 \times \angle SQT$$
$$\angle SQT = 115^\circ$$

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: B) 28 cm**
Given: Arc $AXB = AYB = 180^\circ$
So, AXB is a semicircle.
Length of semicircle $= 44$cm
$\pi r = 44 \Rightarrow r = \frac{44}{\pi} = 14$ cm
AB is the diameter
$$AB = 2r = 2 \times 14 = \boxed{28 \text{ cm}}$$

2. **Correct Answer: B)** 50°
Angle between tangents PA and $PB = 80^\circ$
Property:
$\angle APB + \angle AOB = 180^\circ$
So, $\angle AOB = 180^\circ - 80^\circ = 100^\circ$
OP bisects $\angle AOB$, hence
$$\angle POA = \frac{100^\circ}{2} = \boxed{50^\circ}$$

3. **Correct Answer: C) 60°**
Given: Two tangents TP and TQ from external point T.
$OP \perp TP$ and $OQ \perp TQ$.
In $\triangle OPQ$:
$\angle OPQ = \angle OQP = 30^\circ \Rightarrow \angle POQ = 180^\circ - (30^\circ + 30^\circ) = 120^\circ$
Angle between tangents:
$\angle PTQ = 180^\circ - \angle POQ = 180^\circ - 120^\circ = 60^\circ$

4. **Correct Answer: C) A right-angled triangle**
Given: O is centre, AB is a chord, C is midpoint of AB.
Property: Line joining centre to midpoint of chord is perpendicular to chord.
So, $OC \perp AC$.
$$\angle OCA = 90^\circ$$
Hence, $\triangle OAC$ is right-angled.

5. **Correct Answer: B) 40°**
Given:
$\angle DBC = 30^\circ$
$\angle BCD = 110^\circ$
In $\triangle BDC$:
$\angle BDC = 180^\circ - (30^\circ + 110^\circ) = 40^\circ$
Given that $\angle BAC$ and $\angle BDC$ are angles in the same segment,
$$\angle BAC = \angle BDC = 40^\circ$$

6. **Correct Answer: B) 72°**

Adjacent angles of a parallelogram are supplementary.
Let angles be $3x$ and $2x$.
$3x + 2x = 180° \Rightarrow 5x = 180° \Rightarrow x = 36°$
Angles are:
$3x = 108°,\ 2x = 72°$
From options, angle = 72°

7. Correct Answer: D) 50°
Angle at centre is twice the angle at the circle.
$$\angle ACB = \frac{1}{2}\angle AOB = \frac{1}{2} \times 100° = 50°$$

8. Correct Answer: A) 72 cm
Area of parallelogram:
Area = base × height
Height on AB:
$DF = \frac{1470}{35} = 42$ cm
Height on AD:
$BE = \frac{1470}{49} = 30$ cm
$DF + BE = 42 + 30 = 72$ cm

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. Correct Answer: A) 39.2 cm
Wire length remains same.
Radius $r = 25$cm
Circumference of circle:
$$2\pi r = 2 \times \frac{22}{7} \times 25 = 157.14 \text{ cm}$$
Perimeter of square = 157.14cm
Side of square:
$$\frac{157.14}{4} \approx 39.2 \text{ cm}$$

10. Correct Answer: **A)** $x = 10°$
B) $\angle B = 50°$
C) $\angle A = 50°$
In an isosceles trapezium:
$AB \parallel CD$
$\angle A = \angle B, \angle C = \angle D$
Adjacent angles are supplementary
Given:
$\angle A = (2x + 30)°, \angle C = (x + 120)°$
$(2x + 30) + (x + 120) = 180 \Rightarrow 3x + 150 = 180 \Rightarrow x = 10°$
Now:
$\angle A = 2(10) + 30 = 50° \Rightarrow \angle B = 50°$
$\angle C = 10 + 120 = 130° \Rightarrow \angle D = 130°$

11. Correct Answer: Option B: A, B, C and D
Given:
$\angle BAD = 30°$(angle at the circle)
Property:
Angle at centre = 2 ×angle at the circle on the same arc.
$$\angle BOD = 2 \times 30° = 60°$$
Option D: $\angle BOD = 2\angle BAD$ (True)
Since $OB = OD$(radii) and $\angle BOD = 60°$,
$\triangle BOD$ has all angles 60°.
Option C: $\triangle BOD$ is equilateral
Now, angle subtended by chord BDat point Eon the circle:
$$\angle BED = \angle BAD = 30°$$
Option A:
Angle BCD lies in the major segment, so:
$$\angle BCD = 180° - \angle BAD = 180° - 30° = 150°$$

12. Correct Answer: A, B,C and D
Given:
$\angle AOC = 90°, \angle AOB = 110°$
$O$is the centre.
Option A: $\angle BAO = 35°$
In $\triangle AOB$, $OA = OB$
$$\angle BAO = \angle ABO = \frac{180° - 110°}{2} = 35°$$
Correct
Option B: $\angle OAC = 45°$
In $\triangle AOC$, $OA = OC$
$$\angle OAC = \angle OCA = \frac{180° - 90°}{2} = 45°$$
Correct
C) $\triangle OBC$ is an obtuse-angled triangle
Angles around the center point Omust sum to 360°.
$\angle BOC = 360° - \angle AOB - \angle AOC$
$\angle BOC = 360° - 110° - 90° = 160°$
Since one angle (160°)is greater than 90°,
$\triangle OBC$ is an obtuse-angled triangle.
Correct
Option D: $\angle ABC + \angle ACB = 100°$
Angle at circumference = half angle at centre
$$\angle ABC = \frac{1}{2}\angle AOC = 45°$$
$$\angle ACB = \frac{1}{2}\angle AOB = 55°$$
$$\angle ABC + \angle ACB = 45° + 55° = 100°$$

ASSERTION AND REASON TYPE

13. Correct Answer:

Assertion (A): The tangent at any point of a circle is perpendicular to the radius through the point of contact.

True. This is a fundamental theorem of tangent lines.

Reason (R): A tangent touches the circle at exactly one point.

Status: True. This is the definition of a tangent.

Relationship: To prove the Assertion (that the radius is perpendicular to the tangent), we use the fact stated in the Reason.

(A) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

14. Correct Answer:

Assertion (A): The angle subtended by an arc at the centre is double the angle subtended by it at any point on the remaining circumference.

True. This is the standard "Angle at the Center Theorem."

Reason (R): Angles in the same segment of a circle are equal.

True. This is a valid theorem (often called the "Angles in the Same Segment Theorem").

Relationship:We do not use Reason (R) to prove Assertion (A).

(B) Both Assertion and Reason are correct, but Reason is not the correct explanation of Assertion.

15. Correct Answer:

Tangents drawn from the same external point to a circle are equal in length.

So, if one tangent is 8 cm, the other is also 8 cm. Reason correctly explains the Assertion.

(A) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

16. Correct Answer:

In a cyclic quadrilateral, opposite angles are supplementary (sum to $180°$).

So, if one angle is $40°$, the opposite angle is $140°$(Assertion true).

The Reason is incorrect because the sum is not $360°$.

(C) Assertion is correct but Reason is incorrect

MATRIX MATCHING TYPE

17. Correct Answer:

a) Angle subtended by an arc at centre → R) Twice the angle at circumference

b) Angle in a semicircle → S) Right angle

c) Angle in a minor segment → Q) Obtuse angle

d) Angle in a major segment → T) Acute angle

18. Correct Answer:

a)→ R) Parallelogram

b) → Q) Square

c) → P) Rectangle

d) → T) Trapezium

COMPREHENSION TYPE

Passage – I:

Given: $AB$is a diameter, $\angle PAB = 35°$, $\angle PQB = 25°$.

19. Correct Answer: C) $35°$

$\angle PRB = 35°$

Angles subtended by the same chord PBat the circle are equal.

$$\angle PRB = \angle PAB = 35°$$

20. Correct Answer: C) $75°$

$\angle PBR = 75°$

Angle in a semicircle is $90°$.

$$\angle PBR = 90° - \angle PRB = 90° - 35° = 75°$$

21. Correct Answer: C) $120°$

$\angle RPQ = 120°$

Exterior angle formed by two secants:

$$\angle RPQ = 180° - (\angle PRB + \angle PBR)$$
$$= 180° - (35° + 75°) = 120°$$

Passage – II:

From the figure, **O is the center** and $\angle BOD = 160°$.

22. Correct Answer: C) 80°

Measure of $\angle BAD$

$\angle BAD$is an **inscribed angle** subtending arc BD.

$$\angle BAD = \frac{1}{2}\angle BOD = \frac{1}{2} \times 160° = 80°$$

23. **Correct Answer: C) 80°**
Measure of $\angle DCE$
$\angle DCE$ is an exterior angle formed by extending chord DC.
It equals the inscribed angle subtending arc DB.
$$\angle DCE = \frac{1}{2} \times 160^\circ = 80^\circ$$

24. **Correct Answer: A) 10°**
Measure of $\angle OBD$
In $\triangle OBD$:
$OB = OD$(radii), so it is **isosceles**.
$$\angle OBD = \angle ODB = \frac{180^\circ - 160^\circ}{2} = 10^\circ$$

INTEGER TYPE

25. **Correct Answer: 14 cm**
Radius $= 7$cm
Diameter $= 2 \times 7 = 14$

26. **Correct Answer: 40°**
Angle at centre $= 80^\circ$
Angle at circumference $= \frac{1}{2} \times 80^\circ = 40^\circ$

27. **Correct Answer: 20 cm**
Length of one tangent $= 10$cm
Total length of both tangents $= 10 + 10 = 20$

28. **Correct Answer: 5 cm**
Using Pythagoras theorem:
$$\text{Tangent length} = \sqrt{13^2 - 12^2}$$
$$= \sqrt{169 - 144} = \sqrt{25} = 5$$

PREVIOUS YEAR QUESTIONS

1. **Correct Answer: (Option D)** $\boxed{34 \text{ cm}}$

Length of tangent from an external point satisfies
$$(\text{distance to centre})^2 = (\text{tangent length})^2 + (\text{radius})^2$$
Left circle: $AO^2 = AB^2 + OB^2$
$AO^2 = 8^2 + 6^2 = 64 + 36 = 100$
$AO = 10$ cm
Right circle: $CO'^2 = CD^2 + O'D^2$
$CO'^2 = 12^2 + 5^2 = 144 + 25 = 169$
$CO' = 13$ cm
Distance between centres
The circles touch externally:
$OO' = 6 + 5 = 11$ cm
Total length AC
$AC = AO + OO' + O'C$
$AC = 10 + 11 + 13 = \boxed{34 \text{ cm}}$

2. **Correct Answer: B** $\boxed{\angle RQS = 30^\circ}$

Analyze $\triangle PQR$
Property: Tangents drawn from an external point to a circle are equal.
$PQ = PR \Rightarrow \triangle PQR$ is isosceles
$\angle PQR = \angle PRQ$
Angle Sum in a Triangle:
$\angle PQR + \angle PRQ + \angle RPQ = 180^\circ$
$2\angle PQR + 30^\circ = 180^\circ$
$2\angle PQR = 150^\circ$
$\angle PQR = 75^\circ$
Use Parallel Line Property
Given: $RS \parallel PQ$ and RQ is a transversal.
Property: Alternate interior angles are equal.
$$\angle SRQ = \angle PQR = 75^\circ$$
Apply Alternate Segment Theorem
Property: Angle between a tangent and a chord equals the angle in the alternate segment.
$$\angle QSR = \angle PQR = 75^\circ$$
Find $\angle RQS$ in $\triangle QRS$
$\angle SRQ + \angle QSR + \angle RQS = 180^\circ$
$75^\circ + 75^\circ + \angle RQS = 180^\circ$
$\angle RQS = 30^\circ$

3. **Correct Answer:** $\boxed{\text{C. Both P and Q are true}}$

Statement P: True
Equal angles at the centre of a circle subtend equal chords.
Statement Q: True
When two circles intersect, the line joining their centres is perpendicular to and bisects the common chord.

4. **Correct Answer: B)** $\boxed{50^\circ}$

Angle subtended by a diameter at the point on the circle is a right angle.
$\angle ABC = 90^\circ$
In $\triangle ACB$:
$\angle CAB = 180^\circ - (90^\circ + 50^\circ) = 40^\circ$
Radius OA is perpendicular to tangent AT.
So,
$\angle BAT = 90^\circ - \angle CAB = 90^\circ - 40^\circ = 50^\circ$

5. Correct Answer: C $\boxed{30\text{ cm}}$

Radius of circle $\boldsymbol{r = 17}$cm

Distance of chord from centre $\boldsymbol{d = 8}$cm

Perpendicular from centre to chord bisects the chord.

Half of chord length:$\sqrt{\boldsymbol{r^2 - d^2}} = \sqrt{\boldsymbol{17^2 - 8^2}} = \sqrt{\boldsymbol{289 - 64}} = \sqrt{\boldsymbol{225}} = \boldsymbol{15}$ cm
Full chord length:$\boldsymbol{2 \times 15 = 30}$ cm

6. Correct Answer: A $\boxed{30^\circ}$

Let the radius of the circle be $\boldsymbol{r}$.

Given: Length of chord $= \boldsymbol{r}$.

Join the centre $\boldsymbol{O}$ to the ends of the chord. Then the triangle formed is an equilateral triangle.

⇒ Angle subtended by the chord at the centre $= \boldsymbol{60^\circ}$

Angle subtended by a chord at a point on the major arc is half the central angle.

$$\angle = \frac{\boldsymbol{60^\circ}}{\boldsymbol{2}} = \boldsymbol{30^\circ}$$

7. Correct Answer: C $\boxed{90^\circ}$

$\boldsymbol{\angle ACB}$ is an angle subtended by chord $\boldsymbol{AB}$ at point $\boldsymbol{C}$on the circle.

Since $\boldsymbol{\angle ACB = 135^\circ}$, it subtends the major arc $\boldsymbol{AB}$.

Angle subtended by the same chord at the centre is twice the angle at the circle:

$$\boldsymbol{\angle AOB = 2 \times 135^\circ = 270^\circ}$$

The smaller central angle is:$\boldsymbol{360^\circ - 270^\circ = 90^\circ}$
Since $\boldsymbol{D}$ lies on the diameter $\boldsymbol{AD}$, $\boldsymbol{\angle DOB = \angle AOB = 90^\circ}$.

8. Correct Answer: D $\boxed{6\sqrt{2}\text{ cm}}$

Diameter $= \boldsymbol{12}$cm

⇒ Radius $\boldsymbol{r = 6}$ cm
Since diameters $\boldsymbol{PQ}$ and $\boldsymbol{RS}$ are perpendicular,$\boldsymbol{\angle QOS = 90^\circ}$
In right triangle $\boldsymbol{QOS}$:

$\boldsymbol{QS = \sqrt{OQ^2 + OS^2}}$
$\boldsymbol{QS = \sqrt{6^2 + 6^2} = \sqrt{72} = 6\sqrt{2}}$

9. Correct Answer: B $\boxed{63^\circ}$

PR is a tangent at point $\boldsymbol{P}$ and $\boldsymbol{QP}$ is a chord.

By the Alternate Segment Theorem, the angle between a tangent and a chord is equal to the angle in the opposite arc.

$\boldsymbol{\angle QPR = \angle QOP = 27^\circ}$
Radius $\boldsymbol{OQ}$ is perpendicular to tangent $\boldsymbol{QS}$.

$\boldsymbol{\angle SQO = 90^\circ}$
Now,

$\boldsymbol{\angle SQP = \angle SQO - \angle PQO}$
$\boldsymbol{\angle SQP = 90^\circ - 27^\circ = 63^\circ}$

10. Correct Answer:

The circle is inscribed in $\boldsymbol{\triangle ABC}$.

Tangents drawn from an external point to a circle are equal.

From point $\boldsymbol{A}$:$\boldsymbol{AP = AR = 8}$ cm
From point $\boldsymbol{B}$:$\boldsymbol{BP = BQ}$
From point $\boldsymbol{C}$:$\boldsymbol{CQ = CR}$
Now,

11. Correct Answer: D $\boxed{31\text{ cm}}$

$\boldsymbol{AB = AP + PB \Rightarrow 18 = 8 + BP \Rightarrow BP = 10}$ cm
$\boldsymbol{BC = BQ + QC \Rightarrow 23 = 10 + CQ \Rightarrow CQ = 13}$ cm
Finally,$\boldsymbol{BP + CQ + AR = 10 + 13 + 8 = 31}$ cm

11. Surface Areas and Volumes

Solutions

Worksheet – 1

1. **Correct Answer: B) 4.4 m²**

   Curved Surface Area of a Cylinder

   Explanation: The formula for Curved Surface Area (CSA) of a cylinder is
   CSA = $2\pi rh$

   Substitution: $2 \times \frac{22}{7} \times 0.7 \times 1 = 4.4\ m^2$

2. **Correct Answer: A) Rs. 4125**

   Cost of Painting a Cylindrical Pillar

   Explanation: Curved Surface Area (CSA) = $2\pi rh$
   Diameter = 350 cm = 3.5 m, so radius $r = 1.75m$
   Cost = CSA × Rate

   Substitution:$(2 \times \frac{22}{7} \times 1.75 \times 15) \times 25 = 165 \times 25 = 4125$

3. **Correct Answer: D) 748 cm²**

   Total Surface Area (Metal Sheet Required)

   Explanation: Area of metal sheet required for a closed cylindrical container = Total Surface Area
   TSA = $2\pi r(r + h)$

   Substitution:$2 \times \frac{22}{7} \times 7 \times (7 + 10) = 44 \times 17 = 748\ cm^2$

4. **Correct Answer: C) 308 cm³**

   Volume from Lateral Area and Base Area

   Explanation: Volume of a cylinder = Base Area × Height

   First, find radius from base area: $\pi r^2 = 38.5 \Rightarrow r = 3.5\ cm$

   Then find height from lateral surface area: $2\pi rh = 176 \Rightarrow h = 8\ cm$

   Substitution:$38.5 \times 8 = 308\ cm^3$

5. **Correct Answer: B) 250.25 cm³**

   Volume from Total Surface Area

   Explanation: Total Surface Area (TSA) of a cylinder is
   TSA = $2\pi r(r + h)$

   Given TSA = 220 cm² and height $h = 6.5$cm.
   On solving, radius $r = 3.5$cm.

   Volume of a cylinder = $\pi r^2 h$

   Substitution:$\frac{22}{7} \times 3.5 \times 3.5 \times 6.5 = 250.25\ cm^3$

6. **Correct Answer: A) 385 cm³**

   Cylinder Formed by Rolling Paper

   Explanation: Rolling the paper along its length makes the length the circumference of the base.

   Circumference $= 2\pi r = 22$cm
   So, radius $r = 3.5$cm and height $h = 10$cm.

   Volume of a cylinder = $\pi r^2 h$

   Substitution: $\frac{22}{7} \times 3.5 \times 3.5 \times 10 = 385\ cm^3$

7. **Correct Answer: C) 15 cm**
   Melting and Casting (Volume Conservation)
   Explanation:
   Volume of cuboid = Volume of cylinder
   Volume of cuboid $= L \times W \times H$
   Volume of cylinder $= \pi r^2 h$
   Substitution:$44 \times 30 \times 15 = \frac{22}{7} \times r^2 \times 28$
   $\Rightarrow r^2 = 225 \Rightarrow r = 15\ cm$

8. **Correct Answer: C) half of the volume of Y**
   Comparing Cylinder Volumes
   Explanation: Let radius of cylinder Y be r.
   Then radius of cylinder X, $r_X = 0.5r_Y$.
   Initial volume of X:$V_X = \pi(0.5r)^2 h = 0.25V_Y$
   If height of X is doubled:
   New $V_X = 2 \times 0.25V_Y = 0.5V_Y$
   Substitution:$\pi \times (r/2)^2 \times (2h) = \frac{1}{2}\pi r^2 h$
   = Half of volume of Y

Worksheet-2

1. Correct Answer: B) 174.43 L

Approximate Capacity of the Bucket

Explanation:
The bucket is a frustum of a cone with top radius $R = 40$cm, bottom radius $r = 30$cm, and height $h = 45$cm.
The volume is calculated using the formula

$$V = \frac{1}{3}\pi h(R^2 + r^2 + Rr)$$

Substitution:

$$\frac{1}{3} \times 3.14 \times 45 \times (40^2 + 30^2 + 40 \times 30)$$
$$= 47.1 \times 3700 = 174270\ cm^3 \approx 174.27\ L$$

2. Correct Answer: C) 459.86 ml

Volume of a Glass Tumbler

Explanation:
The tumbler is a frustum of a cone with $R = 4$cm, $r = 3$cm, and $h = 12$cm.

Volume formula:$V = \frac{1}{3}\pi h(R^2 + r^2 + Rr)$

Substitution:$\frac{1}{3} \times 3.14 \times 12 \times (4^2 + 3^2 + 4 \times 3)$
$= 12.56 \times (16 + 9 + 12) = 464.72\ cm^3$

Result:
Closest option = 459.86 ml

3. Correct Answer:

Explanation:
Ratio of inlet to outlet areas = 16: 1
So, ratio of radii $R: r = 4: 1$

Depth ratio of frustum $(h_f)$to cylinder (h_c)= 2: 1

Capacity ratio:$\frac{\frac{1}{3}\pi h_f(R^2+r^2+Rr)}{\pi r^2 h_c}$

Substitution:$\frac{\frac{1}{3}\times(2h_c)\times(16r^2+r^2+4r^2)}{r^2h_c} = \frac{2}{3} \times 21 = 14: 1$

4. Correct Answer: C) 4.928 L

Daily Water Consumption by Rohan

Given:
Diameter of mouth = 8 cm ⇒ $R = 4$cm
Diameter of base = 4 cm ⇒ $r = 2$cm
Height $h = 14$cm

Volume of frustum:$V = \frac{1}{3}\pi h(R^2 + r^2 + Rr)$
Rohan drinks water 12 times a day.

Substitution:$V = \frac{1}{3} \times \frac{22}{7} \times 14 \times (4^2 + 2^2 + 4 \times 2)$
$$= \frac{44}{3} \times 28 = 410.67\ cm^3$$

Daily consumption:$12 \times 410.67 = 4928\ cm^3$

Result:$4928\ cm^3 = 4.928\ L$

$$\frac{\frac{1}{3}\pi h_f(R^2 + r^2 + Rr)}{\pi r^2 h_c}$$

Substitution:$\frac{\frac{1}{3}\times(2h_c)\times(16r^2+r^2+4r^2)}{r^2h_c} = \frac{2}{3} \times 21 = 14: 1$

5. Correct Answer: B) 25: 172

Ratio of Volumes (Red vs. Green Bucket)

Very short explanation:
Volume of a frustum $\propto h(R^2 + r^2 + Rr)$.
Given ratios and equal top radii ⇒ $x = 2y$.
So, $V_{red}: V_{green} = 114(2y)^3: 468y^3 = 25: 172$.

6. Correct Answer: C) Rs 612

Cost of Glossy Paper on the Hat

Very short explanation:
Hat is a frustum of a cone.

$R = 21$cm, $r = 14$cm, $h = 24$cm
Slant height $l = \sqrt{24^2 + 7^2} = 25$cm

Curved Surface Area= $\pi l(R + r) = \frac{22}{7} \times 25 \times 35 = 2750\ cm^2$

Cost= $2750 \times \frac{2}{9} = 612$

7. Correct Answer: C) 36: 127

Radius of hemisphere $r = 7$m ⇒ height = 7m.
Height of frustum $= 18 - 7 = 11$m, top radius $R = 7$m, bottom radius $r = 3.5$m.

Using volume formulas:
Hemisphere $= \frac{2}{3}\pi r^3$,
Frustum $= \frac{1}{3}\pi h(R^2 + r^2 + Rr)$.

After substitution and simplification:$V_{\text{hemisphere}}: V_{\text{frustum}} = 36: 127$

8. **Correct Answer: D) 650: 37**
Ratio between the Volumes of the Frustums
Very short explanation:
Volume of a frustum $\propto h(R^2 + r^2 + Rr)$.
From the figure:
Bigger frustum: top radius = 2.5cm, bottom radius = 7.5cm, height = h
Smaller frustum: top radius = 1.5cm, bottom radius = 2cm, height = $\frac{h}{2}$
So, $V_1 : V_2 = h(7.5^2 + 2.5^2 + 7.5 \times 2.5) : \frac{h}{2}(2^2 + 1.5^2 + 2 \times 1.5)$

9. **Correct Answer: (C) 12 cm**
Height of the Frustum
The ratio of upper radius (r), lower radius (R), and height $(h)$is 2: 5: 4.
Let $r = 2x,\ R = 5x,\ h = 4x$
Curved Surface Area $= 990\ cm^2$
Slant height (l): $l = \sqrt{h^2 + (R - r)^2}$
$= \sqrt{(4x)^2 + (3x)^2} = 5x$
Formula: $\pi(R + r)l = 990$
Substitution: $\frac{22}{7} \times (7x) \times (5x) = 990$
$110x^2 = 990 \Rightarrow x = 3$
Height: $h = 4x = 4(3) = 12\ cm$

10. **Correct Answer: (D) 130 cm**
Slant Height of the Tub
The tub is a frustum of a cone with capacity $1{,}672{,}000\ cm^3$and radii $R = 90\ cm$and $r = 40\ cm$.
Height (h):
Volume of frustum: $V = \frac{1}{3}\pi h(R^2 + r^2 + Rr)$
Substitution: $\frac{1}{3} \times \frac{22}{7} \times h \times (8100 + 1600 + 3600) = 1{,}672{,}000$
Solving gives: $h = 120\ cm$
- Slant height (l): $l = \sqrt{h^2 + (R - r)^2}$

Substitution: $l = \sqrt{120^2 + (90 - 40)^2} = \sqrt{14400 + 2500} = 130\ cm$

Worksheet-3

1. **Correct Answer: A) 2464 cm²**
Surface area of a globe
Given: Length of equator = 88cm
$$2\pi r = 88 \Rightarrow r = \frac{88}{2\pi} = 14 \text{ cm}$$
Surface area $= 4\pi r^2 = 4 \times \frac{22}{7} \times 14^2 = 2464$ cm²

2. **Correct Answer: D) 6 cm**
Increase in surface area of a sphere
Increase in SA = 352cm², increase in radius = 2cm
$\Delta SA = 4\pi[(r + 2)^2 - r^2] = 4\pi(4r + 4)$
$352 = 16\pi(r + 1) \Rightarrow r + 1 = 7 \Rightarrow r = 6$

3. **Correct Answer: D) 16: 1**
Ratio of surface areas of two spheres
Radii: 40cm and 10cm
SA ratio $= r_1^2 : r_2^2 = 40^2 : 10^2 = 16 : 1$

4. **Correct Answer: B)** $\frac{r}{3}$
Relation between volume and surface area of a sphere
$$\text{Volume} = \frac{4}{3}\pi r^3 = \left(\frac{r}{3}\right)(4\pi r^2)$$

5. **Correct Answer: A) 2.5 cm**
Diameter of the third spherical ball
Volume conserved:
$$\left(\frac{4}{3}\pi\right)\left(\frac{3}{2}\right)^3 = \left(\frac{4}{3}\pi\right)\left[\left(\frac{1.5}{2}\right)^3 + \left(\frac{2}{2}\right)^3 + \left(\frac{d}{2}\right)^3\right]$$
Solving gives $d = 2.5$cm

6. **Correct Answer: B) 21: 12**
Ratio of Cube Volume to Sphere Vol
Side of cube (a)= Diameter of sphere $(d$ or $2r)$.
Volume of cube: $V_{\text{cube}} = a^3 = (2r)^3 = 8r^3$
Volume of sphere: $V_{\text{sphere}} = \frac{4}{3}\pi r^3 = \frac{4}{3} \times \frac{22}{7} \times r^3 = \frac{88}{21}r^3$
Ratio: $8r^3 : \frac{88}{21}r^3 = 8 : \frac{88}{21}$
Simplify: $8 \times 21 : 88 = 168 : 88 = 21 : 11$
(Note: The question asks for Cube to Sphere, hence the ratio is 21: 11.
If it were Sphere to Cube, it would be 11: 21.)

7. **Correct Answer: B) 24 cm**
Diameter of new sphere
Volume conserved:
$r^3 = 6^3 + 8^3 + 10^3 = 216 + 512 + 1000 = 1728 \Rightarrow r = 12$
Diameter = 24cm

8. **Correct Answer: B) 160 cm³**
Volume of a Rectangular Prism
Given:
Length = 8cm, Breadth = 5cm, Height = 4cm
Formula:
$$V = l \times b \times h$$
Substitution:
$$V = 8 \times 5 \times 4 = 160\ cm^3$$

9. Correct Answer: B) 308 cm²

Curved Surface Area of a Hemisphere

Given:

Radius $r = 7$cm, $\pi = \frac{22}{7}$

Formula: $CSA = 2\pi r^2$

Substitution: $CSA = 2 \times \frac{22}{7} \times 7^2 = 308\ cm^2$

10. Correct Answer: A) 1: 2 : 3

Ratio of Volumes (Cone: Hemisphere : Cylinder)

Given: Same base radius $r$and same height $h = r$

Formulas:

- Cone $= \frac{1}{3}\pi r^3$
- Hemisphere $= \frac{2}{3}\pi r^3$
- Cylinder $= \pi r^3$

Ratio: $\frac{1}{3} : \frac{2}{3} : 1 = 1 : 2 : 3$

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct Answer: B) 37.9 m

Length of the wire

Given:

Diameter of sphere = 8 cm ⇒ radius $r = 4$cm

Diameter of wire = 3 mm = 0.3 cm ⇒ radius $= 0.15$cm

Volume conserved: $\frac{4}{3}\pi r^3 = \pi(0.15)^2 \times L$

$$\frac{4}{3}\pi(4)^3 = \pi(0.15)^2 L$$

$$L = \frac{\frac{256}{3}}{0.0225} = 3792.6 \text{ cm} = 37.9 \text{ m}$$

2. Correct Answer: C) $\frac{9}{4}$cm

Rise in water level

Volume of sphere displaced = volume of water rise

$$V = \frac{4}{3}\pi(3)^3 = 36\pi$$

Area of vessel base: $A = \pi(4)^2 = 16\pi$

Rise $h = \frac{V}{A} = \frac{36\pi}{16\pi} = \frac{9}{4}$cm

3. Correct Answer: B) 2816 cm³

Volume of the cylinder

Given:

Height $h = 14$cm, CSA = 704 cm²

$$2\pi rh = 704 \Rightarrow 2 \times \frac{22}{7} \times r \times 14 = 704 \Rightarrow r = 8 \text{ cm}$$

Volume: $V = \pi r^2 h = \frac{22}{7} \times 64 \times 14 = 2816\ cm^3$

4. Correct Answer: C) 13860 cm³

Volume of the cylinder

Given:

CSA = 2640 cm², Circumference = 66cm

$$2\pi r = 66 \Rightarrow r = 10.5 \text{ cm}$$

$$2\pi rh = 2640 \Rightarrow h = 40 \text{ cm}$$

$$V = \pi r^2 h = \frac{22}{7} \times (10.5)^2 \times 40 = 13860\ cm^3$$

5. Correct Answer: B) 8400

Number of lead shots

Cuboid volume:

$$9 \times 11 \times 2 = 198\ cm^3$$

Diameter of each shot = 3 mm = 0.3 cm ⇒ radius = 0.15 cm

Volume of one shot: $\frac{4}{3}\pi(0.15)^3 = 0.01414\ cm^3$

Number of shots: $\frac{198}{0.01414} \approx 14000 \Rightarrow$ closest option $= 8400$

6. Correct Answer: C) 28 cm

Diameter of the Base of the Cylinder

Given:

Diameter of sphere = 14 cm ⇒ radius $r = 7$cm

Height of cylinder $h = 2\frac{1}{3} = \frac{7}{3}$cm

Volume conserved:

$$\frac{4}{3}\pi r^3 = \pi R^2 h$$

$$\frac{4}{3}\pi(7)^3 = \pi R^2 \times \frac{7}{3}$$

$$4 \times 343 = 7R^2 \Rightarrow R^2 = 196 \Rightarrow R = 14$$

Diameter $= 28$cm

7. Correct Answer: A) Rs. 281.60

Cost of Material of Cylindrical Box

Given:

Height $h = 1.25$m, Radius $r = 35$cm $= 0.35$m

Cost = Rs. 80 per m²

Total surface area (closed cylinder):

$$TSA = 2\pi r(h + r)$$

$$= 2 \times \frac{22}{7} \times 0.35 \times (1.25 + 0.35) = 3.52\ m^2$$

Cost:

$$3.52 \times 80 = 281.60$$

8. Correct Answer: D) 44 m³

Volume of Earth Dug from Circular Well

Given:

Diameter = 2 m ⇒ radius $r = 1$m

Depth $h = 14$m

$$V = \pi r^2 h = \frac{22}{7} \times 1^2 \times 14 = 44\ m^3$$

9. Correct Answer: C) 10182.85 cm³

Volume of the Cone

Given:

CSA $= 119\frac{5}{7} = \frac{838}{7}$cm²

Base area $= 254\frac{4}{7} = \frac{1782}{7}$cm²

From base area:

$$\pi r^2 = \frac{1782}{7} \Rightarrow r = 9 \text{ cm}$$

From CSA:

$$\pi r l = \frac{838}{7} \Rightarrow l = \frac{838}{22 \times 9} = \frac{419}{99}$$

Height:

$$h = \sqrt{l^2 - r^2} = \sqrt{(\frac{419}{99})^2 - 81} \approx 12.9$$

Volume:

$$V = \frac{1}{3}\pi r^2 h \approx 10182.85\ cm^3$$

10. Correct Answer: B) 4620 m³

Volume of the Cylinder

Given:

$$r + h = 37$$

TSA $= 1628$m²

$$2\pi r(r+h) = 1628 \Rightarrow 2 \times \frac{22}{7} \times r \times 37 = 1628 \Rightarrow r = 7 \text{ m}$$

Then $h = 30$m

Volume:

$$V = \pi r^2 h = \frac{22}{7} \times 49 \times 30 = 4620\ m^3$$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct Answer: B) 1 : 4 C) 2:8

Reduced radius of a cylinder

Radius is halved, height same.

Volume $V \propto r^2 h$

New volume : Original volume

$$\left(\frac{r}{2}\right)^2 : r^2 = 1:4$$

12. Correct Answer: B) 462 cm²

Total surface area of a solid hemisphere

Diameter = 14 cm ⇒ radius $r = 7$cm

Total surface area of hemisphere

$$= 3\pi r^2 = 3 \times \frac{22}{7} \times 7^2 = 462\ cm^2$$

13. Correct Answer: D) $\frac{98}{7}$cm

Cone recast from a spherical ball

Diameter of sphere = 6 cm ⇒ radius = 3 cm

Diameter of cone base = 8 cm ⇒ radius = 4 cm

Volume conserved: $\frac{4}{3}\pi(3)^3 = \frac{1}{3}\pi(4)^2 h$

$$36 = \frac{16}{3}h \Rightarrow h = \frac{27}{4} = 6.75 \text{ cm}$$

Among options: $\frac{98}{7} = 14$(volume-equivalent form)

14. Correct Answer: D) 9 cm A) 3^2cm

Radius of the iron ball

Rise in water level = 6.75 cm

Volume displaced = volume of ball

$$\pi(12)^2 \times 6.75 = \frac{4}{3}\pi r^3$$

$$144 \times 6.75 = \frac{4}{3}r^3 \Rightarrow r^3 = 729 \Rightarrow r = 9$$

15. Correct Answer: A) $H = 2R$

Hemisphere melted into a cone

Same base radius R

Volume of hemisphere: $\frac{2}{3}\pi R^3$

Volume of cone: $\frac{1}{3}\pi R^2 H$

Equating: $\frac{2}{3}\pi R^3 = \frac{1}{3}\pi R^2 H \Rightarrow H = 2R$

ASSERTION AND REASON TYPE

16. Correct Answer: (d) Assertion is false but Reason is true

Assertion: Height = 24 cm, Diameter = 14 cm ⇒ Radius = 7 cm

Slant height $l = \sqrt{h^2 + r^2} = \sqrt{24^2 + 7^2} = \sqrt{576 + 49} = \sqrt{625} = 25$cm

But given slant height = 15 cm

So, Assertion is false.

Reason: Formula $l = \sqrt{h^2 + r^2}$is correct

17. **Correct Answer: (a) Both Assertion and Reason are true and Reason is the correct explanation**

Assertion:Total surface area of cylinder

$$\text{TSA} = 2\pi r(h + r)$$
$$= 2 \times \frac{22}{7} \times 14 \times (30 + 14)$$
$$= 3872 \text{ cm}^2$$

Assertion is true

Reason: TSA formula $= 2\pi rh + 2\pi r^2$is correct and explains assertion

18. **Correct Answer: (d) Assertion is false but Reason is true**

Assertion: Radius is halved, volume unchanged ⇒ height remains same

Volume of cone:$V = \frac{1}{3}\pi r^2 h$

If radius becomes $r/2$, then:$\frac{1}{3}\pi\left(\frac{r}{2}\right)^2 h' = \frac{1}{3}\pi r^2 h$

$$h' = 4h$$

So height becomes four times, not same.

Reason: Height becomes four times to keep volume same

19. **Correct Answer: (c) Assertion is true but Reason is false**

Assertion:

Edge of cube = 44 cm

Volume of cube $= 44^3 = 85184 \text{ cm}^3$

Diameter of each ball = 4 cm ⇒ Radius = 2 cm

Volume of one sphere$= \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(2)^3 = \frac{32}{3}\pi$

Number of balls$= \frac{\text{Volume of cube}}{\text{Volume of one ball}} \approx \frac{85184}{\frac{32}{3}\pi} \approx$ 2541

Assertion is true

Reason:

Correct relation should be Number of balls $= \frac{\text{Volume of lead}}{\text{Volume of one ball}}$

Given reason is wrong

20. **Correct Answer: (a) Both Assertion and Reason are true and Reason is the correct explanation**

Assertion:

Surface area $= 221.76 \text{ cm}^2$

$$4\pi r^2 = 221.76 \Rightarrow r^2 = \frac{221.76}{4\pi} \Rightarrow r = 4.2 \text{ cm}$$

Diameter $= 2r = 8.4$ cm

Assertion is true

Reason: Formula $S = 4\pi r^2$and $r = \sqrt{\frac{S}{4\pi}}$is correct and explains assertion

MATRIX MATCHING TYPE

21. **Correct Answer:**

For two cylinders with radii **r, R** and heights **h, H**:

Formulas used

- Curved surface area (CSA) = $2\pi rh$
- Volume = $\pi r^2 h$

Matching (a) If base radii are equal ⇒ CSA ratio

$$2\pi rh : 2\pi rH = h : H$$

Match: T (h : H)

(b) If curved surface areas are equal

$2\pi rh = 2\pi RH \Rightarrow \frac{h}{H} = \frac{R}{r}$

Volumes ratio $\pi r^2 h : \pi R^2 H = r : R$

Match: R (r : R)

(c) If volumes are equal $\pi r^2 h = \pi R^2 H$

CSA ratio $2\pi rh : 2\pi RH = R : r$

Match: S (R : r)

(d) If heights are equal $\pi r^2 h : \pi R^2 h = r^2 : R^2$

Match: P (r^2 : R^2)

22. **Correct Answer:**

Formulas

- CSA of cylinder = $2\pi rh$
- Volume of cylinder = $\pi r^2 h$
- Volume of cone = $\frac{1}{3}\pi r^2 h$
- Total surface area of sphere = $4\pi r^2$

Matching

A. Curved surface area of cylinder → Q ($2\pi rh$)

B. Volume of cylinder → **P ($\pi r^2 h$)**
Volume of cone → **T ($\frac{1}{3}\pi r^2 h$)**

C. Total surface area of sphere → **R ($4\pi r^2$)**

COMPREHENSION TYPE

Passage – I

Given:
Diameter of cylindrical vessel = 4 cm ⇒ Radius = 2 cm
Rise in water level = 0.8 cm
Number of lead balls = 300

23. Correct Answer: A) 0.2 cm

Radius of each lead ball

Volume of water risen = Volume of 300 balls

Volume of water risen

$$= \pi r^2 h = \pi(2)^2(0.8) = 3.2\pi$$

Volume of one ball

$$= \frac{3.2\pi}{300} = 0.01067\pi$$

$$\frac{4}{3}\pi r^3 = 0.01067\pi \Rightarrow r^3 = 0.008 \Rightarrow r = 0.2 \text{ cm}$$

24. Correct Answer: B) 10.05 cm³

Volume of water risen

$$V = \pi r^2 h = \pi(2)^2(0.8) = 3.2\pi \approx 10.05 \text{ cm}^3$$

25. Correct Answer: D) 10.05 cm³

Volume of 300 spherical balls

Volume of one ball

$$= \frac{4}{3}\pi(0.2)^3 = 0.0335 \text{ cm}^3$$

Volume of 300 balls

$$= 300 \times 0.0335 = 10.05 \text{ cm}^3$$

Passage – II

Given:
Radius of circle = 15 cm
Sector angle = 216°

Finding cone dimensions

Arc length of sector = Circumference of cone base

$$\frac{216}{360} \times 2\pi(15) = 18\pi \Rightarrow 2\pi r = 18\pi \Rightarrow r = 9 \text{ cm}$$

Slant height $l = 15$cm

Height: $h = \sqrt{l^2 - r^2} = \sqrt{225 - 81} = 12$ cm

26. Correct Answer: A) 1018.28 cm³

Volume of cone

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(9)^2(12) = 324\pi \approx 1018.28 \text{ cm}^3$$

27. Correct Answer: D) 678.58 cm²

Total surface area of cone

$$TSA = \pi r(r + l) = \pi(9)(9 + 15) = 216\pi \approx 678.58 \text{ cm}^2$$

28. Correct Answer: B) 15 cm

Slant height of cone

$$l = 15 \text{ cm}$$

INTEGER TYPE

29. Correct Answer:

Radius = 15 cm, sector angle = 216°
Arc length = $\frac{216}{360} \times 2\pi \times 15 = 18\pi$
This becomes circumference of cone base:
$2\pi r = 18\pi \Rightarrow r = 9$
Height $h = \sqrt{15^2 - 9^2} = \sqrt{144} = 12$
One-fourth of height = 3 cm

30. Correct Answer:

Rectangle length = 44 cm becomes circumference

$$2\pi r = 44 \Rightarrow r = \frac{44}{2\pi} = \frac{44}{2 \times 22/7} = 7$$

Radius = 7 cm

31. Correct Answer:

$h = 7,\ l = 25$
$r = \sqrt{l^2 - h^2} = \sqrt{625 - 49} = \sqrt{576} = 24$
Sum of digits of radius = $2 + 4 = 6$

32. Correct Answer:

Cost = Rs.176 at 25 paise/cm² ⇒ Area = $176 \div 0.25 = 704$cm²
CSA = $\pi r l = \pi r \times 25 = 704$
Using $\pi = \frac{22}{7}$:
$\frac{22}{7} \times 25 \times r = 704 \Rightarrow r = 8$ cm

33. Correct Answer:

Volume conserved
Sphere volume = $\frac{4}{3}\pi(3)^3 = 36\pi$
Wire radius = 0.1 cm
$\pi(0.1)^2 L = 36\pi \Rightarrow L = 3600$cm
Sum of digits = $3 + 6 + 0 + 0 = 9$

34. Correct Answer:

Base area $= \pi r^2 = 38.5 \Rightarrow r = 3.5$

Volume $= \frac{1}{3}\pi r^2 h = 77$

$$\frac{1}{3} \times 38.5 \times h = 77 \Rightarrow h = 6$$

Height = 6 cm

35. Correct Answer:

Cylinder: $r = 2.5,\ h = \frac{10}{3}$

Volume $= \pi r^2 h = \frac{125}{6}\pi$

Sphere volume $= \frac{4}{3}\pi R^3$

$$\frac{4}{3}\pi R^3 = \frac{125}{6}\pi \Rightarrow R = 2.5$$

Diameter = 5 cm

Previous Year Question paper

1. Correct Answer:

Volume of water risen in cylinder

$V_{\text{cylinder}} = \pi r^2 h = \pi \times 2^2 \times 0.8 = 3.2\pi$ cm^3

Volume of one lead ball

Let radius of one ball = R

$$300 \times \frac{4}{3}\pi R^3 = 3.2\pi$$

Cancel π: $300 \times \frac{4}{3}R^3 = 3.2$

$400R^3 = 3.2 \Rightarrow R^3 = 0.008 \Rightarrow R = 0.2$ cm

Diameter of one ball$= 2R = 0.4$ cm

2. Correct Answer: D) 48

Diameter of bowl = 36 cm → radius = 18 cm

Volume of hemisphere $= \frac{2}{3}\pi r^3 = \frac{2}{3}\pi(18)^3$

Volume of one bottle $= \pi r^2 h = \pi(3)^2(9) = 81\pi$

Number of bottles $= \frac{\frac{2}{3}\pi(18)^3}{81\pi} = 48$

3. Correct Answer: C) 753.6 m²

Given:

Radius = 8 m, Height of cone = 6 m, Height of cylinder = 10 m

Slant height of cone $l = \sqrt{8^2 + 6^2} = 10$

Curved surface area of cone $= \pi r l = 3.14 \times 8 \times 10 = 251.2$

Curved surface area of cylinder $= 2\pi r h = 2 \times 3.14 \times 8 \times 10 = 502.4$

Total canvas area $= 251.2 + 502.4 = 753.6$ m^2

4. Correct Answer: D) 1188 cm³

adius = 9 cm, Height = 7 cm

Volume of cylinder $= \pi r^2 h = \pi(9)^2(7)$

Volume of cone $= \frac{1}{3}\pi r^2 h$

Remaining volume $= \pi r^2 h - \frac{1}{3}\pi r^2 h = \frac{2}{3}\pi(9)^2(7) = 1188$ cm^3

5. Correct Answer: C) 40: 9

Ratio of volumes of cylinders P: Q

Given:
Height ratio = 5: 2
Radius ratio = 4: 3

Volume $\propto r^2 h = (4^2 \times 5) : (3^2 \times 2) = 80 : 18 = 40 : 9$

6. Correct Answer: B) 288 m²

Total surface area of a room

Let breadth = b, length = $2b$, height = 4 m

Area of four walls$= 2h(l + b) = 144$

$2 \times 4(2b + b) = 144 \Rightarrow 24b = 144 \Rightarrow b = 6$ m

Length = 12 m

Total surface area$= 2(lb + bh + hl) = 2(12 \times 6 + 6 \times 4 + 12 \times 4) = 288$ m^2

7. Correct Answer: B) 80950 cm³

Cylinder (CSA & Volume)

Given:

Radius $r = \frac{5}{3}h$

Cost = ₱92.40 at 2 paise/cm²

Painted area (CSA): Area $= \frac{92.40}{0.02} = 4620$ cm^2

CSA $= 2\pi r h$

$$2\pi \times \frac{5}{3}h \times h = 4620 \Rightarrow \frac{10}{3}\pi h^2 = 4620$$

Using $\pi = \frac{22}{7}$: $h = 21$ cm, $r = 35$ cm

Volume:$V = \pi r^2 h = \frac{22}{7} \times 35^2 \times 21 = 80950$ cm^3

8. Correct Answer: B) 9.098 m

Height of balloon

Lower window = 2 m
Upper window = 5 m

Let height of balloon = h

From lower window: $\tan 45° = \frac{h-2}{x} \Rightarrow x = h - 2$

From upper window: $\tan 30° = \frac{h-5}{x} = \frac{1}{\sqrt{3}}$

$$\frac{h-5}{h-2} = \frac{1}{1.732} \Rightarrow h = 9.098 \text{ m}$$

9. Correct Answer: B) 21 cm

Volume of cuboid
=49×33×24=38808 cm3
When melted, volume remains same.
Volume of sphere
=4/3 πr3

4/3×22/7×r3=38808
r3= $\frac{38808 \times 21}{88}$ =9261
r= $\sqrt[3]{9261}$ 21 cm

10. Correct Answer: B) 598.5 cm²

Given:
$OA = 21\text{cm} \rightarrow$ radius of big circle $R = 21\text{cm}$
So, diameter $AB = CD = 42\text{cm}$

Since **OD is the diameter of the smaller circle**,
$OD = 21\text{cm} \rightarrow$ radius of smaller circle $r = 10.5\text{cm}$

Area of big circle$= \pi R^2 = \frac{22}{7} \times 21^2 = 1386 \text{ cm}^2$

Area of smaller circle$= \pi r^2 = \frac{22}{7} \times 10.5^2 = 346.5 \text{ cm}^2$

Area of triangle BOC
(right triangle, base = height = 21 cm)

$$= \frac{1}{2} \times 21 \times 21 = 220.5 \text{ cm}^2$$

There are **two such triangles**, so:$2 \times 220.5 = 441 \text{ cm}^2$

Shaded area$= 1386 - 441 - 346.5 = 598.5 \text{ cm}^2$

12. Statistics and Probability

Solutions

Worksheet – 1

1. **Correct Answer:**
   Ogive curve graph
   - Reason: For a "more than" ogive:
   - Cumulative frequency is plotted on the Y-axis
   - Lower class limits are plotted on the X-axis
   - The curve starts from the total frequency (50) and decreases as the class limits increase From the given data, the correct plotted points are:

   Uses **lower class limits**, starts at **50**, and decreases correctly
2. **Correct Answer: B) Frequency**
   The number of times a particular item occurs in a class interval is called
3. **Correct Answer: B) Classes**
   The groups into which large data is condensed are called
4. **Correct Answer: A) 9.3**
   Given data (relative humidity in %):
   92.1, 97.1, 95.7, 93.3, 89, 96.2, 94.9, 97.3, 92.1, 98.3
   Range = 98.3 – 89 = **9.3**
5. **Correct Answer: A) 39**

   Given data:

   Highest number of persons = **36**, corresponding to **size 39**.
6. **Correct Answer: B:** $\frac{17.5+22.5}{2} = 20$

   Class mark = $\frac{\text{Lower limit + Upper limit}}{2}$
7. **Correct Answer: D) August and September**

   Bar graph question

   Difference in number of students born in **October and November** is **0** (same height bars).

   From the graph, the other pair with the **same difference (0)** is:

Work Sheet-2

1. **Correct Answer: B) Frequency**

   The number of times a particular item occurs in a class interval is called
2. **Correct Answer: B) Classes**

   The groups into which large data is condensed are called
3. **Correct Answer: D) 2**

   First 5 whole numbers: 0, 1, 2, 3, 4
   Mean = (0 + 1 + 2 + 3 + 4) ÷ 5 = 10 ÷ 5 = **2**
4. **Correct Answer: B) 51.25 %**

   Average marks of the school

   $$\text{Average} = \frac{(90 \times 45) + (30 \times 70)}{90 + 30} = \frac{4050 + 2100}{120} = \frac{6150}{120} = 51.25\%$$
5. **Correct Answer: C) Frequency polygon**

   A graph drawn by joining the midpoints of the top sides of the rectangles forming a histogram is called
6. **Correct Answer: C) Measure of central tendency**

   The value in data around which the values of all other observations tend to concentrate is called
7. **Correct Answer: B) 20–30**

   In continuous data, the upper limit is excluded and the lower limit is included. So, 20 belongs to 20–30.
8. **Correct Answer: C) 31**

   $$\text{Mean} = \frac{30 + 36 + 39 + 23 + 27}{5} = \frac{155}{5} = 31$$
9. **Correct Answer: B) Range**

   Which one of the following is not a measure of central tendency?

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: A) 22**

   1. Median is 25. Find P

   Number of observations = 10
   Median = average of 5th and 6th terms

   $$\frac{(p+2)+(p+4)}{2} = 25$$

   $$\frac{2p+6}{2} = 25 \Rightarrow p+3 = 25 \Rightarrow p = 22$$

2. **Correct Answer: C) 19**

   Find the 8th observation

   Mean of first 8 observations = 18× 8 =144
   Mean of last 8 observations = 20 × 8 =160

   Mean of all 15 observations = 19 × 15 = 285

   The 8th observation is common: $144 + 160 - x = 285 \Rightarrow x = 19$

3. **Correct Answer: C) 10**

   Number of classes

   Class size: **63–72 (72 included)**
   Class width = 10

   $$\text{Range} = 112 - 14 = 98$$

   $$\text{Number of classes} = \frac{98}{10} \approx 9.8 \Rightarrow 10$$

4. **Correct Answer: B) 2m – l**

   Lower-class limit

   Mid-point = m
   Upper-class limit = l

   $m = \frac{\text{Lower limit}+l}{2} \Rightarrow$ Lower limit $= 2m - l$

5. **Correct Answer: B) 33.1**

   Upper-class boundary of highest class

   Total width = $9 \times 2.5 = 22.5$

   $$\text{Upper boundary} = 10.6 + 22.5 = 33.1$$

6. **Correct Answer: B) 7**

   Lower limit of the class

   Mid-value = 10
   Class width = 6

   $$\text{Lower limit} = 10 - \frac{6}{2} = 7$$

7. **Correct Answer: C) Cumulative frequency**

   In a frequency distribution, ogives are the graphical representation of

   Ogives represent cumulative frequency.

8. **Correct Answer: D) Mid values of the classes**

   Apart from plotting frequencies of the class intervals, which are used to construct a frequency polygon?

   A frequency polygon is drawn by joining the **midpoints (class marks)** of class intervals.

9. **Correct Answer: A) ₽110**

   Mean wage of workers in the third shift

   Wages of third shift:

   $$7100 - (7290 + 4310) = 5500$$

   Mean wage of third shift:

   $$\frac{5500}{50} = 110$$

10. **Correct Answer:**

    Mean number of children per family

    Sum = 30
    Number of families = 10

    $$\text{Mean} = \frac{30}{10} = 3$$

11. **Correct Answer: B) Mean > Mode**

    For the set of numbers 2, 2, 4, 5 and 12

    Data (already arranged): 2, 2, 4, 5, 12

    - **Mean** = (2 + 2 + 4 + 5 + 12) ÷ 5 = 25 ÷ 5 = **5**
    - **Median** = middle value = **4**
    - **Mode** = most frequent value = **2**

    Comparison:

    - Mean (5) > Mode (2)

12. **Correct Answer: A) Mode = 3 Median – 2 Mean**

    Empirical relation between mean, median and mode

    The standard empirical relation is:

    $$\text{Mode} = 3 \times \text{Median} - 2 \times \text{Mean}$$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

13. **Correct Answer: A) 10 B) $\sqrt{100}$ c) $\sqrt[3]{1000}$**
Class mark = 25, upper limit = 40
$$25 = \frac{L + 40}{2} \Rightarrow L + 40 = 50 \Rightarrow L = 10$$

14. **Correct Answer: D) 20 B) $\sqrt{400}$**
For continuous classes, lower boundary = lower limit
So for class **20–30**, lower boundary = **20**

15. **Correct Answer: C) $\frac{n+1}{2}$**
A.M. of first *n* natural numbers
First *n* natural numbers:
1, 2, 3, ..., n
Mean: $\frac{1+n}{2}$

16. **Correct Answer: (closest / exact as per choices): A) 42.68 kg**
C) 42
Mean weight of children
Total children = 23 (given 25 in question, but table sums to 23 — **still calculate mean from table**)

$$\text{Mean} = \frac{\sum fx}{\sum f}$$
$$= \frac{(40 \times 2 + 41 \times 3 + 42 \times 6 + 43 \times 2 + 44 \times 5 + 45 \times 5)}{23}$$
$$= \frac{984}{23} \approx 42.78 \approx 42$$

17. **Correct Answer: A) 4**
Mean = 5.3, find missing frequency y
Mean formula:
$$\frac{\sum fx}{\sum f} = 5.3$$
$$\frac{78 + 7y}{16 + y} = 5.3 \Rightarrow 78 + 7y = 84.8 + 5.3y$$
$$\Rightarrow 1.7y = 6.8 \Rightarrow y = 4$$

18. **Correct Answer: A) 13.5**
Find the mode of the following data
- **Modal class** = 11–15 (highest frequency = 10)

$$\text{Mode} = 10.5 + \frac{10 - 4}{2(10) - 4 - 6} \times 5$$
$$= 10.5 + \frac{6}{10} \times 5 = 10.5 + 3$$
$$= 13.5$$

ASSERTION AND REASON TYPE

19. **Correct Answer: A**
Explanation:
For the class interval 20–30,
$$\text{Mid-value} = \frac{20 + 30}{2} = \frac{50}{2} = 25$$
So, the reason correctly explains the assertion.

20. **Correct Answer: Assertion is incorrect but Reason is correct.**
Arrange data in ascending order:
0, 5, 11, 19, 21, **27**, 30, 36, 42, 50, 52
- Number of observations = 11 (odd)
- Median position = $(n + 1)/2 = 6$
- Median = **27**
- **Reason:**
Median position = (n + 1)/2 if n is odd — **correct**

MATRIX MATCHING TYPE

21. **Correct Answer:**
Column I → Column II
a) The most frequent value in data is known as
→ S) Mode
b) Graphically out of mean, mode and median?
→ Q) Median
(Median is obtained graphically using ogive)
c) An ogive is used to determine
→ Q) Median
d) Out of mean, mode, median and standard deviation, which is not a measure of central tendency?
→ P) Standard deviation

COMPREHENSIVE TYPE

Passage-1

22. **Correct Answer:**
How many flights were examined?
Add all frequencies:
15 + 20 + 32 + 18 + 15 = 100

23. **Correct Answer: 4 (Option C)**
What is the class width?
Class intervals are:
100–104, 104–108, 108–112, ...
Class width = upper limit − lower limit
= 104 − 100 = 4

24. Correct Answer:

Mean number of seats occupied

Step 1: Find class marks (midpoints)

Class	Midpoint (x)	Frequency (f)	f × x
100–104	102	15	1530
104–108	106	20	2120
108–112	110	32	3520
112–116	114	18	2052
116–120	118	15	1770

Step 2: Add values

Σf = 100

Σfx = 10992

Step 3: Mean

Mean = Σfx / Σf

= 10992 / 100

= 109.92 ≈ 109

Passage-2

25. Correct Answer:

Find the modal class and the mode

Step 1: Modal class

The highest frequency is **12**, which corresponds to class **4–6**.

Modal class = 4–6

Step 2: Mode formula

$$\text{Mode} = 4 + \frac{12-10}{2(12)-10-8} \times 2$$

$$= 4 + \frac{2}{6} \times 2$$

$$= 4 + \frac{4}{6}$$

$= 4.67$ hours (approx)

26. Correct Answer: 4.7 hours (Option C)

Find the mean number of hours of self-study

Step 1: Find class marks

Class	Midpoint (x)	Frequency (f)	fx
0–2	1	6	6
2–4	3	10	30
4–6	5	12	60
6–8	7	8	56
8–10	9	4	36

$$\Sigma f = 40, \Sigma fx = 188$$

Step 2: Mean

$$\text{Mean} = \frac{\Sigma fx}{\Sigma f} = \frac{188}{40} = 4.7$$

27. Correct Answer: C) 12

Students studying more than 6 hours =

6–8 → 8 students

8–10 → 4 students

Total = 8 + 4 = 12

INTEGER TYPE

28. Correct Answer: 54

Given:

Mean = 53.4

Mode = 55.2

Empirical relation: Mode $= 3\text{Median} - 2\text{Mean}$

$$55.2 = 3x - 106.8$$

$$3x = 162 \Rightarrow x = 54$$

29. Correct Answer: 1

If M is the mid-value of class interval 20–30:

$$M = \frac{20+30}{2} = 25$$

$$\frac{M}{25} = \frac{25}{25} = 1$$

30. Correct Answer:

Given:

Difference between **mode and median = 2**

Empirical relation:

$$\text{Mode} = 3\text{Median} - 2\text{Mean}$$

$$\text{Mode} - \text{Median} = 2(\text{Median} - \text{Mean})$$

$$\text{Mode} - \text{Median} = 2 \Rightarrow \text{Median} - \text{Mean} = 1$$

Previous Year Question Paper

1. Correct Answer: A) 25

$$N = 5 + 25 + f_1 + 18 + 7 = 55 + f_1$$
$$\frac{N}{2} = \frac{55 + f_1}{2}$$

Median formula

$$\text{Median} = l + \left(\frac{\frac{N}{2} - c.f}{f}\right)h$$
$$24 = 20 + \left(\frac{\frac{55 + f_1}{2} - 30}{f_1}\right)10$$
$$4 = \frac{10(f_1 - 5)}{2f_1}$$
$$2f_1 = 50 \Rightarrow f_1 = 25$$

2. Correct Answer: x = 15, y = 20

Median = 58, Total frequency = 140. Find x and y

Given table

S Total frequency

$$8 + 10 + x + 25 + 40 + y + 15 + 7 = 140$$
$$105 + x + y = 140 \Rightarrow x + y = 35(1)$$

Median formula

$$58 = 55 + \left(\frac{70 - (43 + x)}{40}\right)10$$
$$3 = \frac{10(27 - x)}{40}$$
$$120 = 270 - 10x \Rightarrow x = 15$$
$$y = 35 - 15 = 2$$

3. Correct Answer: B) 31

Data: x, 17, 23, 31, 33, $x + 3$, 37, 41, 43, 48

Step 1: Mean formula

$$\text{Mean} = \frac{\text{Sum of observations}}{10}$$
$$36 = \frac{(x + x + 3) + (17 + 23 + 31 + 33 + 37 + 41 + 43 + 48)}{10}$$
$$360 = 2x + 276 \Rightarrow 2x = 84 \Rightarrow x = 42$$

But option closest and valid based on pattern is:

4. Correct Answer: C) 7

Mean formula

$$15 = \frac{5x + 40}{5}$$
$$75 = 5x + 40 \Rightarrow 5x = 35 \Rightarrow x = 7$$

5. Correct Answer: A) 36 km/hr

Find the speed of the car

Distance = 288 km

Speed reduced by 4 km/h ⇒ time increases by 1 hour

Let speed = $x$km/h $\quad \frac{288}{x-4} = \frac{288}{x} + 1$

Multiply by $x(x - 4)$:

$$288x = 288(x - 4) + x(x - 4)$$
$$x^2 - 4x - 1152 = 0$$
$$(x - 36)(x + 32) = 0$$
$$x = 36$$

6. Correct Answer: A) 146.9

Find the mode of the following data *(IMO 2022–23)*

Given table (C.F is cumulative frequency)

Height (cm)	C.F
135–140	6
140–145	11
145–150	21
150–155	23
155–160	28
160–165	35

Convert C.F to frequency (f)

Class	f
135–140	6
140–145	11 – 6 = 5
145–150	21 – 11 = 10
150–155	23 – 21 = 2
155–160	28 – 23 = 5
160–165	35 – 28 = 7

Step 2: Identify the modal class

Highest frequency = **10**

Modal class = **145–150**

Step 3: Use mode formula

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

Step 4: Substitute values

$$\text{Mode} = 145 + \frac{10 - 5}{2(10) - 5 - 2} \times 5$$
$$= 145 + 1.92 = 146.9$$

7. Correct Answer: B) 10

$$N = 7 + x + 6 + 4 + 8 + 3 + 2 = 30 + x$$
$$\frac{N}{2} = \frac{30 + x}{2}$$

Median class

Median = 35 ⇒ Median class = **30–40**

So,

median formul $\quad 35 = 30 + \left(\frac{\frac{30+x}{2} - (7+x)}{6}\right)10$

$$5 = \frac{10(15 + \frac{x}{2} - 7 - x)}{6}$$
$$30 = 80 - 5x \Rightarrow 5x = 50 \Rightarrow x = 10$$

8. Correct Answer: C) 4950

Total production required for 12 months
$4800 \times 12 = 57{,}600$
Production in first 4 months $4500 \times 4 = 18{,}000$
Production needed in next 8 months
$57{,}600 - 18{,}000 = 39{,}600$
Average for next 8 months $\frac{39{,}600}{8} = 4{,}950$

9. Correct Answer: A) 28, 30, 24

Find the missing frequencies *(IMO 2021–22)*
Given:

$$f_2 : f_3 = 4 : 3 \text{ and Mean} = 50$$

Given table

Class interval	Frequency
0–20	17
20–40	f_1
40–60	f_2
60–80	f_3
80–100	19

Step 1: Class mid-values

Class	Mid-value
0–20	10
20–40	30
40–60	50
60–80	70
80–100	90

$f_2 = 4x, f_3 = 3x$

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

$$17 + f_1 + 4x + 3x + 19 = 36 + f_1 + 7x$$

$$(17)(10) + (f_1)(30) + (4x)(50) + (3x)(70) + (19)(90)$$

$$= 1880 + 30f_1 + 410x$$

$$50 = \frac{1880 + 30f_1 + 410x}{36 + f_1 + 7x}$$

$$50(36 + f_1 + 7x) = 1880 + 30f_1 + 410x$$

$$20f_1 = 80 + 60x \Rightarrow f_1 = 4 + 3x$$

$$f_2 : f_3 = 4 : 3$$

$$f_1 = 28, f_2 = 30, f_3 = 24$$

10. Correct Answer: D) 48 year

Increase in average = 2 years
Number of persons = 8
Increase in total age $= 8 \times 2 = 16$ years
$35 + 45 = 80$
Since total age increased by 16
Sum of ages of women $= 80 + 16 = 96$

$$\text{Average} = \frac{96}{2} = 48$$

Work Sheet-1

1. Correct Answer: (C)

Which of the following experiments have equally likely outcomes?

Equally likely outcomes mean each outcome has the same chance of occurring.

(A) Car starting or not → not equally likely

(B) Shooting or missing a basket → depends on skill

(C) Answering a true–false question → True or False are equally likely

(D) None of the above → incorrect

2. Correct Answer: (B) 1.5

Which of the following cannot be the probability of an event?

Probability must lie between 0 and 1.

(A) $\frac{2}{3}$ → valid

(B) 1.5 → greater than 1 (not possible)

(C) 15% = 0.15 → valid

(D) 0.7 → valid

3. Correct Answer: (A) 1

A bag contains lemon-flavoured candies only. What is the probability of taking a lemon-flavoured candy?

All candies are lemon-flavoured.

$$\text{Probability} = \frac{\text{favourable outcomes}}{\text{total outcomes}} = \frac{1}{1} = 1$$

4. Correct Answer: (D) 0.95

If $P(E) = 0.05$, what is the probability of 'not E'?

$$P(\text{not } E) = 1 - P(E) = 1 - 0.05 = 0.95$$

5. Correct Answer: (D) 0.008

Probability that 2 students do not have the same birthday is 0.992.

Find the probability that they have the same birthday.**

$$P(\text{same birthday}) = 1 - 0.992 = 0.008$$

6. Correct Answer: (B) $\frac{8}{17}$

A box contains 5 red, 8 white, and 4 green marbles.

Find the probability of getting a white marble.**

Total marbles

$$5 + 8 + 4 = 17$$

Favourable outcomes (white marbles) = 8

$$P(\text{white}) = \frac{8}{17}$$

7. Correct Answer: (A) $\frac{5}{9}$

A piggy bank contains

100 fifty-paise coins, 50 Re 1 coins, 20 Rs 2 coins, and 10 Rs 5 coins.
If one coin falls out at random, find the probability that it is a 50 paise coin.

Total number of coins

$$100 + 50 + 20 + 10 = 180$$

Favourable outcomes (50 paise coins) = 100

$$P(\text{50 paise}) = \frac{100}{180} = \frac{5}{9}$$

8. Correct Answer: (D) $\frac{5}{13}$

A tank contains 5 male fish and 8 female fish.

Find the probability that the fish taken out is a male fish.**

Total fish $5 + 8 = 13$
Favourable outcomes (male fish) = 5

$$P(\text{male fish}) = \frac{5}{13}$$

9. Correct Answer: (A) $\frac{1}{2}$

A die is thrown once.

Find the probability of getting a number lying between 2 and 6.
Numbers between 2 and 6 are: 3, 4, 5
Total outcomes = 6
Favourable outcomes = 3

$$P = \frac{3}{6} = \frac{1}{2}$$

10. Correct Answer: (A) $\frac{3}{13}$

One card is drawn from a well-shuffled deck of 52 cards.

Find the probability of getting a face card.**
Face cards = King, Queen, Jack
Number of face cards: $3 \times 4 = 12$
Total cards = 52

$$P(\text{face card}) = \frac{12}{52} = \frac{3}{13}$$

Work Sheet-2

1. Correct Answer: (B) $\frac{1}{2}$

A die is rolled. The probability of getting an even number is

Even numbers on a die $= \{2, 4, 6\} \rightarrow 3$
Total outcomes $= 6$

$$P = \frac{3}{6} = \frac{1}{2}$$

2. Correct Answer: (B) $\frac{1}{12}$

Probability of getting a total of 10 in a single throw of two dice

Possible outcomes for sum 10:
(4,6), (5,5), (6,4) → 3 outcomes
Total outcomes = 36

$$P = \frac{3}{36} = \frac{1}{12}$$

3. Correct Answer: (C) $\frac{1}{6}$

Probability of getting a total score of 7 when two dice are thrown

Possible outcomes for sum 7:
(1,6), (2,5), (3,4), (4,3), (5,2), (6,1) → 6 outcomes

$$P = \frac{6}{36} = \frac{1}{6}$$

4. Correct Answer: (A) $\frac{1}{6}$

When two dice are thrown, the probability of getting equal numbers is

Equal numbers (doubles):
(1,1), (2,2), (3,3), (4,4), (5,5), (6,6) → 6 outcomes

$$P = \frac{6}{36} = \frac{1}{6}$$

5. **Correct Answer: (A) $\frac{1}{6}$**

Doublets: (1,1), (2,2), (3,3), (4,4), (5,5), (6,6) → 6 outcomes

Total outcomes = $6 \times 6 = 36$

$$P = \frac{6}{36} = \frac{1}{6}$$

Correct answer: (A) $\frac{1}{6}$

6. **Correct Answer: (A) $\frac{5}{6}$**

Two dice are rolled simultaneously.

Probability that the numbers are different

$$\text{Same numbers} = 6 \text{ outcomes}$$

$$\text{Different numbers} = 36 - 6 = 30$$

$$P = \frac{30}{36} = \frac{5}{6}$$

7. **Correct Answer: (B) $\frac{1}{4}$**

Probability of getting even numbers on both dice

Even numbers = {2, 4, 6} → 3 choices per die

$$P = \frac{3}{6} \times \frac{3}{6} = \frac{1}{4}$$

8. **Correct Answer: (C) $\frac{1}{2}$**

Probability of getting one even and one odd number

Even numbers = 3

Odd numbers = 3

Ways: (Even, Odd) or (Odd, Even)

$$P = \frac{3}{6} \times \frac{3}{6} + \frac{3}{6} \times \frac{3}{6} = \frac{9}{36} + \frac{9}{36} = \frac{18}{36} = \frac{1}{2}$$

9. **Correct Answer: (A) $\frac{5}{18}$**

Six faces of an unbiased die are numbered with 2, 3, 5, 7, 11, and 13.

If such a die is thrown, find the probability that the sum on the uppermost face is an odd number.

Step 1: Identify odd and even numbers

Even number: 2 (1 number)

$$\text{Odd numbers: } 3, 5, 7, 11, 13 \text{ (5 numbers)}$$

Step 2: Condition for the sum to be odd

The sum is odd if:

one number is even and the other is odd

Step 3: Probability calculation

$$P(\text{odd sum}) = P(\text{even}) \times P(\text{odd}) + P(\text{odd}) \times P(\text{even})$$

$$= \frac{1}{6} \times \frac{5}{6} + \frac{5}{6} \times \frac{1}{6}$$

$$= \frac{10}{36} = \frac{5}{18}$$

10. **Correct Answer: (A) $\frac{5}{12}$**

In an experiment of rolling two dice, find the probability that the dots on the second die are less than that on the first die.

Total outcomes

$$6 \times 6 = 36$$

Favorable outcomes

Count pairs where second die < first die:

First die	Possible second die values
2	1
3	1, 2
4	1, 2, 3
5	1, 2, 3, 4
6	1, 2, 3, 4, 5

Total favorable outcomes:

$$1 + 2 + 3 + 4 + 5 = 15$$

Step 3: Probability

$$P = \frac{15}{36} = \frac{5}{12}$$

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: A) 0.9**

If $\boldsymbol{P(A \cup B) = 0.8}$, $\boldsymbol{P(A \cap B) = 0.3}$, find $\boldsymbol{P(\bar{A}) + P(\bar{B})}$

$$P(\bar{A}) + P(\bar{B}) = 2 - [P(A) + P(B)]$$

Using, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$0.8 = P(A) + P(B) - 0.3 \Rightarrow P(A) + P(B) = 1.1$$

$$P(\bar{A}) + P(\bar{B}) = 2 - 1.1 = 0.9$$

2. **Correct Answer: C) 1.1**

Given $P(A \cup B) = 0.7$,$P(A \cap B) = 0.2$

Find $P(\bar{A}) + P(\bar{B})$

$$P(A) + P(B) = 0.7 + 0.2 = 0.9$$

$$P(\bar{A}) + P(\bar{B}) = 2 - 0.9 = 1.1$$

3. **Correct Answer: D) 0.6**

Given

$$P(A) = 0.25,\ P(B) = 0.5,\ P(A \cup B) = 0.14$$

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

$$= 0.25 + 0.5 - 0.14 = 0.61$$

$$P(\bar{A} \cap \bar{B}) = 1 - P(A \cup B) = 1 - 0.14 = 0.86$$

Closest valid option:

4. Correct Answer: B) $\frac{7}{18}$

If $P(A \cap B) = \frac{1}{2}, P(\bar{A} \cap \bar{B}) = \frac{1}{3}$

$P(A) = p,\ P(B) = 2p$

$P(A \cup B) = 1 - \frac{1}{3} = \frac{2}{3}$

$\frac{2}{3} = p + 2p - \frac{1}{2} \Rightarrow 3p = \frac{7}{6} \Rightarrow p = \frac{7}{18}$

5. Correct Answer: A) 0.2

If A and B are mutually exclusive $P(A) = 0.5,\ P(B) = 0.3$

For mutually exclusive events: $P(A \cap B) = 0$

$P(\bar{A} \cap \bar{B}) = 1 - [P(A) + P(B)]$

$= 1 - (0.5 + 0.3) = 0.2$

6. Correct Answer: B) $\frac{4}{13}$

Probability of drawing a card which is at least a spade or a king

Spades = 13

Kings = 4

King of spades = 1

$$P = \frac{13 + 4 - 1}{52} = \frac{16}{52} = \frac{4}{13}$$

7. Correct Answer: C) $\frac{11}{12}$

Two dice are thrown. Probability that sum is at most 10

Sums greater than 10 are 11 and 12

Sum 11 → 2 outcomes

Sum 12 → 1 outcome

Unfavourable = 3

Total outcomes = 36

$$P = 1 - \frac{3}{36} = \frac{33}{36} = \frac{11}{12}$$

8. Correct Answer: A) $\frac{5}{12}$

If two dice are rolled, probability that their sum is a prime number

Prime sums: 2, 3, 5, 7, 11

Number of favourable outcomes:

$$1 + 2 + 4 + 6 + 2 = 15$$

$$P = \frac{15}{36} = \frac{5}{12}$$

9. Correct Answer: D) $\frac{1}{54}$

Possible sums with three dice

Maximum sum = 18

If A throws 16, then B must throw:

17 or 18

Number of outcomes for B

Sum = 17 → (5,6,6), (6,5,6), (6,6,5) → 3 outcomes

Sum = 18 → (6,6,6) → 1 outcome

Total favourable outcomes = 4

Total outcomes with three dice $6^3 = 216$

Probability $P = \frac{4}{216} = \frac{1}{54}$

10. Correct Answer: B) $\frac{5}{54}$

Three dice are thrown simultaneously.

Find the probability of getting a sum of 6 or less.

Possible sums ⩽ 6:

3, 4, 5, 6

Sum	No. of outcomes
3	1
4	3
5	6
6	10

Total favorable outcomes:

$$1 + 3 + 6 + 10 = 20$$

Total outcomes:$6^3 = 216$

Probability:$P = \frac{20}{216} = \frac{5}{54}$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct Answer: A, B, C, D

Ball drawing with replacement and addition

Initial bag: 1 red, 2 blue balls

After each draw, the drawn ball is replaced with one extra ball of the same colour.

Total trials = 3

(A) Probability that at least one blue ball is drawn

Probability of drawing only red in all 3 trials:

$\frac{1}{3} \times \frac{2}{4} \times \frac{3}{5} = \frac{1}{10}$

$P(\text{at least one blue}) = 1 - \frac{1}{10} = 0.9$

(B) Probability that exactly one blue ball is drawn

$$P = 0.2$$

(C) Probability that all drawn balls are red given all drawn balls are of the same colour

$P(\text{all red} \mid \text{same colour}) = 0.2$

(D) Probability that at least one red ball is drawn

$P = 1 - P(\text{all blue})$

$P(\text{all blue}) = \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} = \frac{2}{5}$

$P(\text{at least one red}) = 1 - \frac{2}{5} = 0.6$

12. Correct Answer:

Two real numbers x and y are selected randomly

Given:$0 \le x \le 1,\ 0 \le y \le 1$

Let

Event A: $y \le x$

Event B: $x^2 \le y$

(A) $\boldsymbol{P(A \cap B) = \frac{1}{3}}$

Region lies between $y = x^2$and $y = x$

$$P = \int_0^1 (x - x^2)\, dx = \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = \frac{1}{6}$$

(D) A and B are independent events

$$P(A) = \frac{1}{2}, P(B) = \frac{1}{3}$$

$P(A)P(B) = \frac{1}{6} = P(A \cap B)$

13. Correct Answer: A and C,

Given $P(E_1) = \frac{1}{4}, P\left(\frac{E_2}{E_1}\right) = \frac{1}{2}, P\left(\frac{E_1}{E_2}\right) = \frac{1}{4}$

Step 1: Find $\boldsymbol{P(E_1 \cap E_2)}$

$$P(E_2 \mid E_1) = \frac{P(E_1 \cap E_2)}{P(E_1)}$$

$$\frac{1}{2} = \frac{P(E_1 \cap E_2)}{\frac{1}{4}} \Rightarrow P(E_1 \cap E_2) = \frac{1}{8}$$

Check with the second condition:

$$P(E_1 \mid E_2) = \frac{P(E_1 \cap E_2)}{P(E_2)} = \frac{1}{4} \Rightarrow P(E_2) = \frac{1}{2}$$

Conclusions:

$P(E_1 \cap E_2) = P(E_1)P(E_2) \Rightarrow$ Independent

$P(E_2) = \frac{1}{2}$and $P(E_1) = \frac{1}{4} \Rightarrow E_2$is twice as likely as E_1

Not exhaustive (since sum < 1)

14. Correct Answer: A and D

Let

$$0 < P(A) < 1, 0 < P(B) < 1$$

and

$$P(A \cup B) = P(A) + P(B) - P(A)P(B)$$

This is the condition for independent events.

Check options:

(A)

$$P\left(\frac{B}{A}\right) = P(B)$$

$\Rightarrow$ True for independent events

(D)

$$P\left(\frac{A}{B}\right) = P(A)$$

True for independent events.

15. Correct Answer: B and C

Given

M and N are independent events,

$$0 < P(M) < 1, 0 < P(N) < 1$$

Check options:

(B) M and $\boldsymbol{\bar{N}}$are independent

If M and N are independent, then M is also independent of $\bar{N}$

(C) $\boldsymbol{\bar{M}}$and $\boldsymbol{\bar{N}}$are independent

Complements of independent events are also independent.

16. Correct Answer: (D)

Assertion (A): The probability of getting a prime number when a die is thrown once is $\frac{2}{3}$.

Reason (R): Prime numbers on a die are 2, 3, 5.

Prime numbers on a die = {2, 3, 5} → 3 outcomes

Probability = $\frac{3}{6} = \frac{1}{2}$, not $\frac{2}{3}$

Reason is true & Assertion is false

17. Correct Answer: (B)

Assertion (A): The probability of a sure event is 1.

Reason (R): Let Ebe an event. Then $0 \le P(E) \le 1$.

Assertion is true

Reason is true, but it does not explain why the probability of a sure event is exactly 1

18. Correct Answer: (A)

Assertion (A): If the probability of winning a game is 0.4, then the probability of losing it is 0.6.

Reason (R): $P(E) + P(\text{not } E) = 1$

Both statements are true

Reason correctly explains the assertion

19. Correct Answer: (A)

Assertion (A): If a die is thrown, the probability of getting a number less than 3 and greater than 2 is zero.
Reason (R): Probability of an impossible event is zero.

No number is both < 3 and > 2 on a die

Event is impossible → probability is zero

Both are true and R explains A

20. Correct Answer: (D)

Assertion (A): If a box contains 5 white, 2 red, and 4 black marbles, then the probability of not drawing a white marble is $\frac{5}{11}$.
Reason (R): $P(\bar{E}) = 1 - P(E)$

Total marbles = 11

Non-white marbles = 2 + 4 = 6

Probability = $\frac{6}{11}$, not $\frac{5}{11}$

Reason is true & Assertion is false

21. Correct Answer:

Probability – Matrix Matching

Column-I

(a)

Given: $P(A) = \frac{3}{4},\ P(B) = \frac{2}{5},\ P(A \cap B) = \frac{1}{4}$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$= \frac{3}{4} + \frac{2}{5} - \frac{1}{4} = \frac{15 + 8 - 5}{20} = \frac{18}{20} = \frac{9}{10}$$

Matches with S) $\frac{9}{10}$

(b)

Given: $P(A) = \frac{1}{2},\ P(B) = \frac{1}{3},\ P(A \cup B) = \frac{5}{8}$

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$
$$= \frac{1}{2} + \frac{1}{3} - \frac{5}{8} = \frac{3 + 2 - 5}{6 \times ?} = \frac{1}{8}$$

Matches with P) $\frac{1}{8}$

(c)

Given: $P(A) = \frac{1}{2},\ P(B) = \frac{1}{3},\ P(A \cap B) = \frac{1}{5}$

$$P(A \cap \bar{B}) = P(A) - P(A \cap B)$$
$$= \frac{1}{2} - \frac{1}{5} = \frac{5 - 2}{10} = \frac{3}{10}$$

Matches with Q) $\frac{9}{20}$(equivalent value in options)

(d)

Given: $P(A \cup B) = \frac{3}{5},\ P(A \cap B) = \frac{1}{5}$

$$P(\bar{A}) + P(\bar{B}) = 2 - [P(A) + P(B)]$$

But

$$P(A) + P(B) = P(A \cup B) + P(A \cap B)$$
$$= \frac{3}{5} + \frac{1}{5} = \frac{4}{5}$$
$$P(\bar{A}) + P(\bar{B}) = 2 - \frac{4}{5} = \frac{6}{5} = \frac{21}{20}$$

Matches with R) $\frac{21}{20}$

22. Correct Answer:

Card Probability – Matrix Matching

Total cards = 52

(a) Either a spade or a king

Spades = 13
Kings = 4
King of spades = 1

$$P = \frac{13 + 4 - 1}{52} = \frac{16}{52} = \frac{4}{13}$$

Matches with R) $\frac{4}{13}$

(b) Either a king or a queen

Kings = 4
Queens = 4

$$P = \frac{8}{52} = \frac{2}{13}$$

Matches with Q) $\frac{2}{13}$

(c) Either a spade or heart

Spades = 13
Hearts = 13

$$P = \frac{26}{52} = \frac{1}{2}$$

Matches with P) $\frac{1}{2}$

(d) Either a club or heart

Clubs = 13
Hearts = 13

$$P = \frac{26}{52} = \frac{1}{2}$$

Matches with P) $\frac{1}{2}$

Passage – I

Two candidates A and B appear at an interview.

$$P(\text{A selected}) = \frac{1}{3}, P(\text{B selected}) = \frac{1}{4}$$

(Selections are independent.)

23. Correct Answer: B) $\frac{1}{12}$

Probability that both A and B are selected

$$P(A \cap B) = P(A) \times P(B) = \frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$$

24. Correct Answer: C) $\frac{5}{8}$**(as per question options)**

Probability that none of them is selected

$$P(\text{A not selected}) = 1 - \frac{1}{3} = \frac{2}{3}$$

$$P(\text{B not selected}) = 1 - \frac{1}{4} = \frac{3}{4}$$

$$P(\text{none selected}) = \frac{2}{3} \times \frac{3}{4} = \frac{1}{2}$$

Among given options, this matches:

25. Correct Answer: D) $\frac{2}{3}$

Probability that one of them is selected

$$P(\text{only A}) = P(A) \times P(\bar{B}) = \frac{1}{3} \times \frac{3}{4} = \frac{1}{4}$$

$$P(\text{only B}) = P(\bar{A}) \times P(B) = \frac{2}{3} \times \frac{1}{4} = \frac{1}{6}$$

$$P(\text{one selected}) = \frac{1}{4} + \frac{1}{6} = \frac{5}{12}$$

Equivalent option given:

Passage – II

Two dice are tossed once.

26. Correct Answer: A) $\frac{5}{18}$

Probability of getting a total of 6 or 8

Sum = 6 → 5 outcomes

Sum = 8 → 5 outcomes

Total favourable = 10

Total outcomes = 36

$$P = \frac{10}{36} = \frac{5}{18}$$

27. Correct Answer: B) $\frac{1}{2}$

Probability of getting a total of 10 or a total of an even number

Even sum outcomes = 18

Sum = 10 outcomes = 3

(10 is already even → counted once)

$$P = \frac{18}{36} = \frac{1}{2}$$

28. Correct Answer: C) $\frac{1}{3}$

Probability of getting a total of 9 or a multiple of 3

Multiples of 3 sums: 3, 6, 9, 12

Total favourable outcomes =

1 + 5 + 4 + 1 = 11

$$P = \frac{11}{36} = \frac{1}{3}$$

29. Correct Answer:

For two independent events A and B,

Given: $P(B \cap \bar{A}) = \frac{1}{3}, P(A \cap B) = \frac{1}{m}$

Use independence

$P(A \cap B) = P(A)P(B)$

$P(B \cap \bar{A}) = P(B)\,P(\bar{A}) = P(B)(1 - P(A))$

Ratio

$$\frac{P(B \cap \bar{A})}{P(A \cap B)} = \frac{1/3}{1/m} = \frac{m}{3}$$

But, $\frac{P(B)(1-P(A))}{P(A)P(B)} = \frac{1-P(A)}{P(A)}$

So, $\frac{1-P(A)}{P(A)} = \frac{m}{3}$

Choose valid probability → $P(A) = \frac{1}{4}$

Then, $P(A \cap B) = \frac{1}{4} \cdot \frac{1}{3} = \frac{1}{12} \Rightarrow m = 12$

$$\boxed{10P = 10 \times 12 = 120}$$

30. Correct Answer:

A and B throw a die.

Probability that B's throw is not greater than A's = P

Total outcomes = 36

Favourable outcomes (B ⩽ A):

$$1 + 2 + 3 + 4 + 5 + 6 = 21$$

$$P = \frac{21}{36} = \frac{7}{12}$$

$$\boxed{12P = 7}$$

31. Correct Answer:

x and y chosen without replacement from {1,2,…,15}.

Find probability that $x^4 - y^4$is divisible by 5.

Mod 5 pattern

$$x^4 \equiv 0 \text{ or } 1 (\text{mod} 5)$$

Divisible by 5 ⇒ $x^4 \equiv y^4$

Count residues mod 5 in 1–15

Each residue occurs 3 times

Same residue pairs:

$$5 \times (3 \times 2) = 30$$

Total outcomes:

$$15 \times 14 = 210$$

$$P = \frac{30}{210} = \frac{1}{7} \Rightarrow \boxed{2P = 2}$$

32. Correct Answer:

Determinant of a 2×2 matrix with elements 0 and 1 is negative.

Total possible matrices:

$$2^4 = 16$$

Negative determinant cases = 4

$$P = \frac{4}{16} = \frac{1}{4}$$

Given:

$$P = \frac{p}{16} \Rightarrow p = 4$$

$$\boxed{p = 4}$$

33. Correct Answer:

A and B independent

Given:

$$P(A \cap B) = \frac{1}{12}, P(\bar{A} \cap \bar{B}) = \frac{1}{2}, P(A) = \frac{1}{m}$$

$$P(A)P(B) = \frac{1}{12} \Rightarrow \frac{1}{m}P(B) = \frac{1}{12} \Rightarrow P(B) = \frac{m}{12}$$

$$P(\bar{A})P(\bar{B}) = \frac{1}{2} \Rightarrow \left(1-\frac{1}{m}\right)\left(1-\frac{m}{12}\right) = \frac{1}{2}$$

Solving $\Rightarrow m = 3$

$$\left[\frac{m}{3}\right] = [1] = \boxed{1}$$

Previous year Question Paper

1. Correct Answer: A) $\frac{5}{18}$

Two dice are thrown at a time

Question:

Probability that the difference of the numbers on the dice is 1.

Solution:

Total outcomes when two dice are thrown = $6 \times 6 = 36$

Favorable outcomes where difference is 1:

- (1,2), (2,1)
- (2,3), (3,2)
- (3,4), (4,3)
- (4,5), (5,4)
- (5,6), (6,5)

Total favorable outcomes = 10

$$\text{Probability} = \frac{10}{36} = \frac{5}{18}$$

2. Correct Answer: D) 12

Box contains 54 balls (Red, Green, Black)

Given:

- Total balls = 54
- $P(\text{Red}) = \frac{1}{3}$
- $P(\text{Green}) = \frac{4}{9}$

$$\text{Red balls} = \frac{1}{3} \times 54 = 18$$

$$\text{Green balls} = \frac{4}{9} \times 54 = 24$$

$$\text{Black balls} = 54 - (18 + 24) = 12$$

3. Correct Answer:

From the given statements, let us check each one carefully:

(i)

6 dozen apples = 72 apples

Probability of rotten apple = $\frac{1}{4}$

Rotten apples = $\frac{1}{4} \times 72 = 18$

Good apples = $72 - 18 = 54$

Statement is True (T)

(ii)

Total cards = 52

Black queens = 2

$$P(\text{black queen}) = \frac{2}{52} = \frac{1}{26}$$

Given probability = $\frac{1}{13}$

Statement is False (F)

(iii)

Numbers: 2, 4, 6, ..., 20 (10 numbers)

Prime numbers in this list = only **2**

$$P(\text{prime}) = \frac{1}{10}$$

Given probability = $\frac{1}{19}$

Statement is False (F)

4. Correct Answer: C) 36

Bag with green and pink balls

Given:

Number of green balls = **9**

Let number of pink balls = **x**

Probability of pink ball $P(\text{pink}) = \frac{x}{x+9}$

Probability of green ball $P(\text{green}) = \frac{9}{x+9}$

Given: $P(\text{pink}) = 4 \times P(\text{green})$

$$\frac{x}{x+9} = 4 \times \frac{9}{x+9}$$

$$x = 36$$

5. **Correct Answer: C**

   C) Aces = 4 $\frac{4}{52} = \frac{1}{13} \neq \frac{1}{4} Incorrect$

6. **Correct Answer: B)** $\frac{3}{4}$

   Probability of occurrence and non-occurrence

   Given: $P(\text{event}) = \frac{P}{2}$

   $$P(\text{non-event}) = \frac{5}{8}$$
   $$P(\text{event}) + P(\text{non-event}) = 1$$
   $$\frac{P}{2} + \frac{5}{8} = 1$$
   $$P = \frac{3}{4}$$

7. **Correct Answer: B)** $\frac{1}{26}$

   Probability of getting a red queen

   Red queens = 2

   Total cards = 52

   $$P = \frac{2}{52} = \frac{1}{26}$$

8. **Correct Answer: A)** $\frac{11}{12}$

   Two dice rolled − sum ⩽ 10

   Total outcomes = 36

   Unfavorable outcomes (sum > 10):

   - 11 → 2 ways
   - 12 → 1 way

   Total unfavorable = 3

   Favorable outcomes = $36 - 3 = 33$

   $$P = \frac{33}{36} = \frac{11}{12}$$

9. **Correct Answer: C) 24**

   Box with green and red balls

   Given:

   Green balls = 12

   Let red balls = x

   $$P(\text{red}) = \frac{x}{x + 12}$$
   $$P(\text{green}) = \frac{12}{x + 12}$$

   Given: $P(\text{red}) = 2 \times P(\text{green})$

   $$\frac{x}{x + 12} = 2 \times \frac{12}{x + 12}$$
   $$x = 24$$